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AntifragiCity KPIs and Framework Development

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Abstract

The development of the AntifragiCity KPIs' framework for urban mobility aims to foster environments that thrive amid disruptions, tackling urban degradation, systemic complexities, and limitations in existing assessments lacking adaptability, inclusivity, and holistic scope. Employing a three-stage mixed-methods approach, the research systematically reviewed 2.540 indicators from academic articles, EU projects, and grey literature, complemented by a practitioner survey across eight European cities. Synthesis revealed overlaps in efficiency, safety, sustainability, and equity, but gaps in resilience. Practitioners selected only 31% of indicators, favouring socio-economic and technical metrics over ecological and safety ones, shunning complex antifragility composites due to data constraints.

The resulting framework is adaptive, dynamic, and modular, structured via an IPOO model. It integrates the AntifragiCity Ontology for semantic data management and LCA for sustainability assessments. Validated against Sustainable Development Goals, European Green Deal, and Global

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Plan for Road Safety, it supports data-driven decisions, Mobility Triage, Decision Support Systems, and the SUMA API as an open-source tool for transferable resilience quantification.

Keywords

Key Performance Indicators, indicator framework, urban mobility, resilience, antifragility, sustainability

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R: document, report (excluding periodic and final reports).
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Abbreviations

API	Application Programming Interface
BIM	Building Information Modelling
DSS	Decision Support System
EU	European Union
GDP	Gross Domestic Product
GHG	Green House Gas
IPOO	Input – Process – Output – Outcome framework
JSON	JavaScript Object Notation
KPIs	Key Performance Indicators
LCA	Life Cycle Assessment
MAS	Multi Agent System
PRT	Private Transport
PT	Public Transport
SDG	Sustainable Development Goal
SUMA	Simulator for Urban Mobility Antifragility

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1 Executive summary

The development of the AntifragiCity KPIs' framework for urban mobility, aims to create urban environments that not only withstand disruptions but also improve as a result of them. The motivation for this work stems from the pressing need to address urban degradation, navigate the complex interplay of urban systems, and overcome limitations in current urban mobility assessment practices, which often lack adaptability, inclusivity, and holistic scope.

The research employed a robust three-stage mixed-methods approach. It began with a systematic literature review of academic articles, EU-funded research projects, and grey literature, processing 2.540 indicators to form a foundational pool. This was followed by a comprehensive practitioner survey involving urban mobility experts across eight (8) European cities to gather practical insights and validate relevance. Finally, these theoretical and practical inputs were integrated to define the robust KPIs' framework.

Key Findings and Conclusions:

- **Literature Review Insights:** The review revealed a complex interplay and frequent overlaps among urban mobility facets like efficiency, safety, sustainability, equity, and ecological footprint. While sustainability and efficiency/performance received predominant focus, critical gaps were identified in the underrepresentation of safety, resilience, and equity in existing frameworks. Analysis of terms highlighted a persistent emphasis on quantifiable technical aspects ("number," "vehicle," "time") alongside a growing recognition of human-centric elements ("person," "population"). Distinct priorities emerged across different sources: academic literature focused on theoretical robustness and environmental sustainability, research projects on practical application and social sustainability, and international organizations on overall performance, efficiency, and safety.

- **Practitioner Survey Results:** The survey indicated that practitioners adopt a tailored, context-specific approach to KPIs, commonly selecting only 31% of the available indicators. Categories like "Urban development and growth," "Transport modes, travel times and characteristics," and "Ecological impacts" consistently garnered attention. However, a notable imbalance was observed, with socio-economic and technical metrics overrepresented in practice, while crucial areas like ecological impacts, transport safety, and network performance comparatively to research recommendations, were underrepresented. A significant finding was the practitioners' non-selection of composite indicators from theoretical antifragility models, underscoring a gap between academic theory and practical implementation due to data and resource limitations.

- **The AntifragiCity KPIs' Framework:**
    - **Design:** The proposed framework is comprehensive, flexible, adaptable, and dynamic, moving beyond static measurements to an adaptive approach. It is structured around an IPOO model, integrating diverse quantitative and qualitative data.
    - **Key Components:** It leverages the AntifragiCity Ontology Framework as a semantic backbone for consistent data, event elicitation, and integrating KPIs' concepts. It also

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incorporates LCA methods to evaluate long-term sustainability, environmental footprint, and resource management, enhancing its predictive power for urban stressors.

- **Compliance & Validation:** The framework is rigorously validated against real-world urban data in demo sites and adheres strictly to international standards and regulations, including the Global Plan for Road Safety, Stockholm Declaration, Sustainable Development Goals, New Urban Agenda, and the European Green Deal.

- **Recommendations and Impact:** The AntifragiCity KPIs' framework is designed to serve as a critical instrument for enhancing urban mobility, supporting data-driven decision-making, and facilitating adaptable strategies. It will be integral to Mobility Triage processes, inform a Decision Support System, and refine antifragile traffic control strategies. Crucially, it forms the core analytical foundation for the SUMA API, providing quantifiable metrics for urban mobility resilience. Beyond the project, the framework is envisioned as an open-source, transferable tool for assessing and enhancing urban resilience across various domains (e.g., energy, water, public health), promoting a universal framework for urban resilience and sustainability quantification.

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2 KPIs for urban mobility: state-of-the-art

The following sections delve into the state-of-the-art KPIs' frameworks utilized for assessing urban mobility, acknowledging their pivotal role in decision-making and continuous evaluation of the crucial aspects of efficiency, sustainability, safety, and environmental impacts. First, a concise overview of the current state-of-the-art in urban mobility KPIs reveals a multifaceted landscape, and then a systematic KPIs-oriented literature review across various sources provides a comprehensive understanding of existing indicators' frameworks and their application in diverse contexts of mobility systems.

2.1 KPI-driven frameworks for urban mobility

The adoption of KPIs in the context of urban mobility offers significant advantages to the management and optimization of transportation systems, enabling a more robust evaluation of urban conditions and policy actions [1,2]. The KPI-based frameworks provide quantitative metrics to systematically assess the progress toward specific mobility aspects, such as sustainability [3], efficiency [4,5] and resilience [6,7]. In fact, cities are widely utilizing KPIs' frameworks to evaluate current situations, understand evolutionary trends, and gauge the impact of chosen solutions [1]. This strategic application of KPIs facilitates data-driven decision-making, allowing urban authorities to refine policies and technologies for enhanced performance in areas such as traffic management and accident reduction [2]. Simultaneously, these frameworks support policymakers and planners in making sound decisions regarding efficient resource utilization, the determination of priorities, the procurement of financial resources, and the allocation of funds [8], as well as informed choices in service planning for urban transportation systems [9].

KPIs are instrumental in measuring and comparing the performance of transport systems against their short-term goals [2,4]. They also serve as a valuable tool for comparing mobility levels and impacts across different cities [1], and are essential for international benchmarking, especially for monopolistic systems [9]. Through these evaluation processes, the KPI-driven frameworks identify success criteria and critical indicators for assessing techno-economic, environmental, and social impacts, which is crucial for developing sustainable and realistic urban development policies [10,11]. Moreover, the KPIs' frameworks provide a structured approach for experts to identify deficiency areas and pinpoint areas requiring additional action [1,2].

An important advantage of the KPI-based evaluation frameworks lies in their ability to integrate diverse data sources and methodologies, to generate comprehensive indicator sets, which are transparent and comprehensible by experts from various fields [2]. To this end, these frameworks enhance communication among various stakeholders, thereby supporting the successful design and implementation of urban mobility projects [8]. Specifically, this approach facilitates a pre-emptive impact assessment of potential projects against high-level objectives [2], as well as the post-construction monitoring of project outcomes against predetermined benchmarks [1]. Especially for the mobility systems, monitoring KPIs during the operation phase is crucial for verifying their efficacy and for making necessary adjustments to optimize performance and achieve long-term sustainability goals [10,12].

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Furthermore, the continuous monitoring of KPIs enables dynamic adaptation to unforeseen challenges and emerging trends, ensuring the urban mobility framework remains responsive and effective in mitigating stressors, including potential Black Swan events [10,13]. In addition, this approach supports the establishment of realistic targets and recognizing best practices [9]. By systematically monitoring these performance indicators, cities can not only track progress towards sustainability but also proactively identify areas for improvement and guide resource allocation effectively [14]. Therefore, the KPI-driven evaluation frameworks also offer valuable reference for new urban transportation and mobility systems during their planning and design phase, informing the planning of their operational scheme [15], the optimization of their performance with the integration of intelligent systems [4] and ultimately, the enhancement of their overall resilience and sustainability [16,17].

However, apart from the abovementioned benefits, the implementation and consistent application of KPI frameworks in urban mobility present distinct challenges, including data acquisition, standardization, and the dynamic nature of urban environments [2]. Firstly, many limitations stem from the availability, measurability, reliability, and feasibility of data [3]. The selection of urban mobility indicators often depends on the existence of specific data sources, which can hinder objective and comparable assessments, especially when such data is unavailable [18]. This issue is further compounded by the lack of robust data collection and management systems and data ownership, leading to disparities in data availability, fragmentation, and incompatibility, which hinder the establishment of consistent metrics [4,19].

Secondly, despite the extensive number of defined mobility indicators and the relevant claims to assess the transportation system as a whole, there is a preference to focus primarily on public transportation and private motorized vehicles, often overlooking active modes like walking and cycling [3]. The comprehensive accounting for sustainability is frequently overlooked, resulting to limited scope and focus of the current KPIs' frameworks [3,20,21]. Furthermore, the high variability in indicator use and the fragmentation of the field - with many new indicator systems introduced with limited reference to previous work - lead to inconclusive and incomparable assessments, hinder the accumulation of knowledge, and ultimately impede the development of universally applicable and comprehensive assessment tools [3].

Therefore, the difficulties in consistent measurement and comparison is a persistent limitation pertinent to all KPI-based assessment frameworks and especially, those applied to urban mobility where diverse data streams, stakeholder inputs, and system interdependencies complicate uniform evaluation [9,10]. Additionally, operationalizing and implementing certain indicators can prove overly complex and challenging to measure in practice, for example, the percentage of public transport ticket transactions utilizing electronic payment technologies [4]. Consequently, some KPIs may be too specific to particular technological solutions or lack relevance across a wide range of member states [4], while others may focus on easily measurable objectives, which can lead to the exclusion of more subjective and qualitative aspects of sustainability, such as social equity and well-being [3]. Moreover, existing frameworks often lack governance dimensions critical for implementation in developing regions, rarely incorporate citizen perceptions alongside objective measurements, and frequently require data inputs unavailable in resource-constrained environments, thereby limiting their applicability and generalizability [10].

These challenges highlight the need for a more adaptable and inclusive framework that can accommodate varying data availability, integrate qualitative and quantitative inputs, and provide a holistic assessment of urban mobility [3,18,21]. An effective KPI framework needs to address these limitations and provide a unified representation of urban dynamics, thereby facilitating a more robust and comprehensive assessment of the urban pulse. For this purpose, a systematic literature

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review is critically needed to synthesize existing knowledge, identify these inherent deficiencies, highlight promising practices, and ultimately guide the development of more adaptable, inclusive, and robust KPI frameworks. These frameworks can provide a comprehensive assessment of the sustainability, resilience, and antifragility of urban mobility.

The research conducted in Task 2.7 particularly examined the capacity of KPI-based frameworks to capture two key aspects: the urban pulse and Black Swan events. Specifically, the term *"urban pulse"* captures the continuous and varying activity that shapes the urban environment, including the dynamic properties of cities, the rhythms of urban practice, the spatiotemporal variations, as well as the aspect of urban vitality. In contrast, *"Black Swan events"* refer to unpredictable, high-impact occurrences that deviate significantly from normal expectations, often requiring entirely novel responses to mitigate their disruptive effects on urban systems.

2.2 Review methodology

The literature review followed a systematic approach to ensure transparency, comprehensiveness, and replicability. Given the variety of sources in the field of urban mobility, comprising research articles, operational reports, policy documents, and conference proceedings, a multi-faceted search strategy was employed to capture the breadth of contemporary research and practical applications. This approach ensured a broad exploration of scholarly and grey literature, facilitating a holistic understanding of existing KPIs' frameworks and their practical implementation in the context of urban mobility.

The review included three (3) pools of sources. First the primary academic literature, comprising peer-reviewed journal articles and conference papers on KPI-based assessments of urban transport and mobility systems. Second, EU funded research projects that developed specific KPI frameworks were also considered, recognizing their direct relevance to policy implementation and practical application within European urban contexts. Finally, the grey literature from technical reports, white papers, and publications by international organisations in the wide field of urban transport and mobility, such as the International Transport Forum (ITF) and the World Bank.

The review process, which is graphically presented in **Figure 2.1**, encompasses four (4) consecutive steps. The initial step involved defining the search criteria, specifically publications after 2015 and the appropriate set of keywords. The keywords were strategically chosen to capture the multidisciplinary nature of urban mobility, combining the terms *mobility / transportation* with the terms *efficiency / safety / sustainability / equity / ecological footprint*, aligning with the core dimensions of the urban pulse framework. The search additionally included the terms *indicators / KPIs* to ensure the identification of relevant assessment frameworks.

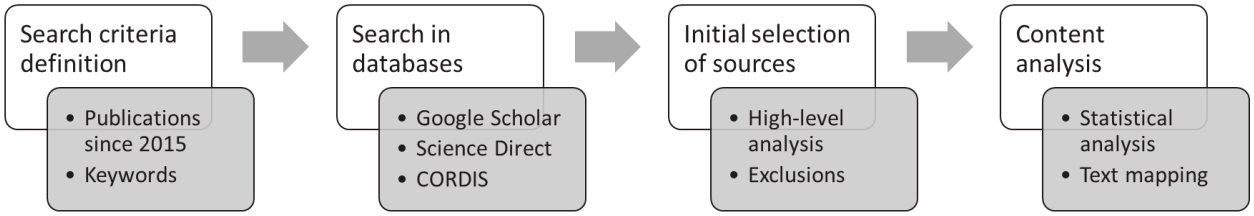

Figure 2.1. Steps of the literature review process

The next step of the review process involved searches in the appropriate databases. The databases selected for the systematic review of the academic literature included ScienceDirect and

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Google Scholar, based on their extensive coverage of scientific literature. The database for the review of the research projects' reports was CORDIS as the official repository for EU-funded initiatives. Regarding the search for the reports from international organisations, a broader set of search engines and organizational websites were employed to encompass the diverse range of reports and policy documents. This meticulous selection process aimed to maximize the comprehensiveness of the review, addressing the limitations inherent in single-database searches.

As shown in **Figure 2.1**, after the initial screening, which involved the application of the exclusion criteria, the selected documents underwent a high-level analysis of a preliminary assessment of titles and abstracts. This step was crucial for identifying the most pertinent studies, filtering out those that did not directly address KPI frameworks in urban mobility. Subsequently, a full-text review of the remaining documents was conducted to extract detailed information regarding their identified KPIs and contextual applications, ensuring a thorough understanding of each framework's contributions and limitations. The content of the remaining documents was further subjected to statistical analysis and text-data mapping, which included bibliometric analysis and qualitative content analysis to identify co-occurrent terms, emerging themes, and prevalent indicators within the literature. The statistical analysis was implemented in MS Excel workbooks, while the text-data mapping was performed using VOSviewer. The content analysis led to the determination of the thematic areas and consequently the formation of the indicator groups that underpin the comprehensive framework for assessing the urban transport and mobility performance under consideration of multiple critical dimensions. This rigorous, multi-stage review methodology thus enabled a comprehensive synthesis of the current state-of-the-art in urban mobility KPIs' frameworks, identifying key trends, gaps, and best practices across diverse sources to inform the development of a novel, more robust and comprehensive framework.

An overview of the sources identified in these three (3) pools is illustrated in **Figure 2.2**. This visualization will delineate the distribution of various publication types, such as academic publications, research projects, and international organisations' reports, providing insight into the primary avenues of knowledge dissemination within this domain. More specifically, a total of 63 academic articles were included in the final pool, in which 77 indicator groups and totally 796 indicators were identified, representing the highest number of individual metrics documented across all source categories. The pool of research projects included 15 EU-funded research projects from different funding programmes, 13 closed and 2 ongoing, from which 223 indicators were extracted. Conversely, the grey literature encompassed reports from 10 international organizations, contributing an additional 177 unique indicators to the cumulative dataset. The complete lists of the examined reports form research projects and international organizations that were included in this review, are presented in Appendix A.

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Figure 2.2. Overview of the selected sources

In total, the review process identified 2.540 indicators, including 1.713 KPIs from the academic literature, 460 from funded research projects, and 367 from organisational reports. Due to the extensive number of indicators and the inherent overlaps identified, the indicators were further analysed as context-specific groups, overlooking slight differences in assessment or measurement.

2.3 Literature review findings

While the review sought a comprehensive understanding of urban mobility indicators by targeting the key areas of efficiency, safety, sustainability, equity, and ecological footprint, the findings revealed that the complex interplay among these facets often results in varied interpretations and overlaps across the examined thematic areas. In fact, the content analysis revealed that certain indicator groups frequently cover multiple thematic areas, as indicatively shown in the excerpt from **Table 2.1** highlighting the inherent interconnectedness of urban systems.

Indicator Group	Thematic Area
Accessibility – Land use mix (Number of job opportunities and commercial services within 30-minute travel distance)	Economic Sustainability / Equity
Affordability (Portion of household expenditures devoted to transport by 20% lowest-income households)	Equity / Economic Sustainability
Appropriate use of land	Environmental Sustainability
Balancing family costs in transportation	Economic Sustainability / Equity
Citizen involvement	Social Sustainability
Climate change emissions (Per capita fossil fuel consumption, CO_2 and other climate change emissions)	Ecological Footprint
Community livability	Social Sustainability
Commute access	Economic Sustainability / Performance
Control of emission toxic substance and harmful	Ecological Footprint

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Damaged area (from disturbances)	Performance/Resilience
Diversifying public transportation methods	Social Sustainability / Performance
Environment (as a dimension of sustainable transportation)	Environmental Sustainability
Equity – disabilities	Equity
Equity – fairness	Equity / Economic Sustainability
Evolution of performances (time period for improvement, comparison of levels, effect of disturbance)	Performance/Resilience
Extent of congestion (miles of roadway where average journey times are 30% longer than unconstrained)	Performance/Efficiency
GDP, economic efficiency, income distribution, unemployment rate	Economic Sustainability
Increasing Access to public transportation	Equity / Social Sustainability
Increasing the share of transportation in economic well-being	Economic Sustainability
Level of Service	Performance/Efficiency

Table 2.1. Identification of thematic areas for given indicator groups

This analytical complexity underscores the necessity for a nuanced approach to the discrimination of the thematic aspects that allows for a holistic understanding of urban dynamics rather than a reductionist view. For this purpose, the identified thematic aspects are framed as distinct yet potentially intersecting areas, rather than isolated silos. In line with the current research trends [3,14], this integrated perspective emphasizes how interconnected various aspects of urban life are, making it essential to adopt a comprehensive assessment framework that captures these intricate relationships rather than treating each aspect in isolation.

Although the original distinctions among these diverse aspects remain evident, the sections within these areas form specific subareas that facilitate a more granular analysis of urban pulse components, which are illustrated in Figure 2.3a. For example, within the performance facet, some indicators focus more on the efficiency of mobility systems, while others address the resilience of the relevant infrastructure and the served community, potentially extending also to safety aspects. Simultaneously, within this conceptual approach, broader aspects like sustainability can effectively be specified in terms of the environmental, social, and economic dimensions that permeate various sub-areas, as frequently observed in studies evaluating urban mobility systems [17,22].

(a) Landscape of thematic areas (b) Indicators per thematic area

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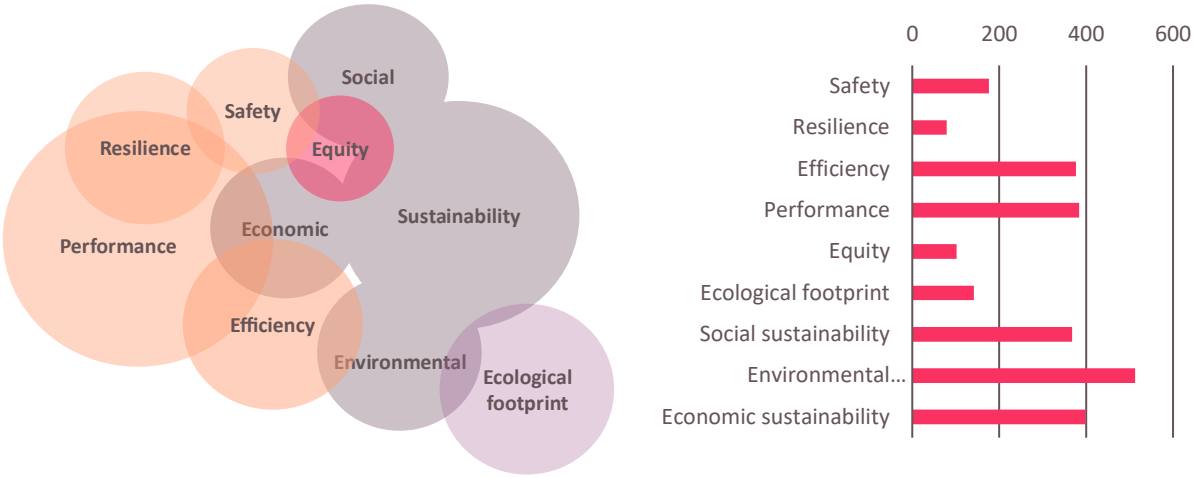

Figure 2.3. Landscape of thematic areas (a) and distribution (b) of indicators

According to the evidence of Figure 2.3b, the statistical analysis of the retrieved indicators reveals that the broader aspect of sustainability that integrates the environmental, economic, and social dimensions is predominantly represented across numerous frameworks, particularly when considering its environmental and economic dimensions. The integrated sustainability aspect accounts for 50,39% of the total 2540 collected indicators, with the environmental (20,20%), economic (15,71%), and societal (14,49%) dimensions each making a substantial contribution. The second most prevalent category of indicators encompasses two interrelated aspects, namely performance (15,12%) and efficiency (14,84%) that are core to the technical assessment of transportation systems. This category constitutes 29,96% of the total indicators, reflecting a significant focus on operational effectiveness and system output in planning and operating urban mobility systems.

The remaining thematic areas, namely safety, resilience, equity, and ecological footprint, each account for less than 7,00% of the indicators. This distribution underscores areas that warrant greater integration into holistic urban pulse frameworks, despite their critical importance for comprehensive urban sustainability assessments. Particularly the thematic area of resilience has the smallest contribution of 3,11%, followed by the area of equity with 4,02%. This finding aligns with broader literature indicating a general underrepresentation of resilience and equity metrics in urban sustainability frameworks, which often prioritize environmental and economic dimensions [23].

The categorization of indicators within thematic areas and their emergent intersections informed a statistical clustering process of indicator groups, based on conceptual relationships and reference frequencies in the literature, to identify prevailing thematic areas and subareas. Figure 2.4 presents a detailed breakdown of these groups by thematic areas and subareas, along with their occurrence frequencies, highlighting the dominant ones essential for a comprehensive KPIs' framework.

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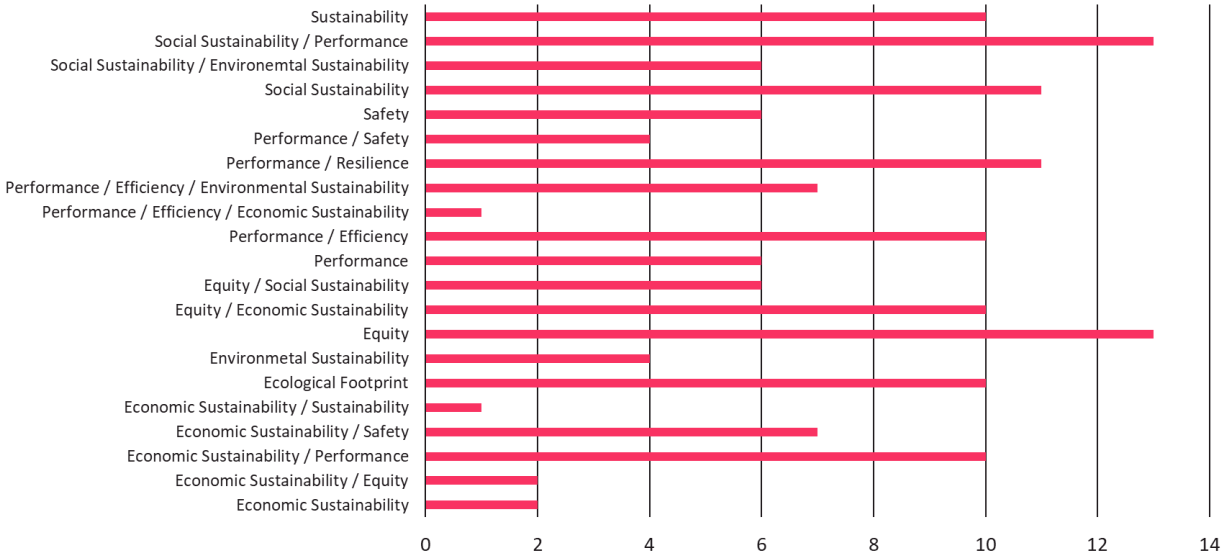

Figure 2.4. Distribution of indicators in thematic areas and subareas

A closer examination of the definitions of indicators collected from diverse research and policy sources on urban mobility reveals the assessment approaches used to quantify them. The relevant definitions were analysed in a text-data mapping process with VOSviewer software, which categorized them based on the co-occurrence of terms employed and their association. Figure 2.5 presents the mapping of the most co-occurring terms with binary counting and normalized based on their association strength. This visualisation demonstrates by the nodes' size the number of co-occurrences and by the edges' weight the interrelation between these terms. Specifically, the term number, vehicle and time are the most frequently employed metrics in the examined KPIs, whereas the term number plays a pivotal role in quantifying various aspects of urban systems, from vehicle counts to population densities, often serving as a fundamental unit of measurement across diverse indicators.

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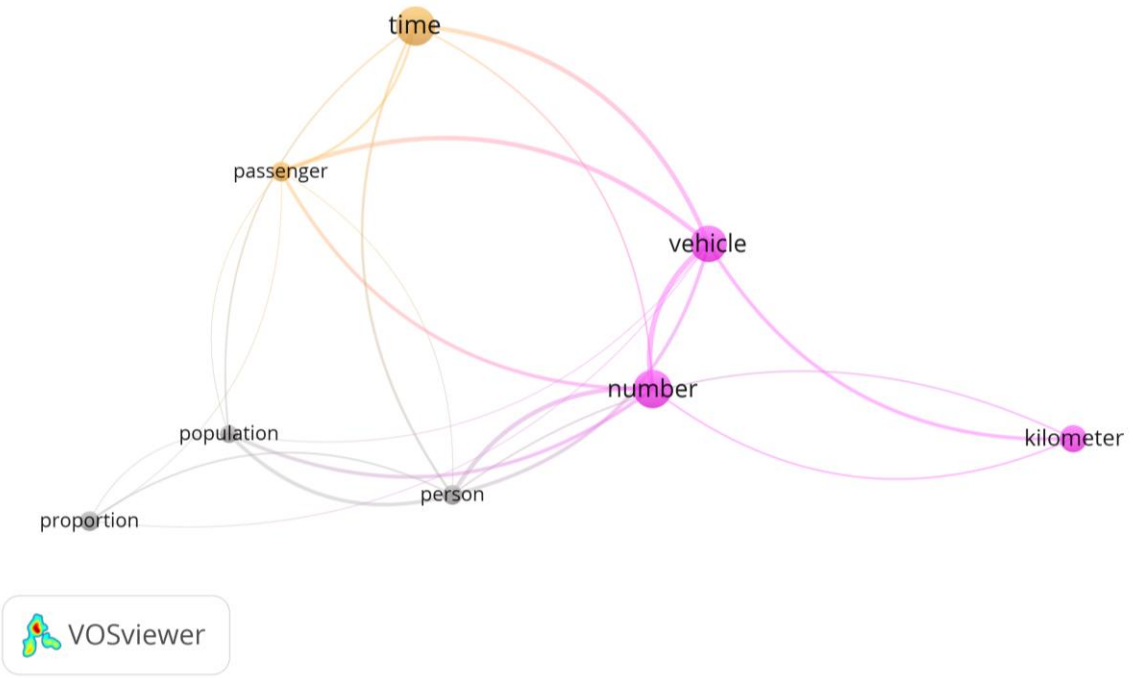

Figure 2.5. Text-data map of the terms in KPIs' definitions

The map of Figure 2.5 also demonstrates three (3) discrete clusters of most associated terms, specifically the first cluster, depicted in magenta, primarily focuses on terms related to *number*, *vehicle*, and *kilometre*, indicating a strong emphasis on traffic flow and infrastructure metrics. The second cluster (ochre) highlights *time*, and *passenger*, suggesting a focus on the mobility flows and patterns within urban environments. Finally, the third cluster (grey) is defined by the terms *person*, *population*, and *proportion*, which are indicative of demographic and social dimensions, thereby emphasizing the human-centric aspects of urban mobility. This categorization underscores the variations in measurement approaches and the differing priorities within urban mobility assessments.

It is also important to examine the findings derived from the three (3) different source categories for a more comprehensive understanding of how different types of literature contribute to the overall landscape of urban mobility KPIs. Based on the results in Figure 2.6, it is evident that the different sources, i.e. academic papers, research projects' reports, and policy documents, despite their common trends, tend to emphasize distinct facets of urban mobility, reflecting their inherent objectives and target audiences. For instance, the aspect of resilience is more frequently discussed in academic papers and research projects, while organisational reports almost omit it. Moreover, the facets of equity, and ecological footprint are meticulously examined in the academic literature, whereas policy documents and research project reports tend to give them less attention in favour of other aspects. It is also important to note that the international organizations propose a balanced distribution of KPIs among the sustainability dimensions, whilst the academic researchers emphasize the dimension of environment, and the research projects prioritize the social aspect.

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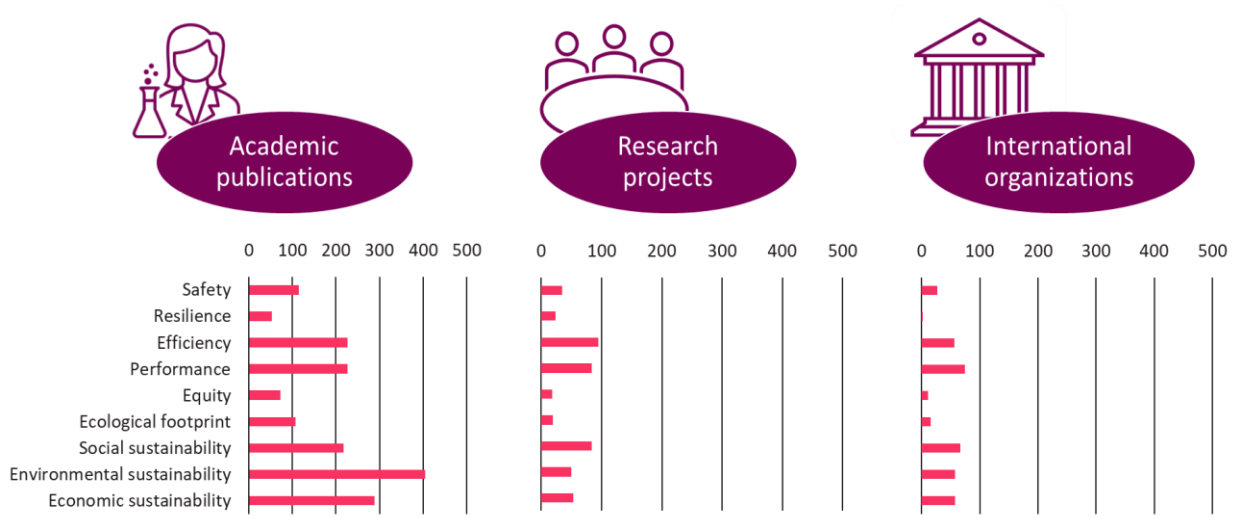

Figure 2.6. Distribution of indicators per pool category and thematic area

This nuanced distribution of emphasis across various source categories highlights the need for a comprehensive KPIs' framework that integrates insights from diverse perspectives to achieve a holistic understanding of urban mobility assessment. To this end, the subsequent subsections 2.3.1 - 3 examine in greater detail the characteristics of the KPIs' frameworks identified in each of the three (3) source categories, namely the academic literature, the research projects' reports, and the international organizations reports.

2.3.1 Academic literature

The academic literature produced the highest number of indicators per thematic area and the most comprehensive and theoretically robust frameworks for urban mobility assessment. These frameworks frequently incorporate sophisticated methodologies for data collection and analysis, often leveraging advanced statistical techniques and modelling to capture complex urban dynamics [10]. Additionally, academic contributions often explore novel indicators, integrating interdisciplinary perspectives to address emerging challenges in sustainable and resilient urban mobility [18,24]. This scholarly focus on methodological rigor and innovative metrics positions academic literature as a critical foundation for developing comprehensive urban pulse frameworks, contributing significantly to both constitutive and conceptual advancements in responsible urban mobility systems [12]. Specifically, these academic frameworks delve into the intricacies of indicator selection, often emphasizing criteria such as understandability, utility, comprehensive coverage of categories, and strong conceptual grounding in existing literature to ensure relevance and applicability [8,25]. Furthermore, a significant portion of academic research prioritizes the development of indicator systems over composite indices, largely due to the methodological complexities and inherent subjectivity associated with index construction, such as weighting schemes and normalization methods [14,18].

Given the high number of KPIs derived from academic publications, the statistical analysis of their frameworks reflects the primary trends in the overall distribution. The charts of Figure 2.7 highlight the slight differences between the research trends expressed from the academic literature and the findings from the overall statistical analysis of KPIs. This underscores the academic sector's role in pioneering detailed and exhaustive indicator sets that often become foundational for broader

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applications in planning and policy-making for urban mobility. This evidence indicates that, while the academic literature develops mobility indicators across all thematic areas, it clearly prioritizes those related to environmental sustainability over the other two sustainability dimensions and all the remaining aspects. This emphasis on environmental indicators within academic discourse is particularly evident in studies focusing on urban transportation sustainability assessments, which often review extensive frameworks and indicators for measuring progress towards sustainable transport systems[3].

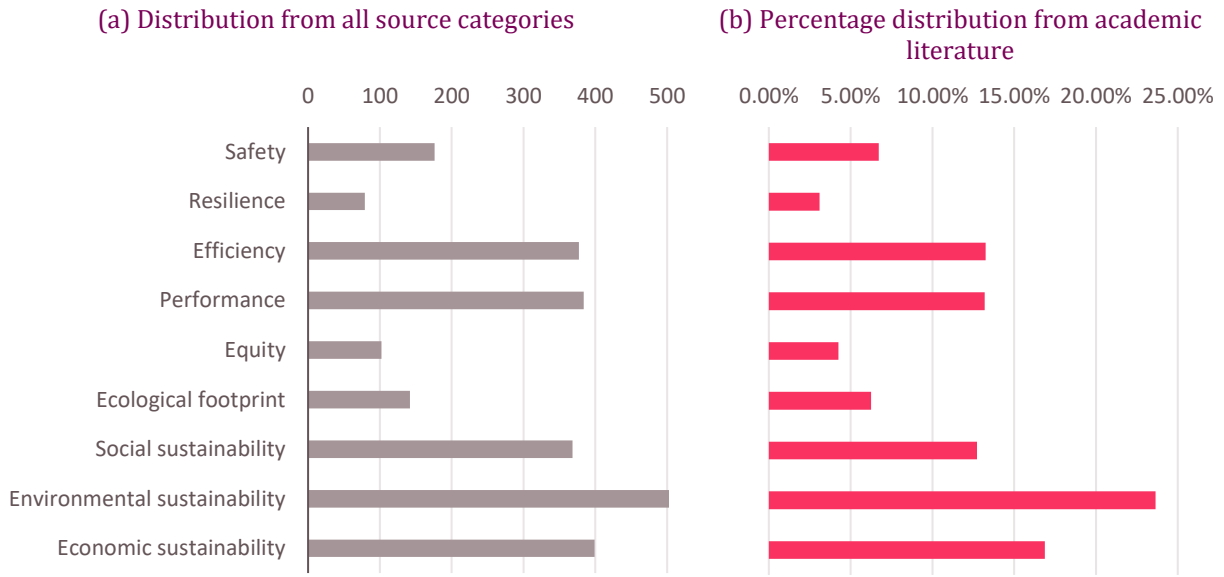

Figure 2.7. Distribution of indicators from all source categories (a) and from academic literature (b)

The findings also suggest that the KPIs' frameworks from the academic sources tend to under-represent the area of social sustainability, however the aspect of equity is consistently addressed as a discrete dimension. This trend suggests a gap in the comprehensive integration of social aspects beyond equity within academic frameworks, despite the recognized importance of social dimensions for holistic urban sustainability. Moreover, the technical aspects of performance and efficiency are steadily addressed, often through indicators pertaining to traffic flow, congestion, and operational throughput, thereby reflecting a persistent focus on infrastructural and logistical optimization within urban mobility studies, as Karjalainen and Juhola[3] also suggest. In contrast, while discussions of resilience, safety, and ecological footprint are present and relevant to infrastructural aspects, they often receive less attention compared to sustainability discussions in academic frameworks. This divergence underscores a need for further academic inquiry into comprehensive social sustainability metrics and integrated resilience and ecological impact assessments within urban mobility frameworks.

The most frequently occurring terms employed in the academic discourse on indicator-based mobility assessment are demonstrated in the text-data map of Figure 2.8. According to this mapping, the term *time* is key to the academic research on mobility KPIs and often appears in conjunction with concepts such as travel time reliability, commuting duration, and real-time traffic monitoring, highlighting its centrality to understanding urban mobility dynamics. Furthermore, the cluster of terms *vehicle*, *emission*, *capita* that demonstrate strong association among each other, reflect the

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indices of transportation systems, which are often analysed in terms of their environmental impact and per vehicle and per capita usage.

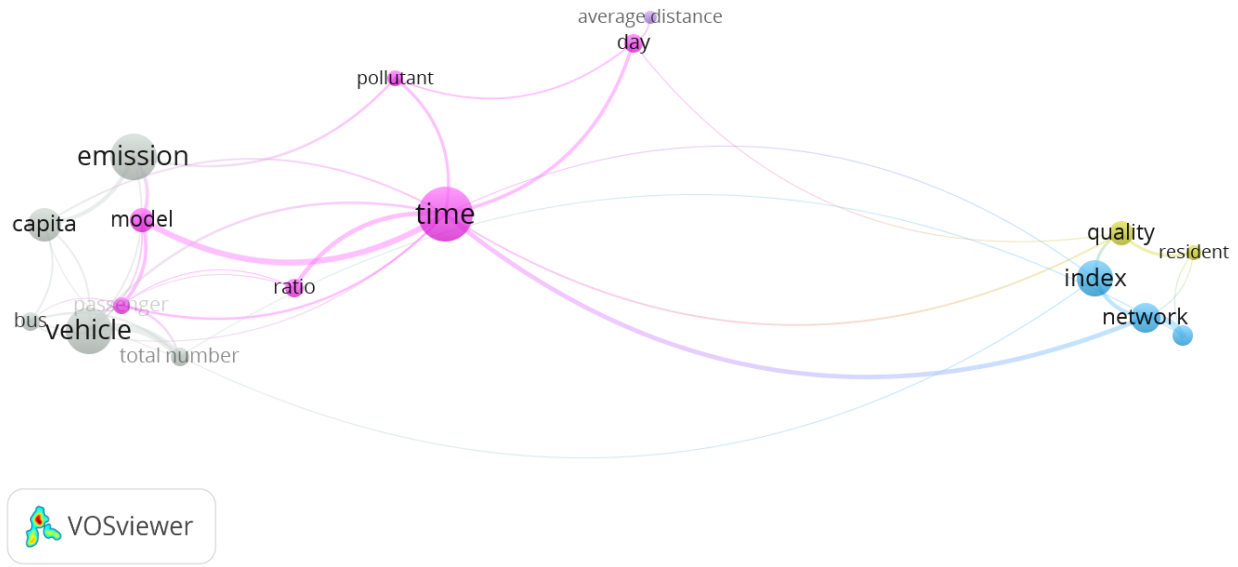

Figure 2.8. Text-data map of the terms from academic literature

Frequently co-occurring terms that derive from different research strands, are displayed in the right-hand side of the map. The term index and network are central to a discrete set of academic frameworks focused on complex urban system modelling and network analysis, often involving sophisticated graph theory applications to evaluate connectivity and efficiency. This suggests a strong academic inclination towards quantitative, composite, and readily measurable indicators that aim to capture the dynamic interplay of urban mobility systems. Moreover, the separate cluster of the terms *quality* and *resident* indicates that there is a different strand focusing on the human-centric aspects of urban mobility, emphasizing how transportation quality directly influences the lived experience and satisfaction of citizens, particularly in relation to accessibility and user preferences.

This dual focus highlights a dichotomy in academic research: some studies emphasize technical performance and environmental impact, while others delve into the social dimensions and lived experiences of urban inhabitants. This disparity in focus suggests a critical area for future interdisciplinary research, aiming to bridge the gap between technical, environmental performance metrics and the nuanced social and experiential dimensions of urban mobility, thereby fostering more holistic and integrated mobility assessment frameworks.

2.3.2 Research projects' reports

The research projects' reports represent a significant practical application of theoretical frameworks discussed in academic literature, frequently providing empirical validation and showcasing the implementation of urban sensing technologies to inform policy and planning decisions. Statistically they provide the 18.14% of the collected mobility and transportation indicators. These reports often translate complex theoretical models into actionable metrics, enabling cities to monitor and improve transportation efficiency, sustainability, and overall urban liveability, frequently incorporating advancements in smart city technologies and data analytics, like in Allmer et al. [26] or in Dionelis et al. [11].

The comparison between the distribution of these indicators per thematic area and all identified KPIs in Figure 2.9 reveals a distinctively different prioritization. Specifically, the technical aspects of performance and efficiency are given paramount importance in research project reports, often focusing on optimization strategies, real-time traffic management, and infrastructure utilization. This emphasis aligns with practical imperatives to enhance urban functionality and mitigate immediate operational challenges within urban mobility systems [1]. In contrast to academic discourse, which often highlights environmental sustainability, these reports underscore the pragmatic need for quantifiable metrics that directly inform infrastructure development and operational improvements, like for instance in the work of Allmer et al. [26] This practical orientation often translates into a focus on developing urban mobility indices, which condense numerous indicators into a single, comprehensive score for simplified assessment and decision-making.

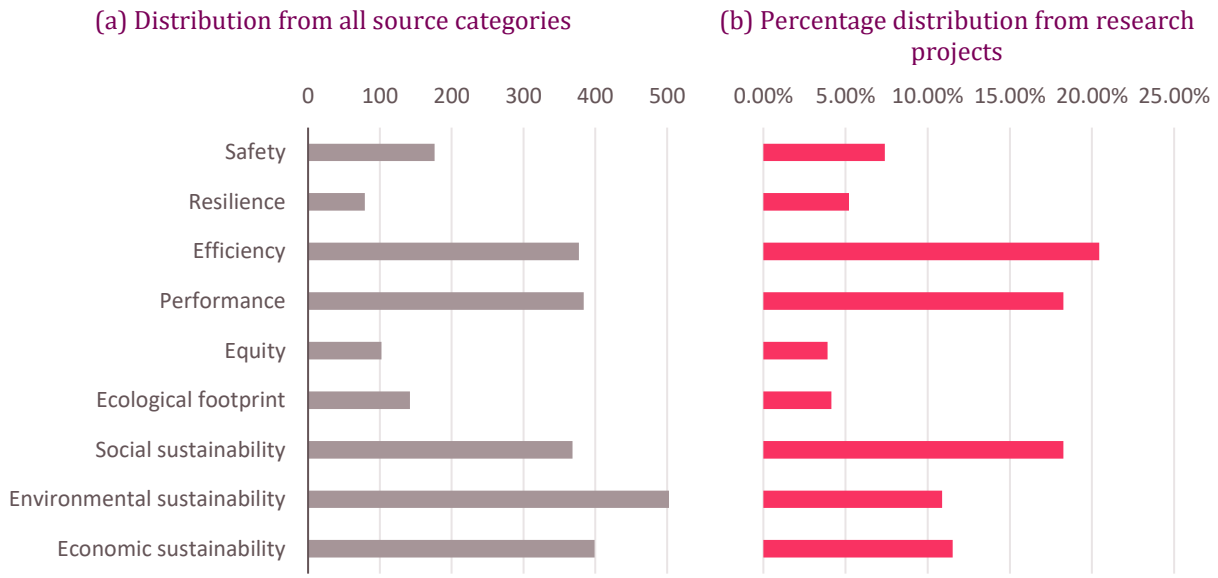

Figure 2.9. Distribution of indicators from all source categories (a) and from research projects (b)

According to the evidence of Figure 2.9, research projects' KPI frameworks seem to integrate a higher percentage of indicators in the thematic area of resilience. This observation suggests a pragmatic emphasis on systems capable of withstanding disruptions and adapting to unforeseen circumstances, as indicated by the funding priorities and policy directives often associated with these projects. Furthermore, the focus safety of users and citizens is another critical aspect that gains prominence in the field of EU-funded research on urban mobility. This prioritization of safety aligns with broader policy goals aimed at ensuring well-being and efficient operation of urban systems, often emphasizing equity in accessibility and infrastructure provisions to enhance safety and efficiency. Ultimately, the thematic area of ecological footprint, on the other hand, is under-represented in this source category, probably owing to the contextual specificities of the engineering discipline.

The text data map of the research projects' indicator-based frameworks of Figure 2.10 showcases a lower number of frequently occurring terms and a broader distribution of unique terminology, reflecting the diverse and often highly specialized nature of individual project objectives and methodologies, thereby exhibiting less thematic convergence compared to academic

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literature. The identified terms are also characterized by looser and less complex, almost linear association patterns, indicating that while projects often share common high-level goals, their detailed operational approaches and specific indicator selections can vary significantly.

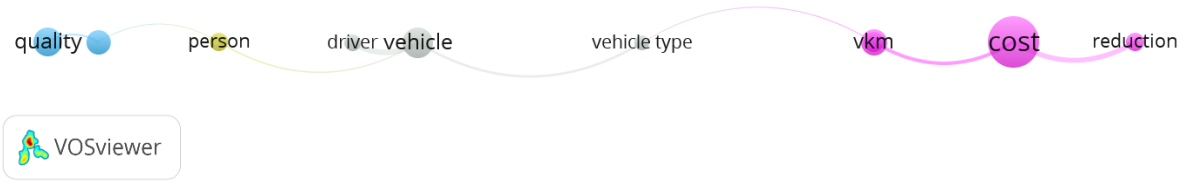

Figure 2.10. Text-data map of the terms from research projects

The most frequently used and co-occurring term in these indicators' frameworks is *cost*, underscoring the pervasive economic considerations that drive the implementation and evaluation of urban mobility solutions within practical research initiatives. The assessment of KPIs related with the terms *vehicle* and *driver* are also popular in this research strand, highlighting the continued focus on individual transportation modes, traffic aspects, and human factors in operational efficiency and safety. Furthermore, the prevalence of these terms suggests a persistent emphasis on traditional transportation paradigms, potentially overlooking multimodal integration and sustainable alternatives. Moreover, the discrete cluster of the terms *quality* and *availability* underlines the importance of service levels and accessibility, which are crucial for user satisfaction and system effectiveness and also for the facet of social sustainability. This indicates a persistent focus on tangible operational aspects and immediate user experience within project-based research, often at the expense of broader environmental impacts or long-term urban resilience strategies [27].

2.3.3 International organisations' reports

According to Figure 2.11, the KPIs' distribution in the reports from international organizations demonstrates a significant emphasis on the aspect of performance and efficiency, which aligns with their mandate to promote robust and effective urban systems globally. This is evident in the frequent appearance of terms related to operational throughput and infrastructural optimization in their indicator frameworks, reflecting a prioritization of measurable outcomes and systemic functionality. However, it is interesting to note that the aspect of performance is more pronounced compared to efficiency, suggesting a focus on overall output and achievement rather than the input-output ratio. This distinction highlights a strategic emphasis on the attainment of specified goals and service delivery benchmarks, which are critical for demonstrating progress toward global sustainability agendas.

Additionally, reports from the examined international organisations pursue a balanced exploration of the sustainability dimensions, with a slight prioritisation of the societal dimension. This focus reflects in practice the principles of sustainable development and the imperative to consider social aspects alongside environmental and economic factors in urban planning and policy of the urban mobility systems. Simultaneously, the findings from the collected policy documents of Figure 2.11 suggest a significant interest towards the parameter of safety. This fact underscores a commitment to safeguarding citizens and infrastructure, often manifesting in regulations and initiatives aimed at accident prevention and risk mitigation in urban environments.

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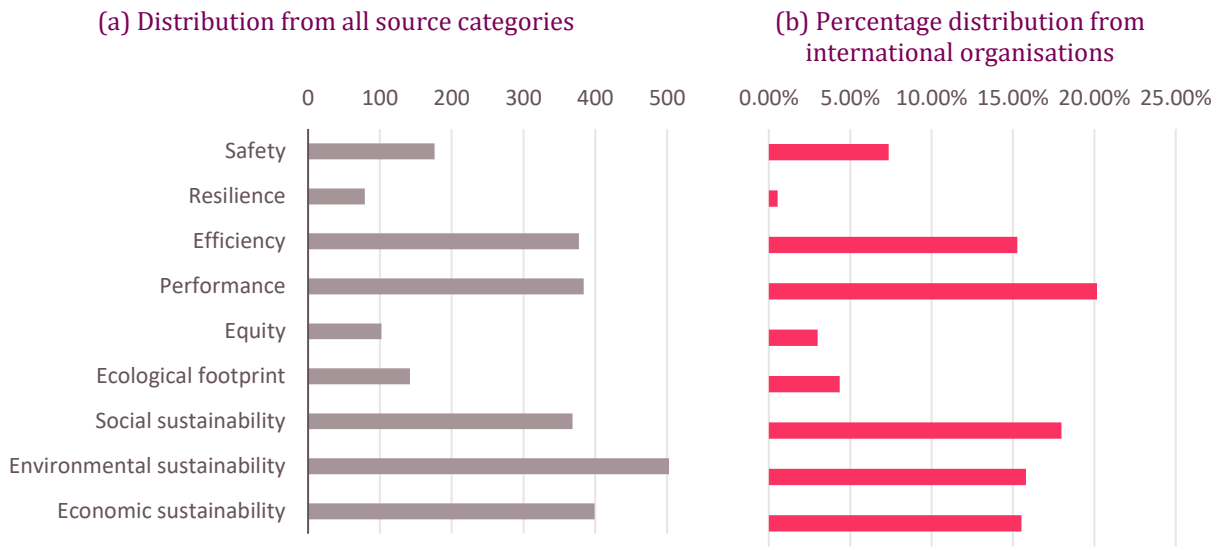

Figure 2.11. Distribution of indicators from all source categories (a) and from international organisations (b)

Conversely, the minimal emphasis on resilience and the relatively lower emphasis on ecological footprint within these international reports suggests a potential area for enhancement, particularly in integrating comprehensive environmental impact assessments such as LCAs more prominently into their indicator frameworks [28]. Moreover, the attention to equity is also reduced in these reports, indicating a potential oversight in addressing disparities in access to urban services and infrastructure that should be addressed [29].

Regarding the definitions of the transport and mobility indicators identified in the organisational reports, Figure 2.12 demonstrates the map of the most used terms and their associations. A notable observation is the structure of the map, where more than one terms have a central position, suggesting a multifaceted understanding of key performance indicators rather than a singular dominant focus. While the number of terms employed is significant, there is a convergence of frequently used terms and their interrelations that signify common priorities across diverse international organizations. Specifically, the terms *number* and *service* are centrally located within this network, indicating a strong emphasis on quantifiable service delivery metrics as foundational elements in assessing urban mobility performance.

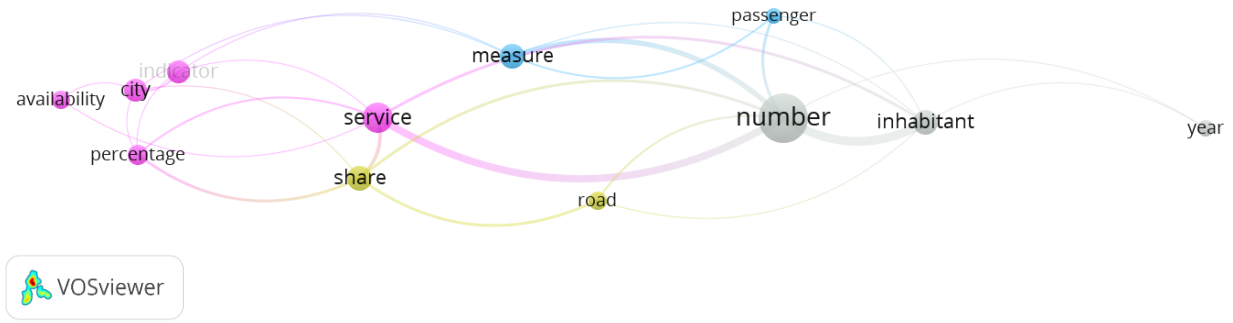

Figure 2.12. Text-data map of the terms from international organisations

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The text data map of Figure 2.12 confirms that the attention of international organisations revolves around the quantification and standardisation of given performance-related aspects pertaining both to technical and social dimensions of urban mobility systems, which is crucial for benchmarking and comparative analysis across different urban contexts. Among the most co-occurring terms from different clusters are inhabitant, passenger, city, highlighting a user-centric approach to urban mobility planning, emphasizing user experience, and the intricate relationship between urban dwellers and their transportation systems [27].

2.4Discussion

The adoption of KPIs significantly benefits urban mobility and transport management by enabling data-driven decision-making, evaluating urban conditions, assessing policy impacts, and facilitating international benchmarking across aspects like sustainability, efficiency, and resilience. However, the effectiveness of these frameworks is often constrained by limitations in data availability, methodological inconsistencies in KPI selection, and the inherent complexities of urban systems, which frequently lead to an oversimplification of multifaceted issues. Despite their advantages, KPI framework implementation faces challenges related to data acquisition (reliability, standardization), a common focus on motorized transport over active modes, and fragmented, incomparable assessments due to varied indicator use and lack of comprehensive sustainability scope.

The literature review underscores the variety of indicators used globally demonstrating high activity in defining and implementing urban mobility KPIs' frameworks to assess and improve the efficiency, sustainability, safety, equity, and ecological footprint of urban mobility systems. The literature review underscores the variety of indicators used globally demonstrating high activity in defining and implementing urban mobility KPIs' frameworks to assess and improve the aspects of efficiency, sustainability, safety, equity, and ecological footprint of urban mobility systems. Despite that the review process addressed directly these thematic areas, the findings demonstrated that the complexity and the overlap among these topics necessitate a nuanced approach to their individual measurement and collective impact. Therefore, the proposed conceptual approach introduces nine (9) main overlapping thematic areas, as well as their secondary, intersecting subareas of common interest. The main thematic areas include the following facets:

- Sustainability
    - Economic sustainability
    - Environmental sustainability
    - Social sustainability
- Ecological footprint
- Equity
- Performance
- Efficiency
- Resilience
- Safety

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This active development reflects a global commitment to address pressing urban challenges and stressors such the ecological degradation of urban environments and the interplay between human well-being and the environment through quantifiable metrics. The broader aspect of sustainability with its economic, environmental, and social dimensions, and the technical aspects of performance and efficiency are consistently prioritized across various frameworks and sources. This trend is also reflected by the prevalence of generic and versatile terms for the definitions of these indicators and the convergence of different research aspects, namely the infrastructure, the serviceability and the citizenry. Key terms frequently used in KPI definitions include *number*, *vehicle*, *time*, *passenger*, and *cost*, reflecting a focus on quantifiable metrics related to traffic flow, infrastructure, and economic considerations. However, social aspects like *quality* and *resident* also feature, indicating a growing emphasis on human-centric dimensions. This alignment suggests a growing consensus on the critical components necessary for holistic urban mobility assessment, encompassing both the physical urban fabric and the human experience within it [10,18,30,31].

The review also highlighted that different sources prioritize distinct facets: academic literature often provides robust theoretical frameworks emphasizing environmental sustainability, research projects focus on practical application, technical performance, and social sustainability, and international organizations prioritize overall performance, efficiency, and safety. Despite the direct alignment in many areas, the variability in emphasis across these different sources underscores the challenge of developing a universally applicable and comprehensive set of urban mobility KPIs. The academic publications address all the examined thematic areas, but place greater emphasis on environmental sustainability, ecological footprint, resilience, safety, and equity. Following the same trends, the research projects focus more on efficiency, resilience, and social sustainability of urban mobility systems, offering practical insights into implementation and impact. Conversely, international organizations and standardization bodies tend to prioritize broader metrics related to overall system sustainability, and especially the social dimension, systems' performance, and safety, often encompassing economic and governance dimensions.

All the abovementioned observations highlight the need for a more adaptable, inclusive, and holistic KPI framework that can integrate diverse data sources, accommodate qualitative and quantitative inputs, and bridge the existing gaps between technical, environmental, and social dimensions to provide a comprehensive assessment of urban dynamics in the planning and management of mobility systems. The current research on urban mobility necessitates the exploration of the dynamics and complexities of spatial and functional interactions in a transdisciplinary setting, whereas the operational perspective of research projects mandates a coherent baseline to effectively track, compare, and support the progression towards sustainable transportation systems in urban areas, ensuring that new assessment tools and smart technologies incorporate additional dimensions, such as LCA. At the same time, the established organisations and standardization bodies, such as ITF, the World Bank, and EU, emphasize the necessity of robust indicator systems that facilitate continuous monitoring and evaluation for effective governance over time in compliance with existing regulatory frameworks.

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3 Requirements specification and KPIs' definition

This section presents the methodological approach for developing a comprehensive framework capable of sensing the urban pulse, predicting stressors, and defining key performance indicators through a multi-faceted research design. Findings from a specially designed survey enabled the identification and integration of urban authorities' requirements with relevant research trends and those of the AntifragiCity project, yielding an initial catalogue of pertinent KPIs.

3.1 Research design

This research identifies links between current practices in mobility systems assessment and the innovative antifragility methodological framework by developing a comprehensive KPI framework. This framework will also inform context-specific dashboards that capture the pulse of each city. To ensure its applicability, insights from real cities are essential to tailor KPIs to their specific needs and contexts. Furthermore, the research supports indicators for the proposed equilibrium modelling approach presented in D2.3; given their composite nature, it is crucial to assess how easily they can be derived from readily available city data and to adjust the methodological framework as needed.

The specification of the relevant requirements and consecutively, the development of a comprehensive set of urban mobility KPIs for the purpose of the AntifragiCity project necessitates a carefully designed methodology that integrates insights from multiple stakeholders and leverages established frameworks for indicator development. This research employs a three-stage mixed-methods approach, which is presented in the flowchart of Figure 3.1. This three-stage approach involves: a systematic review of existing literature a survey among urban mobility practitioners, and the integration of literature- and practice-based inputs to support the definition of the AntifragiCity KPIs' framework. This structured approach ensures that the resulting KPIs are not only theoretically sound but also practically relevant and aligned with the diverse needs of urban stakeholders, particularly managers and planners of urban mobility systems.

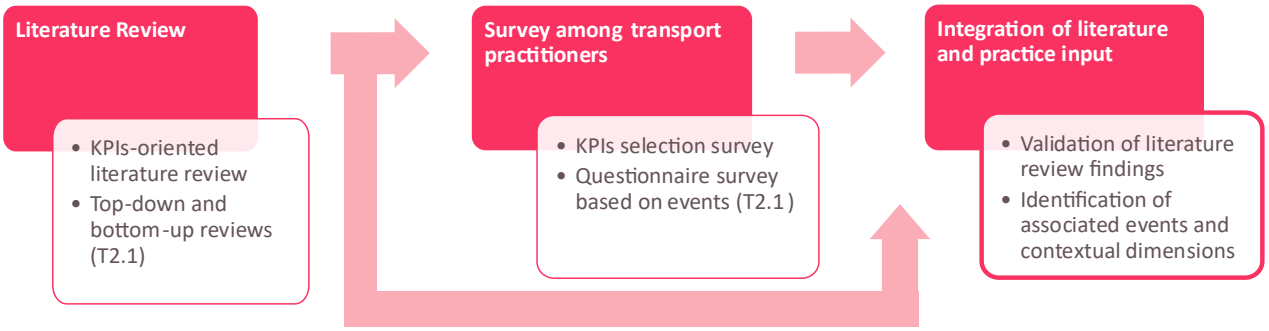

Figure 3.1. Methodology of requirements specification and KPIs' definition

The initial stage focused on the synthesis of the findings from 2 independent systematic literature reviews, namely the systematic review on urban mobility events of D2.1 and the specified KPIs-oriented literature review in the context of Task 2.7. The initial review in D2.1 provided a broad

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overview of urban systems' dynamics during diverse events including the various types of disruptive events, their corresponding impacts, and types of coping strategies. The subsequent, more focused review within Task 2.7 then distilled this understanding into specific indicators and metrics relevant for measuring resilience and sustainability of urban mobility systems. The findings from both systematic reviews collectively formed a foundational pool of potential indicators, providing a comprehensive base for subsequent refinement and selection.

According to Figure 3.1, the following stage of the research on the development of the AntifragiCity KPIs' framework comprised a survey among urban mobility practitioners, including local government authorities, to elicit their requirements and perspectives on crucial performance indicators for urban mobility management. This participatory approach aimed to bridge the gap between academic theory and practical application, ensuring that the developed KPIs are both robust and operationally viable for real-world urban planning and management. More specifically, this stage involved compiling a preliminary catalogue of KPIs, creating a specially designed questionnaire, and distributing it to the project's partners, as shown in Figure 3.2. The outcomes of this survey were subsequently analyzed to refine the KPI set, namely the KPIs' catalogue that underlined the practical insights and priorities of urban mobility professionals.

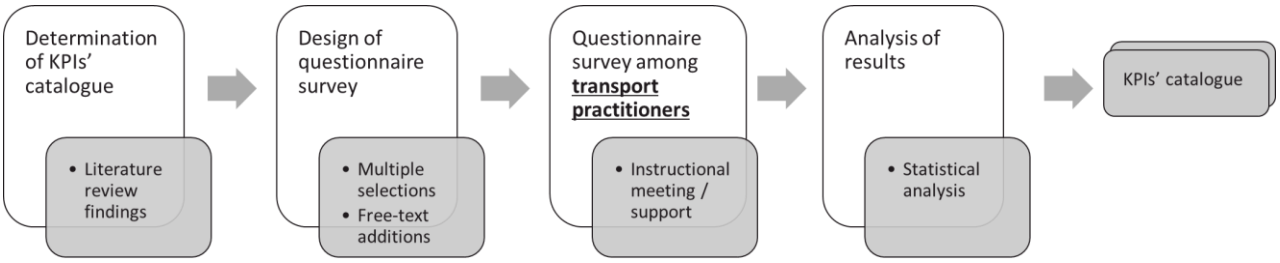

Figure 3.2. Steps of the KPIs' survey process

It is important to note that in order to ensure a larger and more representative sample of urban mobility and transport practitioners the survey was addressed to all the partners from the four AntifragiCity pilot cities, as well as partners who had access to the consultation of their local authorities. The given pilots (Larissa, Bratislava, Odessa, and AHEPA Hospital form Thessaloniki) given their specificities, can only offer limited insights for the generalization of the framework to cities throughout Europe. Consequently, provided the collaboration with numerous transportation experts within the AntifragiCity project, the survey was additionally expanded to more cities. This expansion was critical for gathering a diverse range of perspectives on the assessment practices of urban mobility and transportation from various European contexts, thereby enriching the framework's applicability and generalizability.

The final stage of the proposed methodology for the development of the AntifragiCity KPIs framework shown in Figure 3.1 involved integrating the insights gleaned from the systematic literature reviews and the practitioner survey to finalize a robust and comprehensive set of indicators. This integration process ensured that the selected KPIs were not only theoretically grounded but also practically relevant and reflective of the diverse needs and priorities of urban mobility stakeholders. Moreover, the practice-derived inputs delineate the current capacity and availability in urban environments for data collection and analysis, thereby guiding the feasibility of implementing the proposed KPIs and ultimately, the comprehensive antifragility framework. This iterative process of refinement and validation, encompassing state-of-the-art research, established policies, and stakeholder feedback, culminated in a KPI framework that is both academically rigorous

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and directly applicable to urban planning challenges, including the capacity for cities to measure and improve resilience and sustainability of urban mobility systems.

3.2 Determination of KPIs' catalogue

Building on the literature review, the determination of the KPIs' catalogue followed a structured, multi-step process to ensure comprehensive coverage and practical utility. The KPIs collected during the initial literature review were categorized based on their definitions into groups of identical or similar indicators, refining the initial set of 2.540 indicators into 198 groups. This categorization facilitated the identification of redundant or overlapping indicators, enabling a streamlined and non-repetitive framework. The 11 indicators proposed by the AntifragiCity Equilibrium Framework in D2.3 were then added to this collection, yielding a total of 209 discrete indicator groups. The complete catalogue of refined indicator groups and their definitions is provided in Appendix B.

In order to increase the readability, clarity, and overall utility of the extensive KPI catalogue, a meticulous categorisation process was undertaken. To enhance the readability, clarity, and overall utility of the extensive KPI catalogue, a meticulous categorization process was undertaken. Specifically, the KPIs were categorized according to predominant themes identified in the literature review, including system performance, ecological impacts, transport characteristics, safety, accessibility, infrastructure, and economic development. This yielded seven (7) main thematic groups, each further subdivided into subgroups following the same principle, as shown in Figure 3.3. The final thematic groupings were also validated by mobility experts to ensure logical coherence and comprehensive coverage of urban mobility facets.

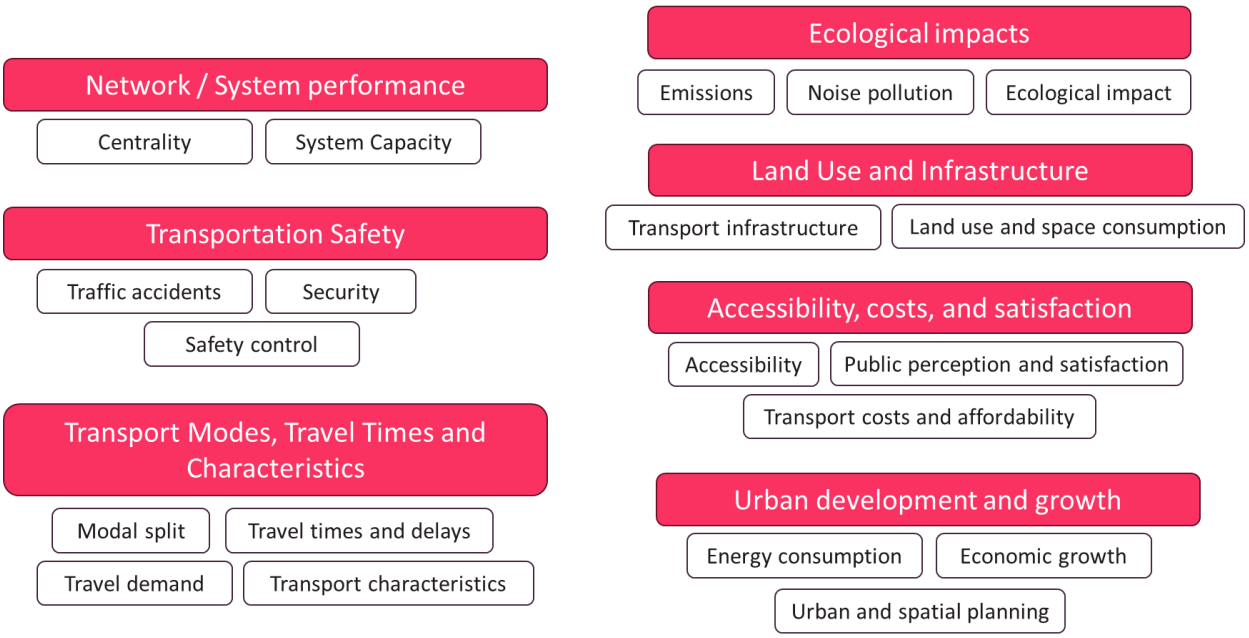

Figure 3.3. Thematic groups and subgroups of the KPIs' catalogue

As illustrated in Figure 3.3, the seven main indicator groups are each subdivided into two (2) to four (4) contextual subgroups, reflecting the complexity and multifaceted nature of the respective urban mobility domain. This hierarchical organization facilitates a granular analysis of urban

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mobility aspects, allowing for a detailed understanding of specific challenges and opportunities within each thematic area. Moreover, the number of KPIs per subgroup varies with the indicator variability of each thematic area, as shown in the pie chart of Figure 3.4. This variable distribution highlights the diverse range of measurable metrics within each domain, from a high concentration in "Transport modes, travel times and characteristics" and “Accessibility, costs and satisfaction“ to a more focused set in "Network / system performance," reflecting the differing complexity and data availability across urban metrics.

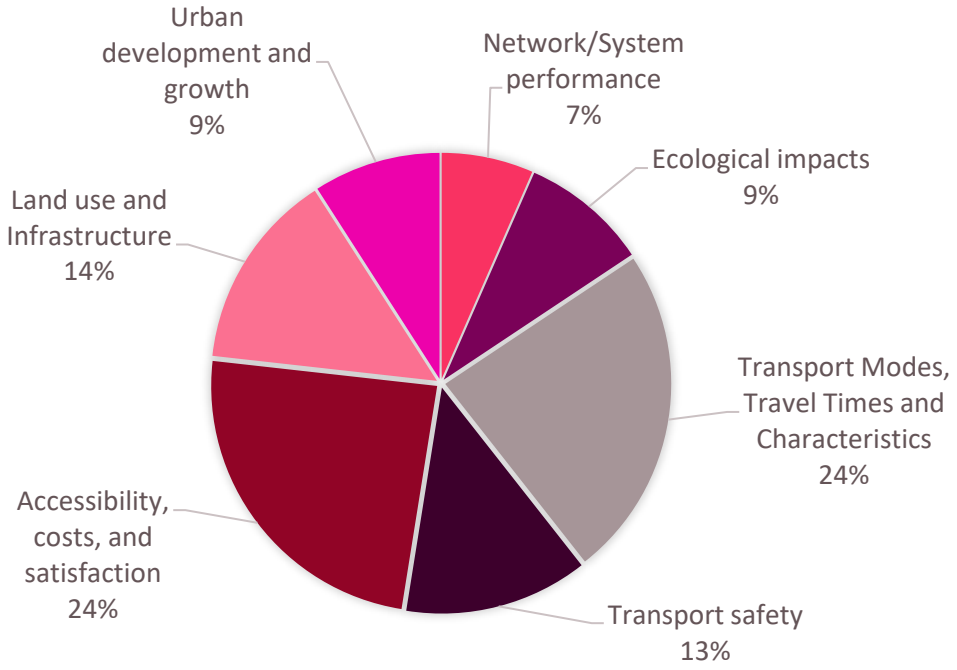

Figure 3.4. Distribution of KPIs per thematic group

Although the “Accessibility, costs and satisfaction” group accounts for one of the highest proportions of KPIs at 24% and “Ecological impacts” one of the lowest at 9%, both contain only three (3) subgroups. This evidence confirms that the number of the identified subgroups is not relevant to the number of KPIs in the respective domain. Rather, the categorization process prioritized thematic relevance and practical applicability, ensuring that each subgroup accurately represents a distinct facet of urban mobility assessment. This structured approach, moving from a broad initial collection to a refined and thematically organized catalogue, mirrors established methodologies in sustainable urban mobility indicator development.

The thematic areas and subareas identified in the KPI-oriented literature review are reflected in the groups of the KPIs’ framework, as shown in Figure 3.5. This correspondence ensures a coherent and comprehensive framework for assessing urban mobility by incorporating diverse aspects, ranging from ecological footprint to transportation system factors and users' accessibility and satisfaction.

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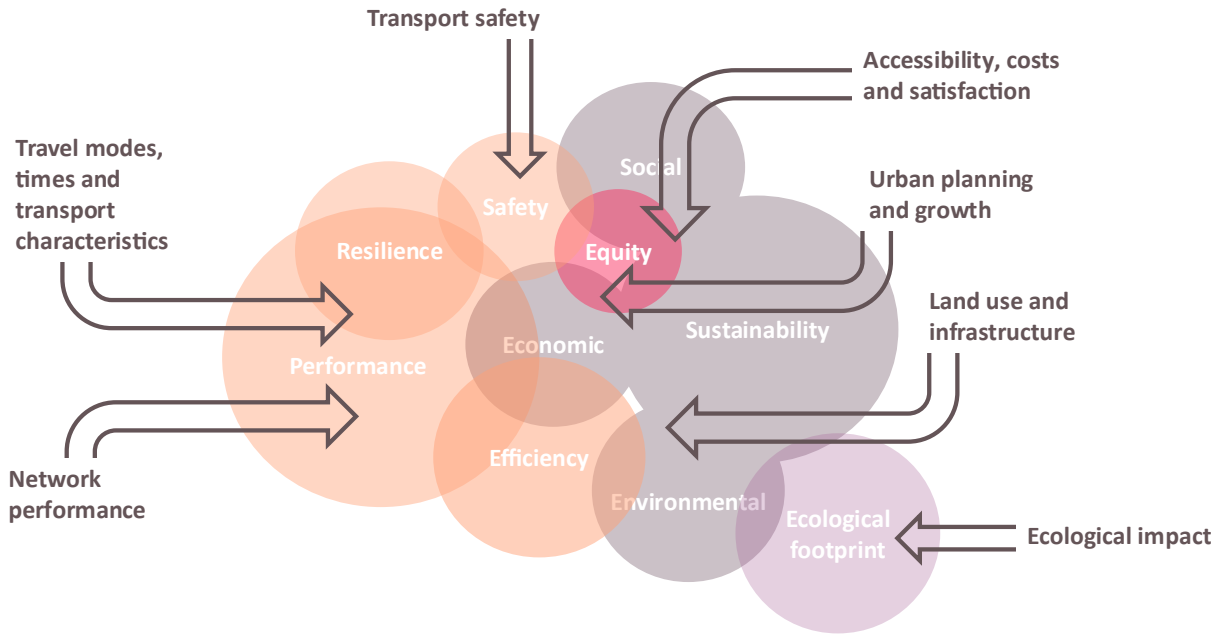

Figure 3.5. Correspondence of KPIs' groups with the identified thematic areas and subareas

All identified KPI groups correspond to specific subareas within the conceptualization of the AntifragiCity KPIs' framework, ensuring that the framework provides a holistic perspective on urban mobility's efficiency, resilience, and sustainability. For instance, these metrics offer a comprehensive view of a city's progress toward climate targets and quality of life enhancements, serving as quantifiable measures of success that guide strategic decisions for more resilient and efficient urban transformations, and particularly for the development of the antifragility assessment framework.

3.3 Questionnaire survey

The questionnaire survey was based on the KPIs' catalogue developed through the comprehensive literature review and refined through expert consultation, forming the empirical foundation for gathering mobility practitioners' perspectives on indicator relevance and feasibility. This survey aimed to capture the perceived importance and applicability of each KPI within various urban contexts, thereby validating the framework's practical utility. The survey also aimed to elicit insights into established practices for KPI selection and application, bridging empirical perspectives with real-world implementation strategies. This approach mirrors methods where experts rate and select proposed indicators for relevance and feasibility, followed by analysis to remove redundancies and identify the most suitable metrics [2,32].

The questionnaire was developed in an MS Excel sheet, a software familiar to all the participants, thereby facilitating broad participation and efficient data collection from diverse experts. Additionally, this familiar user interface enabled participants to provide open-ended responses and comments ensuring their qualitative feedback could be systematically gathered alongside quantitative ratings. The catalogue's entries are organised in a hierarchical structure, reflecting the thematic groupings and subgroups identified during the categorization process. Each KPI within this structure was presented with a clear definition, a description of its intended measurement, and the relevant questions.

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The specific questionnaire presented the entire KPIs' catalogue, organised into the identified groups, indicatively shown in Figure 3.6 and also provided participants with the option to openly suggest additional indicators or offer qualitative feedback on the existing ones. This iterative process of expert consultation and feedback collection, a common practice in developing robust indicator sets, ensures the framework's adaptability and relevance to diverse urban mobility challenges [2].

	Ecological impacts				
	Emmissions				
20	**GHG Emissions**	The primary environmental impact being studied, referring to greenhouse gas emissions from the freight sector, mainly CO2. (Also listed as 'GHG Emissions' for Scope 1, 2, 3 and 'Annual greenhouse gasesa by transportation per capita', 'Annual greenhouse gases by transportation per area', 'Annual greenhouse gases by transportation per total energy used by transportation', 'Annual on-road greenhouse gases per total annual VKT', 'GHG emissions in transportation per capita', 'Share of GHG emissions in the transportation sector from total GHG emissions')			
21	**CO2**	Undesirable output variable, consumption calculated using CO2 emission factor. (Also listed as 'Carbon dioxide emissions', 'CO2 emissions from transportation', 'Carbon emissions per capita', 'Greenhouse gas emissions from transport per capita', 'Per capita emissions of greenhouse gases', 'Carbon emissions', 'Carbon dioxide gas emissions', 'Transport CO2 Emissions')			
22	**Pollutant emissions from transport**	Analyses transport emissions of nitrogen oxides, non-methane volatile organic compounds, and particulate matter. (Also listed as 'Nitrogen oxide emissions', 'Total motor vehicle pollutant emissions', 'Air polluting emissions', 'Annual air pollution emissions by transportation per capita', 'Annual air pollution emissions by transportation per area', 'Annual air pollution emissions by transportation per total energy used by transportation', 'Annual on-road air pollution emission per total annual Vehicle Kilometers Traveled', 'Per capita emissions of global air pollutants', 'Air pollution', 'Air pollution rate')			
23	**Average CO2 emissions per kilometer from new passenger cars**	Measures the average emissions of carbon dioxide per kilometer for new passenger cars.			
24	**Energy consumption and polluting emissions**	Amount of energy used and pollutants released.			
25	**Average density of pollutants generated by transport**	Concentration of pollutants from transport.			
26	**Average exposure to transport-generated pollutants**	Average contact with pollutants originating from transport.			
27	**Average exposure to overall pollutants**	Average contact with all types of pollutants.			
28	**Average duration of the exposure to the pollutants .**	Average length of time exposed to pollutants			
29	**Total polluting emissions**	Overall amount of pollutants released			
30	**NOx emissions**	Emissions of nitrogen dioxide pollution			

Figure 3.6. Indicative caption from the questionnaire survey

Furthermore, the questionnaire included three (3) questions designed to gather insights into the indicators' application practices, specifically about the current use, the applied metric, and the application context, as presented in Figure 3.7. This multi-faceted approach allowed for a thorough understanding of current indicator utilization and the identification of potential gaps or areas for further development within the proposed framework.

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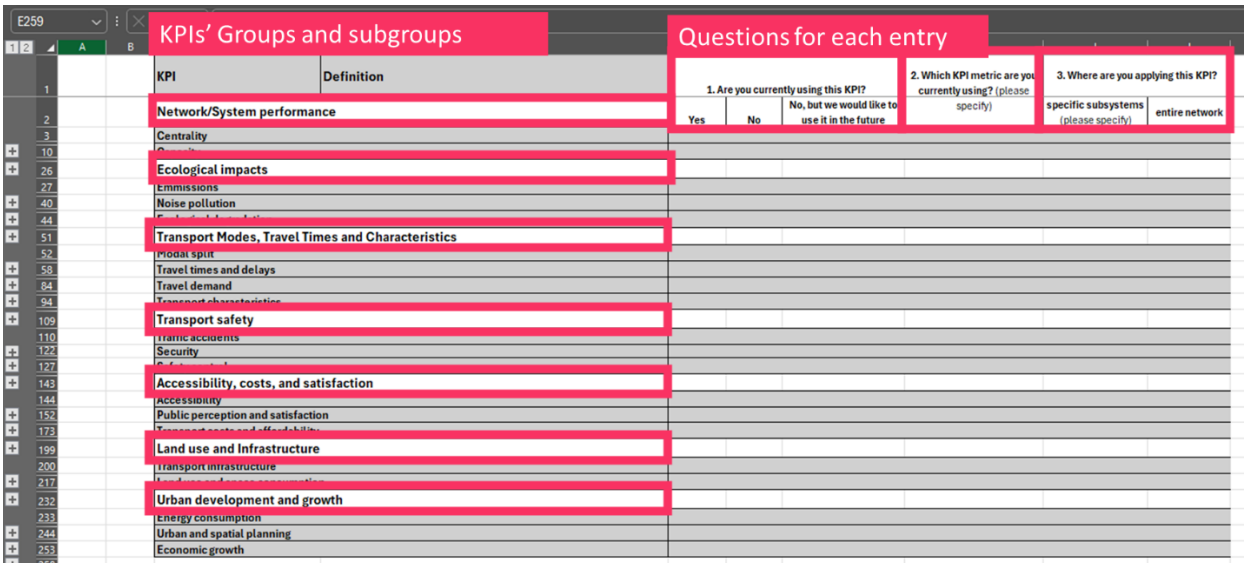

Figure 3.7. Overview of the questionnaire survey

The questions corresponded to each KPI entry in the provided catalogue, as demonstrated in Figure 3.7. The first question aimed to ascertain whether the participant organizations currently utilize the proposed KPIs in their urban planning and management activities, or whether they are planning of incorporating them in the near future, in the context of the AntifragiCity project. The second question delved into the specific metrics and methodologies employed for measuring these KPIs, seeking to understand the practical implementation aspects across different urban contexts. The third question aimed to identify the specific urban planning and management contexts in which these KPIs are applied, thereby revealing their scope, scale, and versatility—for instance, across the entire urban system or within specific subsystems.

The questionnaire survey was distributed to all the AntifragiCity partners who could access urban mobility practitioners. The primary respondents were the four (4) Demo Sites, specifically the partners from the cities of Bratislava, Larissa, Odessa and Thessaloniki. The survey was then extended to university or business partners involved in the project, who could use their pre-acquired experience or leverage their networks to engage additional experts from local mobility and transport authorities. The additional responses involved another four (4) cities, specifically Athens, Cardiff, Delft, and Zurich yielding a total of eight (8) participating cities throughout Europe. This broad distribution strategy ensured a comprehensive collection of expert opinions, reflecting diverse geographical and operational perspectives on urban mobility indicators, which is crucial for the framework's generalizability and robustness.

3.4Findings on current practice

The subsequent analysis of the collected data focused on synthesizing these diverse expert perspectives to identify common trends, significant discrepancies, and areas of consensus regarding KPIs' utility and implementation. This analytical phase involved both quantitative statistical assessments of KPIs' selections and qualitative thematic analysis of open-ended feedback, thereby providing a comprehensive understanding of current practices and future needs in urban mobility performance measurement.

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Out of the eight (8) cities surveyed, six (6) successfully completed the entire questionnaire, yielding a satisfactory response rate of 75%. The respondents represent cities spanning a wide geographical and socio-economic diversity across Europe ensuring a robust and representative sample for analysing current practices in urban mobility KPIs' utilization. It is worth noting that one of the respondents was the city of Odessa that given its geopolitical situation, provided perspectives on the unique challenges and priorities for monitoring mobility performance in its post-war future.

At a first glance, the collected responses revealed significant variations in the prioritization and application of sustainable urban mobility indicators, highlighting the challenge of establishing universally applicable KPIs without contextual adaptation. The chart in **Figure 3.8** illustrates the selection rates of various KPIs across the surveyed cities, showing which ones are commonly monitored and customized for data collection and management in their practices. Conversely, it reveals which cities prioritize a more focused subset of indicators, thereby underscoring the necessity for flexibility in indicator frameworks to accommodate diverse urban contexts and strategic objectives. This heterogeneity aligns with observations in other European urban areas, where a common mandatory approach for Sustainable Urban Mobility Plan indicators has yet to be uniformly defined, leading to country-specific or even city-specific variations in KPI sets [1].

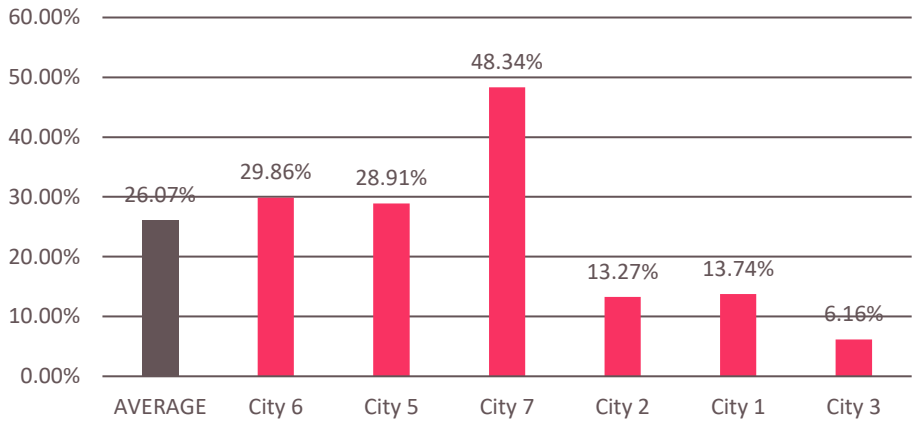

Figure 3.8. KPIs' selection rates per respondent city and average

These results demonstrate that the average selection rate of KPIs, while highly variable, generally ranges around 26%, indicating a selective approach to urban mobility monitoring rather than a comprehensive adoption of all available indicators, as underlined by AECOM[4]. According to Figure 3.8 , City 7 selected more than 48% of the provided KPIs across diverse aspects, while City 3 selected only about 6% of KPIs. Simultaneously, Cities 5 and 6 selected approximately the average rate of KPIs and Cities 1 and 2 reached approximately 13%, respectively, showcasing a spectrum of engagement with the comprehensive indicator set. This selective adoption suggests that cities often tailor their indicator sets to specific local priorities and data availability, rather than employing a one-size-fits-all methodology. This variability emphasizes the importance of adaptable frameworks that can accommodate diverse urban planning objectives and socio-economic contexts, moving beyond rigid, generalized approaches to sustainable urban mobility assessment.

Given the diversity in the number of KPIs per group in the original KPIs' catalogue of the survey, it is necessary to examine both the number of the selected indicators (**Figure 3.9**a) and the selection rates (**Figure 3.9**b) by normalizing them against the total number of KPIs within each respective group to gain a more accurate understanding of cities' preferences and strategic foci. This

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normalization process ensures that comparisons between cities, despite their differing engagement levels with the overall indicator catalogue, accurately reflect their specific prioritization of various aspects of urban mobility. The respective charts of **Figure 3.9** allow a direct comparison of the relative importance given to different mobility dimensions, thereby offering insights into their tailored monitoring and assessment approaches.

Figure 3.9. Average number of selected KPIs' per group (a) and average KPIs' selection rate per group (b)

This evidence indicates that certain KPI groups are selected more frequently than others across the surveyed cities, irrespective of the total number of selected indicators, suggesting a broad consensus on their critical role in assessing urban mobility performance. For instance, the "Urban development and growth" group demonstrated an average number 7,5 of selected KPIs, scoring an impressive selection rate of 41,67%, which highlights its significant perceived relevance in urban planning and sustainability initiatives. This observation is further supported by the fact that this KPI group contains one of the smallest numbers of indicators to select from, highlighting its high importance to the city stakeholders despite its limited size. Conversely, the group of "Accessibility, costs and satisfaction" reached also 12 selected KPIs on average, but it yielded a lower selection rate of 25.00%. This finding highlights that despite the significant number of indicators available within this group — specifically, the most populated one with 48 KPIs — cities exhibit a more discerning approach to their adoption, prioritizing a subset that aligns directly with their strategic objectives and data collection capabilities. It seems that some categories, regardless of size, consistently capture urban mobility practitioners' attention due to their intrinsic link to core urban development goals, others, despite their numerous components, are selectively engaged with, reflecting a nuanced, context-dependent approach to urban mobility assessment.

According to the results of Figure 3.9, the KPIs' groups can be discerned based on their relation between their average number of selected indicators and their selection rate, revealing varying degrees of importance and utility assigned by urban stakeholders. For example, the groups

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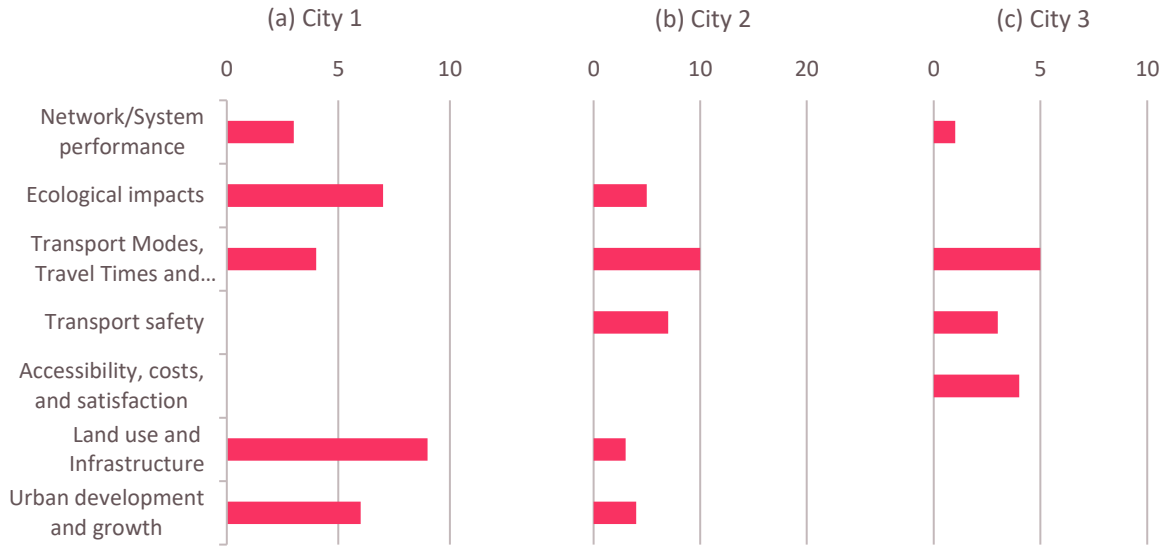
of “Network/system performance”, “Ecological impacts” and "Urban development and growth" exhibit from low to average number of selected indicators and a relatively higher selection rate, reflecting a focused prioritization of these areas in urban mobility assessment. On the other hand, the groups of “Transport modes, travel times and characteristics” and “Accessibility, costs and satisfaction” tend to have a higher number of selected indicators but a lower selection rate, which indicates a broader but less intensive engagement with these dimensions, possibly due to the extensive number of available KPIs within these categories. This suggests that while cities acknowledge the importance of comprehensive monitoring in these areas, practical limitations often necessitate a more strategic, targeted approach to data collection and analysis, particularly for categories with a large number of potential indicators. Furthermore, the groups of “Transport safety“ and “Land use and infrastructure“ demonstrate a balance between the number of selected indicators and their selection rates, implying a consistent and moderate emphasis on these aspects of urban planning and mobility.

3.4.1 Diversity among cities

Figure 3.10 demonstrates analytically the dynamics of KPIs’ selection across the six different urban contexts, illustrating how cities prioritize certain indicators over others based on their unique developmental goals and challenges. These results illustrate the diversity in data availability, methodologies and the localized focus that often guides the KPI-driven evaluation of sustainable urban mobility. This nuanced approach to KPI selection underscores the need for flexible frameworks that can accommodate diverse urban contexts and strategic priorities, allowing for both broad coverage and targeted deep dives into specific areas of interest, as also highlighted by researchers [3,10,14,32]. This contextual adaptability is crucial for frameworks aiming to capture the "urban pulse" comprehensively, allowing for the integration of both universally recognized metrics and those tailored to specific local conditions and stakeholder priorities.



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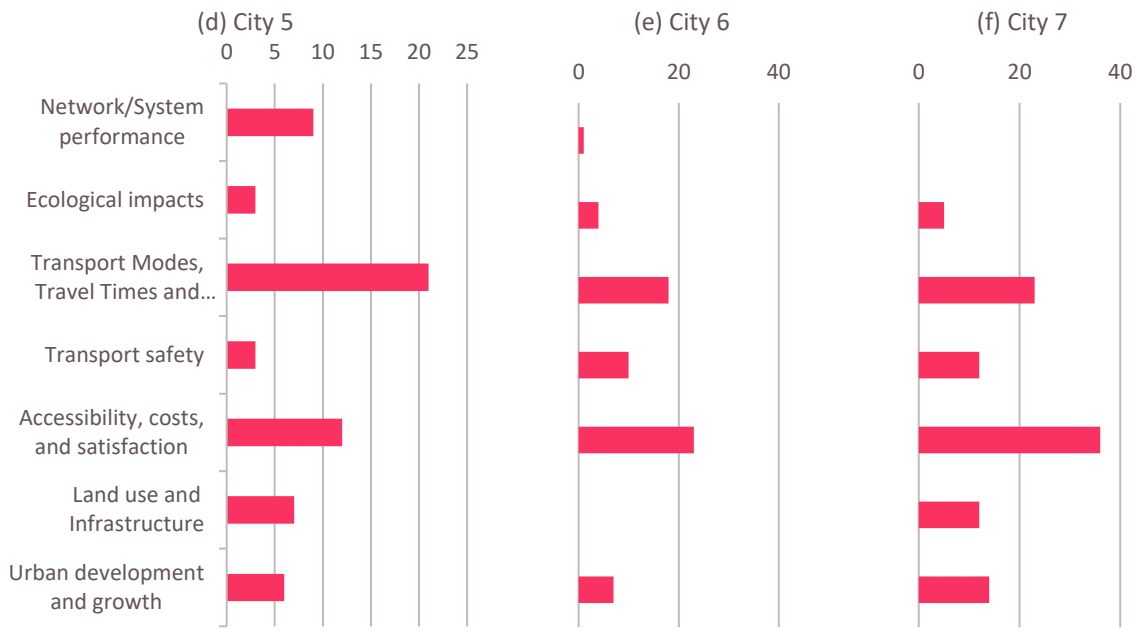

Figure 3.10. Distribution of selected KPIs per group among the respondents (a - f)

The results presented in charts (a) to (f) of **Figure 3.10** reveal a high degree of diversity in the selection of individual KPIs within each group across the surveyed cities. These results demonstrate that even within broadly prioritized categories, specific indicators are chosen based on the unique challenges and opportunities present in each urban environment, whereas some other groups were not selected at all. For example, Cities 2 and 7 excluded completely the group "Network / system performance", while Cities 1 and 2 did not include "Accessibility, costs and satisfaction" in their frameworks, emphasizing the context-specific nature of urban mobility assessments. The only KPIs' group that was consistently selected by all the respondents was the "Transport modes, travel times and characteristics". This consistent selection underscores its fundamental importance in almost all urban mobility analyses, suggesting it serves as a baseline for understanding the operational dynamics of urban transport systems. The second most frequently selected groups were the "Ecological impacts', "Transport safety" and "Urban development and growth" groups, highlighting their pervasive relevance in contemporary urban planning and sustainability initiatives.

The charts in Figure 3.11 demonstrate the distribution of KPIs' selection per specific subgroup in the three most frequently chosen thematic groups, offering a more granular perspective on how particular metrics within these areas are prioritized. Specifically, the most frequently selected groups, ranked in descending order of selection rate, were "Urban development and growth" (Figure 3.11a), "Transport modes, travel times and characteristics" (Figure 3.11b), and "Ecological impacts" (Figure 3.11c), with detailed breakdowns revealing the specific indicators that consistently garner attention within these critical domains.

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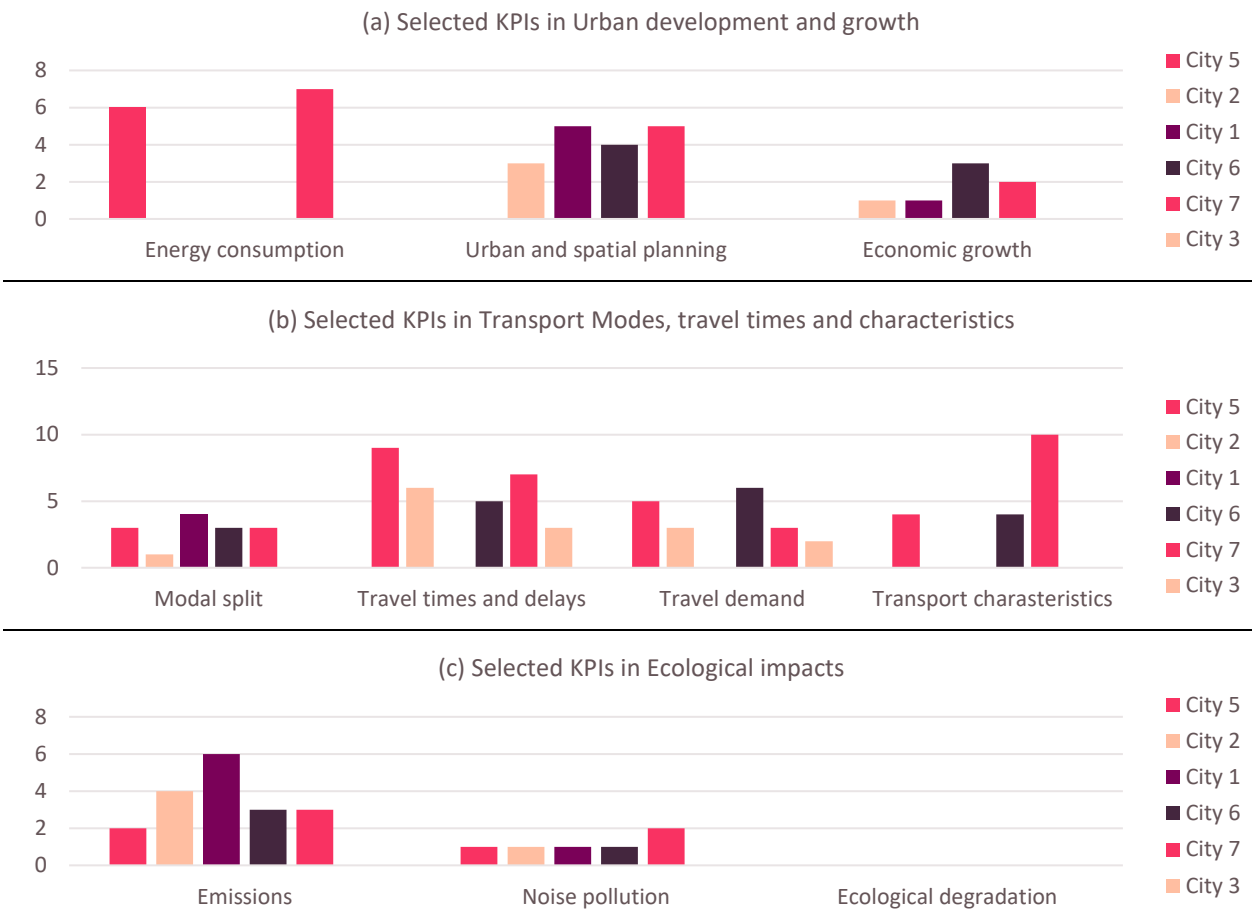

Figure 3.11. Distribution of selected KPIs in subgroups from the groups of "Urban development and growth" (a), "Transport modes, travel times and characteristics" (b), and "Ecological impacts" (c)

In this case, the selection of KPIs followed a similar trend across all cities, although the specific indicators within each category varied based on local priorities and data availability. In the "Urban development and growth" group, for instance, cities rarely selected indicators related to the popular domain of energy consumption. Instead, they most frequently chose KPIs from the urban and spatial planning subcategory — specifically, urban density and green space area. Based on Figure 3.11a, cities also consistently selected KPIs reflecting economic growth, particularly GDP per capita and financial metrics of urban growth per sector. This indicates a strategic emphasis on quantifiable measures of urban expansion and economic prosperity, often prioritizing these over direct energy consumption metrics within the urban development discourse.

On the other hand, the subgroup of "Transport modes, travel times and characteristics" of Figure 3.11b exhibited a strong convergence of the respondents towards the KPIs related to modal split, travel times and delays, particularly with reference to the public transport, as well as travel demand with special focus on metrics about transport and motorized travel demand. However, the subgroup of transport characteristics showed a more diversified selection, reflecting varied approaches to assessing public transport ridership, trip numbers, and speeds.

Conversely, the "Ecological impacts" group demonstrated according to Figure 3.11c, a consistent focus on indicators about diverse emissions, particularly CO_2, GHG, NOx and pollutants

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from transport, underscoring an almost universal concern for environmental sustainability in urban mobility. This pervasive emphasis aligns with broader global initiatives urging the development of indicator systems to monitor progress towards sustainable development goals, as evidenced by efforts from organizations like the United Nations and the Organization for Economic Co-operation and Development [1,18]. Moreover, there is observed a consistent selection of KPIs about noise pollution and specifically, population exposure to transport noise, reflecting an increasing recognition of the health and well-being impacts of urban transport systems. However, surprisingly the subgroup about ecological degradation for the overall assessment of the status of the ecological systems in the urban contexts was completely overlooked, indicating a potential gap in current urban sustainability frameworks regarding a holistic assessment of environmental health beyond direct emissions and noise.

The charts of Figure 3.12 present the analytical KPIs' distributions of two moderately selected groups, namely "Land use and infrastructure" (Figure 3.12a) and "Accessibility, costs and satisfaction" (Figure 3.12b). These groups and their respective subgroups exhibit a higher divergence in their selection patterns, indicating that cities adopt varied approaches to assess and monitor these specific aspects. The "Land use and infrastructure" group, for instance, often includes metrics related to spatial features and particularly length and density of the infrastructure network for the various modes, selected by almost all the respondents according to their infrastructural specifications. However, within this group, the subgroup of land use and space consumption was inconsistently chosen, suggesting a nuanced approach to evaluating the spatial efficiency of urban development beyond mere infrastructure provision. In fact, among the 13 available KPIs, only the population density was frequently cited.

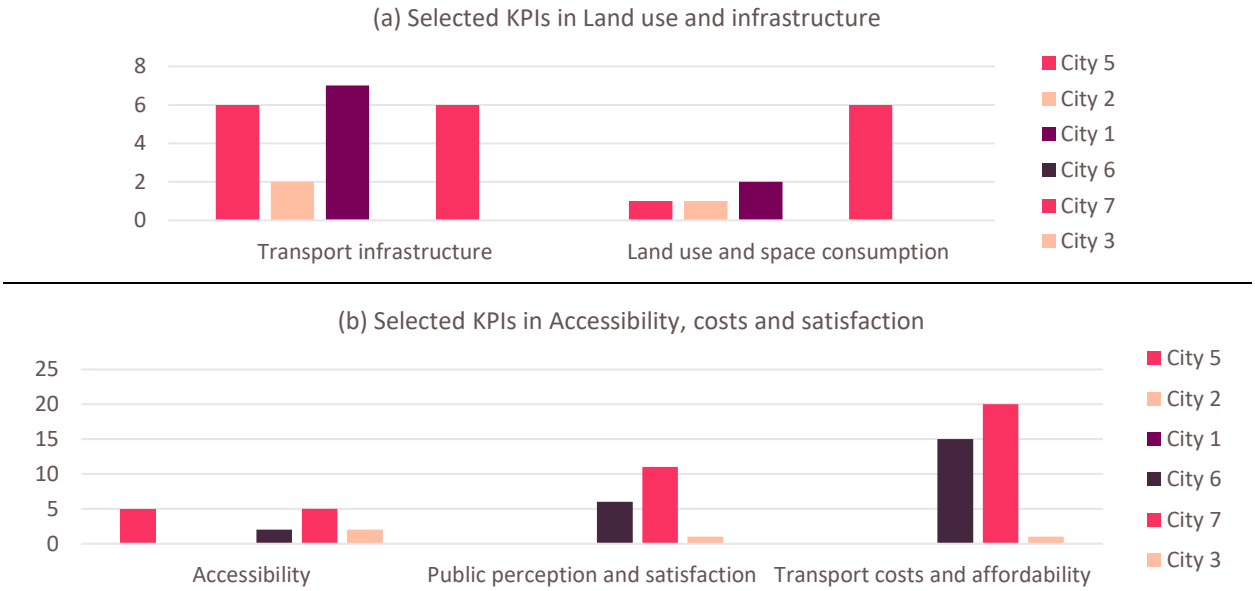

Figure 3.12. Distribution of selected KPIs in subgroups from the groups of "Land use and infrastructure" (a), and "Accessibility, costs and satisfaction" (b)

Similarly, the "Accessibility, costs and satisfaction" group (Figure 3.12b) showcased varied priorities, with some cities emphasizing accessibility for all citizens, while others focused on perceived satisfaction like overall mobility satisfaction and quality of life, as well as affordability of

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transport services, like travel costs and expenditures. This divergence highlights the contextual nature of KPIs' selection, wherein local socio-economic conditions and policy objectives significantly influence the prioritization of specific indicators within broader thematic categories. It must also be noted that several respondents overlooked entire KPIs' subgroups, suggesting an incomplete or inconsistent adoption of a comprehensive framework for urban pulse monitoring, particularly concerning aspects such as the promotion of non-motorized travel and the integration of technological solutions for sustainable urban transport.

The groups of "Transport safety" and "Network / system performance" shown in **Figure 3.13**, represent the least frequently selected categories, underscoring a potential oversight in comprehensively assessing the resilience and efficiency of urban systems. Although the subgroup of traffic accidents in "Transport safety" (**Figure 3.13**a) was deemed important by many practitioners, the indicators for security and safety control were largely neglected, revealing a critical gap in addressing potential risks and vulnerabilities due to regulatory deficiencies or inadequate awareness.

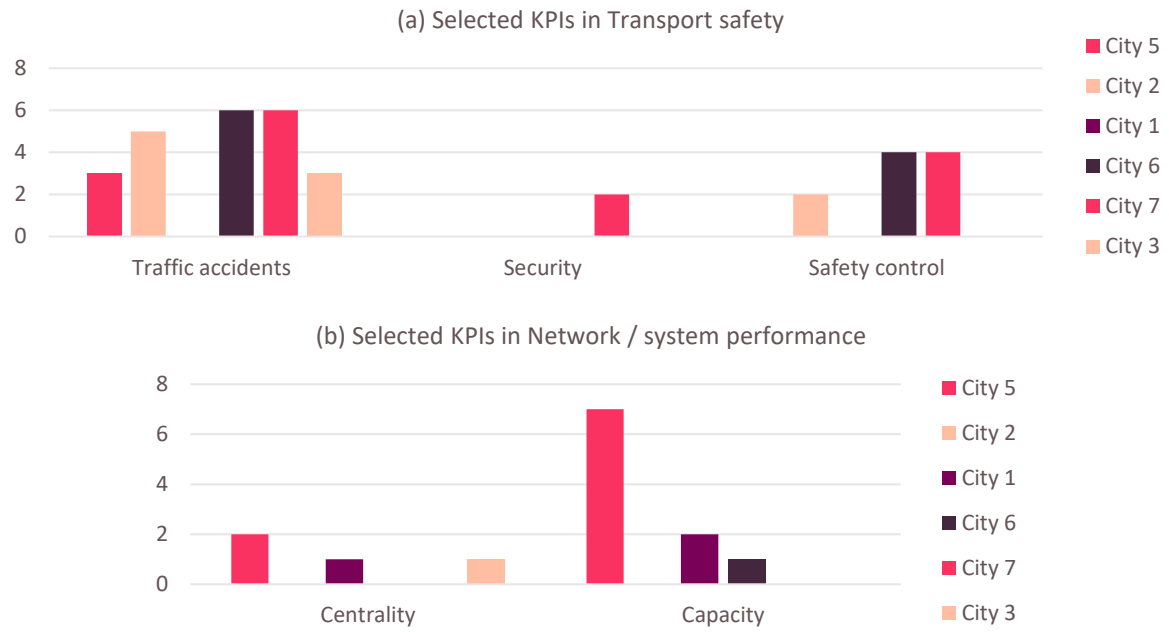

Figure 3.13. Distribution of selected KPIs in subgroups from the groups of "Transport safety" (a), and "Network / system performance" (b)

Regarding the group of "Network / system performance" in Figure 3.13b, the selection frequencies are highly divergent, indicating a lack of consensus on key metrics for evaluating the operational efficiency and reliability of urban transport networks. Although it would be reasonable for the practitioners to select numerous of the KPIs available in the technical subgroups of centrality and capacity, it is noted that the number and frequency of these selections was surprisingly low, suggesting a potential difficulty in the operational application of these metrics. The most selected KPIs from this domain were the commuter flow centrality and the mobility throughput, both of which are critical for understanding the dynamism and throughput of urban transportation systems and both requiring sophisticated data collection and analytical methods. This underscores the necessity for advanced analytical frameworks that can integrate diverse data streams, ranging from traditional

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traffic sensors to real-time social media feeds, to provide a holistic view of urban dynamics and potential disruptions.

As a general comment, it must be noted that the surprising peaks in the selection rates in specific subgroups derive from practitioners that either have prioritized detailed analyses within their core areas of expertise, or they are currently undertaking specific projects that necessitate a deeper dive into particular facets of urban mobility.

3.4.2 Commonly selected KPIs

The survey's results revealed the most used KPIs across the different urban mobility aspects. Provided that the participants responded completely independently, the analysis of their combined responses offers a robust indication of prevailing priorities in urban mobility monitoring, transcending individual biases and revealing a broader consensus on essential indicators. To this end, the KPIs that were cited by more than two (2) participants were considered as commonly selected indicators, providing a foundational set of metrics for a comprehensive urban pulse framework. **Table 3.1** presents an overview of these commonly selected KPIs, organized into their respective groups and subgroups from the original KPIs' catalogue.

A. Network/System performance			
A.1 Centrality			
Commuter Flow Centrality			
A.2 Capacity			
Mobility throughput			
B. Ecological impacts			
B.1 Emissions			
GHG Emissions	CO2 Emissions	Pollutant emissions from transport	NOx emissions
B.2 Noise pollution			
Noise pollution	Population exposed to noise		
C. Transport Modes, Travel Times and Characteristics			
C.1 Modal split			
Modal split	Personal travel mode split	Freight mode split	Cross-layer synergy
Travel times and delays	Vehicle travel times	Average commute travel time	Travel time
Average daily trip duration per person	Average trip duration per kilometre	Delays in the transport system	Congestion and delays
C.2 Travel times and delays			
Vehicle travel times	Average commute travel time	Travel time	Average daily trip duration per person

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Average trip duration per kilometre	Delays in the transport system	Congestion and delays	
C.3 Travel demand			
Transport demand	Non-motorized transport demand	Motorized transport demand	PRT Travel demand
PT demand			
C.4 Transport characteristics			
Average public transport speed	Quantity of PT trips / Number of daily PT trips	Freight ton-km	Ridership
D. Transport safety			
D.1 Traffic accidents			
Traffic accidents	Traffic conflicts	Accident rate	People dead in road accidents
Annual traffic fatalities per capita	Crash Count	Traffic related injuries and fatalities	
D.2 Safety control			
Laws and regulations	Collision Type		
E. Accessibility, costs, and satisfaction			
E.1 Accessibility			
Transport accessibility	Land use accessibility	Accessibility for emergence rescue	Destination Accessibility
E.2 Public perception and satisfaction			
Feeling of safety during traveling	Mobility satisfaction	General satisfaction	Overall quality of life in the city
Satisfaction with traffic flow on major city streets		Satisfaction with on-street bicycle accessibility	
E.3 Transport costs and affordability			
Passenger waiting costs	Vehicle running and depreciation costs	Personnel costs	Running costs
Travel costs	Parking fees	Fuel price	Daily parking ticket costs
Residents' parking ticket expenditure	Expenditure for public transportation	Private transportation cost	Annual infrastructural investments
Annual transportation expenditures per capita	Annual public transportation revenues per public transportation expenditures		
F. Land use and Infrastructure			
F.1 Transport infrastructure			

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Infrastructure spatial features	Length of total operational roads	Ratio of roadways of specific standards	Network density / Total roads length per area
Public transport stops	Public transport route network	Public transport efficiency	Public transport availability
Total highways length per area	Urban roads length per area	Total highways length per capita	Bus route length per area
F.2 Land use and space consumption			
Population density			
G. Urban development and growth			
G.1 Energy consumption			
Energy Utilization	Renewable Energy Utilization	Fossil fuel consumption in transportation	Energy consumption in transportation
G.2 Urban and spatial planning			
Urban density	Density of public services within all area services	Greening coverage of built-up areas	Green space area
Green space area per capita			
G.3 Economic growth			
GDP per capita	Urban growth generated by transport sector	Net public finance	

Table 3.1. Overview of the most selected KPIs

The analytical distribution of the most selected KPIs per thematic group is demonstrated in **Figure 3.14**. This evidence confirms the originally observed imbalance in comprehensive urban monitoring, where certain thematic areas consistently receive more attention or encompass a wider variety of indicators than others. Notably, a significant emphasis is placed on “Accessibility, costs and satisfaction“ and "Transport and land use integration", underscoring the growing recognition of the interconnectedness between urban planning, mobility, and the quality of life for city residents. This prioritization aligns with the increasing focus on human-centric urban development and the understanding that effective transportation systems must serve the diverse needs of the populace, encompassing not only efficiency but also equity and environmental sustainability [31,33].

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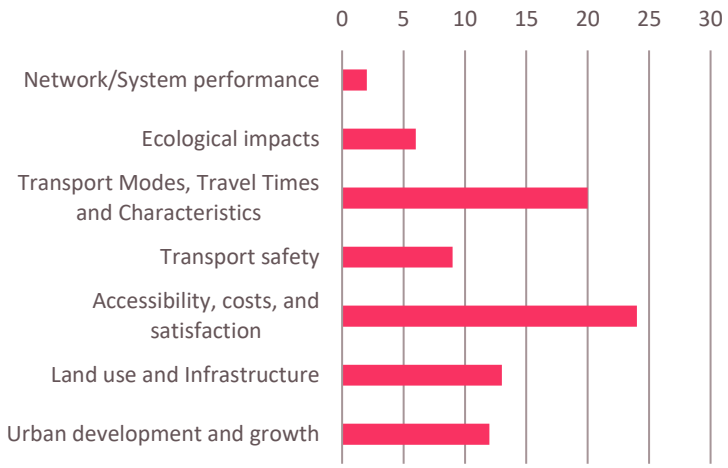

Figure 3.14. Distribution of most selected KPIs per group

Conversely, the relatively lower emphasis on "Network/system performance" and "Ecological impacts" highlights a potential gap in the holistic assessment of urban environments, suggesting that while the immediate human experience of mobility is prioritized, the underlying operational robustness and long-term environmental consequences receive comparatively less attention. This disparity suggests an opportunity to integrate more sophisticated metrics for network efficiency and ecological footprint into urban monitoring frameworks, ensuring a more balanced and forward-looking approach to urban development [34,35]. A more comprehensive framework would necessitate the inclusion of robust indicators for network resilience, system antifragility, and environmental sustainability, moving beyond direct human-centric metrics to encompass the broader ecological health and operational homeostasis of the urban system.

Although the most selected KPIs represent a comprehensive set of widely applicable indicators, their number of 64 is notably extensive, potentially posing challenges for practical implementation without the application of a specific selection method. To address this issue, a manageable subset of the top selected KPIs can be derived from the practitioners' survey. This refined set, encompassing fewer but highly relevant indicators, facilitates a more targeted approach to urban pulse monitoring, aligning with the need for computationally tractable and operationally implementable metrics, especially when the operational capacity of urban stakeholders is limited. This curated selection of KPIs, therefore, represents a pragmatic balance between comprehensive coverage and operational feasibility in cases where resource constraints or the immediacy of decision-making necessitates a streamlined approach to urban analysis, as exemplified by the current conditions in Odessa.

In practice, the top selected KPIs include the indicators that were commonly selected by at least four out of the six survey respondents. **Table 3.2** demonstrates the set of the top selected KPIs, which numbers a total of 22 items from the different thematic groups. For clarity, this table omits groups and subgroups without any selected indicators.

B. Ecological impacts	
B.1 Emissions	
CO2 Emissions	Pollutant emissions from transport
B.2 Noise pollution	

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Noise pollution		
C. Transport Modes, Travel Times and Characteristics		
C.1 Modal split		
Modal split	Personal travel mode split	Freight mode split
C.2 Travel times and delays		
Vehicle travel times	Travel time	Congestion and delays
C.3 Travel demand		
Transport demand	Motorized transport demand	
D. Transport safety		
D.1 Traffic accidents		
Traffic accidents	Accident rate	Annual traffic fatalities per capita
E. Accessibility, costs, and satisfaction		
E.1 Accessibility		
Transport accessibility	Land use accessibility	
E.3 Transport costs and affordability		
Travel costs		
F. Land use and Infrastructure		
F.1 Transport infrastructure		
Infrastructure spatial features	Length of total operational roads	
F.2 Land use and space consumption		
Population density		
G. Urban development and growth		
G.2 Urban and spatial planning		
Urban density	Green space area	Green space area per capita
G.3 Economic growth		
GDP per capita		

Table 3.2. Overview of the top selected KPIs

This refined set of indicators, while more manageable, magnifies the previously identified imbalance, emphasizing transport modes, travel times and characteristics over all the other aspects, as presented in the diagram of **Figure 3.15**. According to this distribution, the remaining aspects of the indicators appear to be represented quite equally, with the notable exception of system performance, which is entirely omitted. This omission highlights a critical gap in current urban monitoring practices, as robust indicators of system performance, including structural redundancy, diversity, and antifragility, are crucial for assessing the long-term resilience and adaptive capacity of urban infrastructure.

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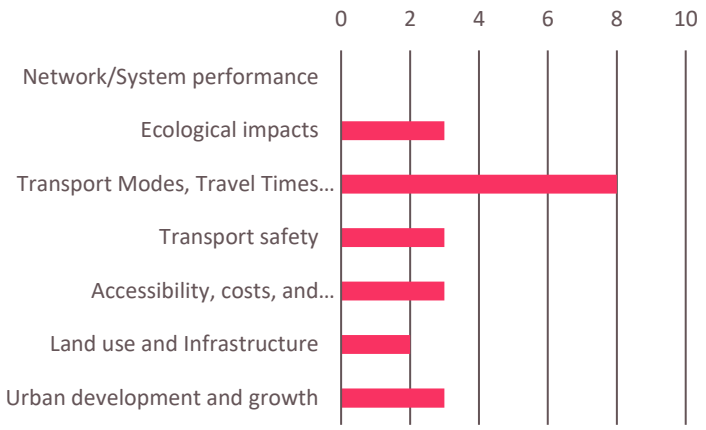

Figure 3.15. Distribution of top selected KPIs per group

3.4.3 Inputs for the Equilibrium Modelling Framework

The Equilibrium Modelling Framework (D2.3) provided a structured approach to integrate diverse urban data streams and derive actionable insights for predicting and mitigating the impacts of potential disruptions within the urban environment. In the context of this approach 11 composite indicators were defined to synthesize complex information from the examined urban system. These indicators were also included in the extensive KPI catalogue and presented to the urban mobility practitioners in the questionnaire survey.

The survey results revealed that the majority of the composite indicators of the Equilibrium Framework were not selected by the practitioners. This suggests a divergence between theoretical constructs for urban equilibrium and the practical priorities of urban stakeholders, who may favour more direct, immediately actionable metrics for urban management. Another reason for this omission is the complexity of the composite indicators, which often require extensive data integration and sophisticated analytical techniques that might not be readily available or easily interpretable by practitioners in operational settings.

Although the proposed composite indicators integrate various components that were individually selected by the transportation experts. For example, mobility throughput reflects the efficiency of the mobility system, by incorporating the efficiency of modal spit and travel time, which were individually prioritised by the practitioners. This evidence suggests that while practitioners may not explicitly prioritize composite indicators, the constituent elements of these indicators are indeed considered vital. Additionally, although urban authorities might hesitate to engage with complex assessment approaches, they are willing to provide all the required inputs for the Equilibrium Modelling if presented in the simple and familiar format of their selected KPIs.

On the other hand, there is one composite indicator that despite its computational complexity, was favoured by practitioners: the "Cross-layer synergy" indicator, which reflects the interconnectedness and cooperative functioning of different urban mobility subsystems. This finding underscores the recognition among practitioners of the critical importance of systemic integration and interdependencies within urban infrastructure, even when faced with data-intensive metrics. Therefore, the value of the proposed indicators' set is recognised but the practical constraints of day-to day operations may limit their immediate adoption, highlighting the need for user-friendly interfaces and streamlined data pipelines to bridge the gap between theoretical models and practical application.

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Furthermore, the potential disconnect between theoretical advancements in urban resilience and antifragility and their practical application in urban planning and policy, highlights the need for enhanced knowledge transfer and implementation strategies in the context of the AntifragiCity project. This suggests a critical need to bridge the gap between academic conceptualizations of urban resilience, such as those embedded in the AntifragiCity framework, and the pragmatic requirements of urban practitioners, through more intuitive presentation of complex metrics in the demonstrative use cases.

3.5 Integration of literature and practice inputs

This section synthesizes findings from the extensive literature review with empirical insights gleaned from practitioners' surveys to formulate a robust set of requirements for the KPIs' framework for urban pulse monitoring, thereby bridging theoretical constructs with practical application. This integrated approach ensures that the identified requirement specifications for the AntifragiCity KPIs are not only academically sound but also directly relevant and actionable for urban planners and policymakers.

In general, there is a shared understanding among researchers, practitioners, and international organizations regarding the critical components necessary for a holistic urban mobility assessment, balancing both the physical urban fabric and the human experience within it. There are several points of convergence between the examined literature and the survey's findings, particularly in the **prioritization of sustainability and efficiency**. Both streams of insight indicate a consistent prioritization of the broader aspect of sustainability (including its economic, environmental, and social dimensions) and the technical aspects of performance and efficiency. These areas are consistently highlighted across various frameworks and sources, such as the evaluation framework proposed by Fanati Rashidi et al. [36] and the ranking approach developed by Al-lami and Török[37]. This alignment underscores the critical need for urban frameworks to incorporate both environmental considerations and operational effectiveness as core tenets.

Another key convergence point between academic literature and practical experience is the **focus on quantifiable metrics**. This emphasis reflects a shared understanding that objective, measurable indicators are essential for effective monitoring, evaluation, and evidence-based decision-making in urban management, aligning with advancements in urban sustainability assessment that integrate real-time, data-driven approaches. For instance, key terms frequently used in KPIs' definitions, such as *number, vehicle, time, passenger,* and *cost*, reflect a common focus on traffic flow, infrastructure, and economic considerations.

However, not all the aspects of urban mobility are easily captured by quantitative metrics and especially since there is a clear need for a **comprehensive framework that incorporates both quantitative and qualitative data**. While technical and environmental aspects are strong, there is a growing recognition in both areas of the importance of social aspects. Terms like "quality" and "resident" appearing in KPI definitions indicate an increasing focus on human-centric dimensions and the quality of life for city. This fact underlines a **growing emphasis on human-centric dimensions**, highlighting the need to integrate qualitative data, such as sentiment analysis from social media, to capture the nuanced experiences and perceptions of urban dwellers.

A critical point of convergence between scholarly discourse, policies, and practical application also emerges in the acknowledgment of **interconnectedness of urban planning and mobility**. The survey results underscore a growing recognition of the interconnectedness between urban planning, mobility, and the quality of life, which aligns with theoretical discussions in the

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literature about comprehensive urban pulse frameworks [1,10,16] . This integrated perspective emphasizes that urban mobility extends beyond mere transportation logistics, encompassing broader aspects of urban design, social equity, and environmental stewardship, thereby promoting a holistic understanding of urban systems. This necessitates a framework that can flexibly adapt to emerging urban challenges and opportunities, enabling dynamic prioritization of KPIs based on evolving societal and environmental imperatives [2,35].

Furthermore, both sources of findings underscore the importance of **strategic adaptability and context-specificity** of urban mobility KPIs' frameworks. This suggests a need for frameworks that can accommodate diverse urban contexts and strategic objectives. The literature review highlights the variability in emphasis across different sources, while the survey reveals that cities tailor their indicator sets to specific local priorities and data availability, rather than employing a one-size-fits-all methodology. This necessitates a robust, adaptable framework capable of integrating diverse data streams and adjusting to the unique requirements of various urban environments.

Nevertheless, there are some discrepancies between the academic discourse and practical implementation, particularly concerning the integration of advanced analytical techniques and the availability of data required in the assessment processes. While scholarly works often advocate for sophisticated real-time monitoring and predictive analytics, practitioners frequently face limitations in data infrastructure, interoperability, and the resources needed to implement such advanced methodologies. This is one of the major reasons for the lack of consensus on a definitive order of importance for KPIs, particularly given the varying strategic transport policies and urban management systems across different global regions. These data and resource limitations also explain the limited number of KPIs effectively applied in real-life practice, as shown through the survey outputs in Table 3.3. This gap highlights the necessity for a framework that not only specifies ideal KPIs but also provides adaptable strategies for their implementation, accounting for varying levels of data availability and technological infrastructure across different urban contexts.

Groups and subgroups	Original KPIs	Commonly selected KPIs	Selection rate
A. Network/system performance	**19**	**2**	*11%*
A.1 Centrality	5	1	*20%*
A.2 Capacity	14	1	*7%*
B. Ecological impacts	**18**	**6**	*33%*
B.1 Emissions	11	4	*36%*
B.2 Noise pollution	2	2	*100%*
B.3 Ecological degradation	5	0	*0%*
C. Transport Modes, Travel Times and Characteristics	**49**	**20**	*41%*
C.1 Modal split	4	4	*100%*
C.2 Travel times and delays	24	7	*29%*
C.3 Travel demand	8	5	*63%*
C.4 Transport characteristics	13	4	*31%*
D. Transport safety	**27**	**13**	*48%*
D.1 Traffic accidents	10	7	*70%*
D.2 Security	3	2	*67%*
D.3 Safety control	14	4	*29%*

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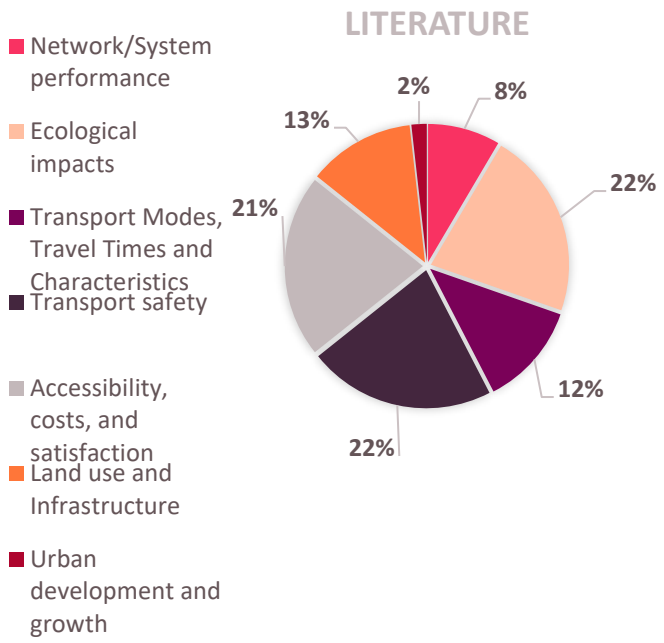
E. Accessibility, costs, and satisfaction	**49**	**24**	*49%*
E.1 Accessibility	6	4	*67%*
E.2 Public perception and satisfaction	19	6	*32%*
E.3 Transport costs and affordability	24	14	*58%*
F. Land use and Infrastructure	**28**	**13**	*46%*
F.1 Transport infrastructure	15	12	*80%*
F.2 Land use and space consumption	13	1	*8%*
G. Urban development and growth	**4**	**12**	*63%*
G.1 Energy consumption	9	4	*44%*
G.2 Urban and spatial planning	7	5	*71%*
G.3 Economic growth	3	3	*100%*

Table 3.3. Originally available and commonly selected KPIs per group and subgroup

The selection rates for the commonly selected KPIs in Table 3.3 indicate that experts effectively utilize a smaller yet representative set of indicators, tailored to their specific contexts. In fact, the results indicate that, out of the total 209 available KPIs, only 64 were commonly selected, representing 31% of the entire catalogue. This adaptive selection process reflects a practical approach to urban management, where data availability and local priorities often dictate the feasibility of KPI implementation rather than a universally prescribed set. Moreover, the evidence of Figure 3.16 demonstrate the differences in the distribution of KPIs among the diverse thematic groups between the original KPIs' catalogue and the commonly selected KPIs.

(a) Distribution of KPIs based on literature



(b) Distribution of KPIs based on practice

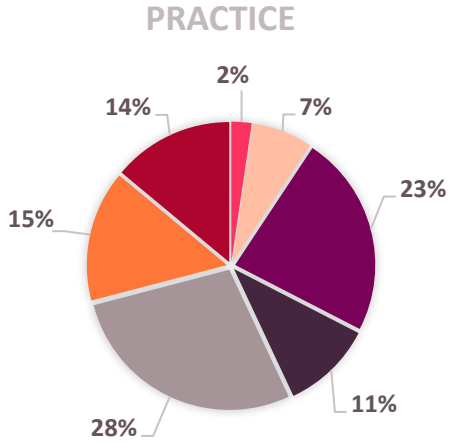


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Figure 3.16. Distribution of KPIs from original KPIs' catalogue (a) and survey's commonly selected KPIs (b)

The findings of Figure 3.16 demonstrate that the commonly applicable KPIs' frameworks are also imbalanced among the different dimensions, with a notable overrepresentation of socio-economic and technical metrics. In contrast, KPIs related to ecological impacts, transport safety, and network performance are underrepresented in practice compared to researchers' recommendations. This disparity emphasizes the need for a framework that actively promotes the integration of underrepresented yet crucial dimensions, thereby moving beyond a purely technical or economic focus towards a more holistic assessment of urban sustainability. All these observations are essential for the development of the AntifragiCity KPIs' framework, which will integrate the capacity for contextual adaptation, enabling cities to prioritize and implement indicators most pertinent to their unique socio-economic, environmental, and infrastructural landscapes.

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4 Development of KPIs' framework and metrics

The development of the AntifragiCity KPIs framework leverages insights from the integration of findings from the literature review and the practitioners' survey to establish a robust and adaptable set of metrics. This section presents the conceptual approach and architecture of the proposed KPIs' framework, its inputs and review processes, as well as the connection with the established AntifragiCity Ontology Framework.

4.1 Methodological approach

This framework aims to bridge the gap between theoretical ideals and practical implementation by incorporating mechanisms for dynamic KPI selection and adaptation based on data availability and specific urban challenges, thereby addressing the discrepancies identified between academic discourse and real-world application. Given the identified requirements, the proposed methodological approach is designed to be both comprehensive and flexible, allowing for the quantification of urban mobility and the reflection of the urban pulse through a harmonized set of indicators. Instead of defining a specific set of KPIs, the proposed framework emphasizes a process for KPI definition that is flexible enough to allow for continuous evolution and customization according to varying urban contexts and strategic objectives. This approach ensures that the framework can dynamically adapt to emerging urban challenges and opportunities, enabling the prioritization of KPIs based on evolving conditions and events. This dynamic adaptability is crucial for maintaining the relevance and efficacy of the AntifragiCIty framework in addressing the multifaceted and continuously changing nature of urban environments under various stressors, including unforeseen Black Swan events.

Building on the evidence derived from the practitioners' survey, the proposed KPIs' framework will facilitate the selection of relevant KPIs, allowing cities to customize their measurement approaches based on their unique characteristics, such as size and economic development status, as well as their operational and disruptive scenarios. This customized selection process is integral to ensuring that the framework remains pertinent and effective across a diverse range of urban settings, promoting a more nuanced understanding of urban performance, resilience and especially antifragility. This approach acknowledges that a "one-size-fits-all" solution is inadequate for evaluating the complex dynamics of urban systems and instead advocates for a modular KPI framework that can be either extended as required by domain experts or kept concise in line with common practices of the operational authorities. This framework also supports the evolutionary development of KPIs, enabling cities to refine their assessment metrics over time to better reflect long-term dynamics and operationalize antifragility metrics.

The process presented in Figure 4.1 outlines a systematic methodology for defining and refining context specific KPIs, ensuring alignment with urban policy objectives and data-driven insights while considering the contextual relevance and applicability of each indicator. The process encompasses five consecutive steps and, if required, can also occur cyclically. The initial step involves the delineation of the strategic vision and overarching directions of the urban development, which subsequently informs the identification and prioritization of relevant thematic areas and subareas, including the spectrum of sustainability, performance, efficiency, resilience, equity, and ecological

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footprint. This comprehensive delineation ensures that the subsequent selection of KPIs is directly aligned with the city's aspirations and challenges, fostering a targeted and impactful assessment framework. For this purpose, an AHP-like method could be employed to weight the importance of various thematic areas and extract the most critical ones for focused KPI development [10,37].

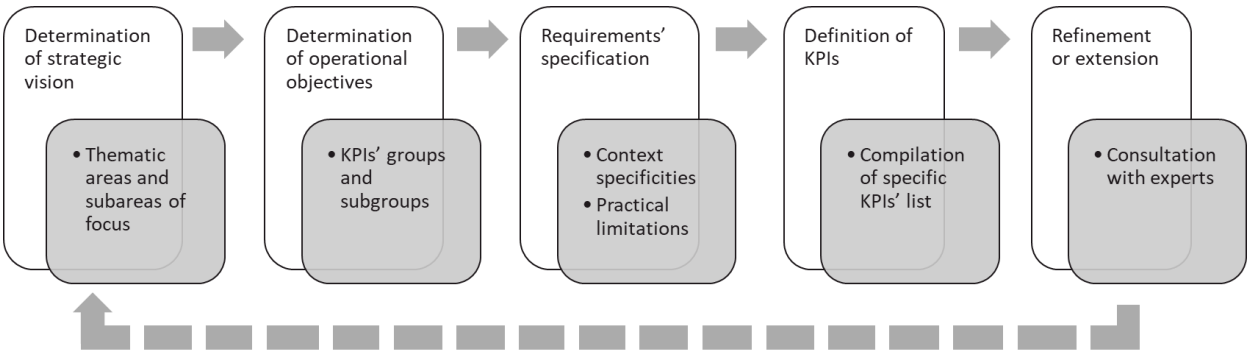

Figure 4.1. Steps for development of the AntifragiCity KPIs' framework

According to Figure 4.1, the next step involves establishing more detailed, context-specific operational objectives for the framework, including specific targets and benchmarks to guide the selection of appropriate mobility indicators. This entails a thorough examination of the technical dimensions and the particularly, the established KPIs' groups and subgroups that reflect the diverse facets of urban mobility, ranging from transport characteristics and accessibility to safety and ecological impact. Again, the prioritization of the examined KPIs' groups could derive through an AHP-like approach. This systematic process ensures that the selected KPIs are not only relevant to the city's strategic vision but also granular enough to capture the intricate dynamics of urban mobility and its impact on the urban pulse, thereby enhancing the framework's analytical precision.

Following this, the proposed methodology progress to the requirements' specification. This entails a thorough examination of current urban practices, existing constraints, and operational limitations, like resource deficiencies or data unavailability, to ensure that the chosen indicators are both relevant and actionable. At this stage it is also important to develop the operational scenarios under which the framework will operate, including disruptive events, to thoroughly test the robustness and adaptability of the selected KPIs. These requirements will serve as exclusion criteria or filters for the subsequent selection of appropriate indicators from the pool of the KPIs' catalogue. This case-specific approach allows for the development of a tailored indicator pool for each unique urban system, ensuring that the selected KPIs are meaningful and directly applicable to the local context.

As described in Figure 4.1, the methodology progresses to the systematic identification and selection of pertinent indicators within each prioritized thematic area and group, drawing upon the extensive KPIs' catalogue. The selection process is guided by established criteria pertinent to the scope, technical specifications, and operationalization of the indicators' set. The algorithm of the selection procedure prioritizes indicators based on their relevance to the strategic vision, reference to the prioritised aspects, feasibility, and potential for integration with existing data streams, alongside their capacity to provide actionable insights for urban authorities. The product of this process is a refined set of KPIs that accurately reflect the unique characteristics and strategic objectives of the urban environment, ensuring a robust foundation for monitoring and evaluation of the city's pulse.

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The final step of this process includes the procedures of refinement or extension of the previous preliminary KPIs' set. This refinement stage critically evaluates the chosen indicators for redundancy and interdependence, ensuring a parsimonious yet comprehensive set of metrics. Simultaneously, the option for extension of the KPIs' set ensures the long-term applicability and adaptability of the framework to evolving urban challenges and emerging data sources. This final step is crucial for validating the indicator selection through expert focus groups and stakeholder engagement, thereby enhancing the credibility and practical utility of the framework for assessing urban performance. Furthermore, this step can initiate the entire process again, fostering a cyclical, adaptive approach to urban pulse monitoring. The iterative nature of this methodology allows for continuous improvement and recalibration of the KPI system, ensuring its ongoing relevance and effectiveness in capturing the multifaceted dynamics of urban mobility and overall development.

The AntifragiCity framework employs a comprehensive, mixed-methods approach for KPI selection, combining several logic types to ensure adaptability, inclusivity, and robustness. Essentially, the KPIs' framework selection logic can be described as a combination of criteria-based filtering and multi-criteria decision analysis, ontology driven backbone, and expert input. In more detail, the framework integrates a criteria-based selection process that prioritises indicators based on their relevance to the strategic vision, operational objectives, feasibility, and potential for integration with existing data streams. Furthermore, an ontology-driven backbone provides a structured representation of urban phenomena and their interrelationships, facilitating the identification of appropriate indicators that align with predefined categories and concepts. Crucially, the framework incorporates input from urban mobility practitioners through a comprehensive survey, validating the relevance and applicability of indicators to real-world urban contexts and tailoring them to specific needs. This multifaceted approach allows the framework to dynamically adapt to emerging urban challenges and opportunities, ensuring the selected KPIs are not only theoretically sound but also practically relevant and actionable for urban planners and policymakers.

4.2 Integration of data source types

The framework integrates diverse data source types, encompassing both quantitative and qualitative data, to provide a comprehensive understanding of urban dynamics. The following subsections 4.3.1 and 4.3.2 describe the quantitative and qualitative inputs, respectively, required for the AntifragiCity KPIs' framework. These varied data streams are essential for a multi-layered validation and evaluation framework, ensuring the comprehensive assessment of urban systems and the detection of potential stressors.

4.3.1 Quantitative input

The quantitative input of the proposed KPIs' framework encompasses a diverse array of measurable data, including environmental parameters, economic factors, traffic patterns, and infrastructure performance metrics, all of which are critical for objective assessment of urban dynamics. **Table 4.1** presents these types of data, detailing their descriptions with relevant examples.

Type of Quantitative Input	Description/Examples
Environmental data	Raw data streams and resources related to the urban environment and ecological systems, for example air quality, noise levels, and meteorological conditions.

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Traffic data	Raw data streams and resources related to urban traffic flows and conditions, such as data on public transport availability (e.g. frequency of service, ridership, etc.), service quality (e.g. travel time, delays, etc.), and safety (e.g. number of injuries, fatalities, etc.).
Infrastructure data	Raw data streams and resources pertaining to urban infrastructure and especially related to transport, e.g., road pavement conditions, bridge structural integrity, traffic signal operations, and public transport facility maintenance status.
Economic data	Raw data streams and resources pertaining to urban economic indicators, such as GDP contributions from mobility sectors, employment rates in transport, cost of congestion, and economic impacts of disruptions.
Land use data	Raw data streams and resources pertaining to land use patterns and urban spatial organization, such as zoning classifications, land cover types, building density, and green space distribution.
Sensed events data	Data from diverse data streams that are semantically analyzed and classified by the AntifragiCity Ontology, e.g., traffic accidents, road closures, public transport delays, and social media-reported disruptions.

Table 4.1. Types of quantitative input

Table 4.1 exhibits input data from diverse data streams that are semantically analyzed and classified by the AntifragiCity Ontology and KPIs' Framework, divided into six (6) distinct types, namely environmental, traffic, infrastructure, economic, land use, and sensed events, collectively providing a holistic view of urban functionality. The data sources could include both publicly available datasets and information from private operators, are systematically integrated to ensure a robust foundation for KPI calculation [32]. This multi-source integration is pivotal for understanding interdependent processes across environmental, technical, and socio-economic subsystems. Such a comprehensive data-driven analysis moves beyond simplistic linear cause-effect relationships to decipher how local operational conditions and decision-making processes contribute to emergent behaviors within the urban transport system.

The quantitative input derives from various collection methods, including on-site measurements, census data, geospatial information, and aggregated sensor data, all contributing to a robust understanding of urban systems. These data streams, often sourced from IoT sensors, GPS traces, mobile phones, and other urban monitoring systems, provide granular insights into the operational characteristics and performance of the city. Moreover, historical data can be leveraged to establish baselines and identify long-term trends, which is essential for predictive modeling and scenario analysis. Similarly, in some cases of data scarcity, synthetic data generation techniques could be employed to augment existing datasets, thereby enhancing the robustness of the analytical framework. The integration of such diverse quantitative inputs necessitates sophisticated data fusion techniques to harmonize disparate data formats and ensure consistency across the analytical pipeline.

Another important aspect of the quantitative inputs is the calibration and validation of the various data against established benchmarks to maintain data integrity and reliability. Especially the sensor-based data, which can be susceptible to calibration drifts and environmental interference, requires rigorous quality assurance protocols to ensure accuracy and trustworthiness [34,38].

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Furthermore, the cost of data acquisition and processing for certain types of quantitative indicators, such as those derived from extensive sensor networks, must be carefully weighed against their potential benefits. Consequently, resource allocation for data infrastructure and analytical capabilities becomes a pivotal consideration in the holistic design and implementation of the urban pulse framework.

4.3.2 Qualitative input

The qualitative input complements the quantitative data by providing rich contextual information and insights into human perceptions, behaviours, and subjective experiences within the urban environment and particularly concerning aspects of mobility and transport. These data, often gathered through surveys, focus groups, and sentiment analysis of social media, capture the nuances that numerical metrics might miss [3,29]. Such qualitative indicators are particularly useful for assessing aspects like public satisfaction with transport services, perceptions of safety, and community engagement in urban planning initiatives, thereby offering critical insights into the social dimensions of urban performance [15,39].

Table 4.2 presents a detailed overview of various qualitative data types, their descriptions, and relevant examples, emphasizing their role in providing a comprehensive understanding of urban dynamics. Moreover, these qualitative data streams are instrumental in identifying emergent issues or nascent societal trends that might not be immediately apparent through purely quantitative analyses. For instance, qualitative indicators can be expressed as scoring systems based on predefined criteria, which allows for systematic assessment of subjective qualities, such as the effectiveness of public transport services or the perceived quality of green spaces. This approach allows for a holistic understanding of urban mobility systems by incorporating subjective human experiences alongside objective numerical measurements.

Type of Qualitative Input	Description/Examples
Human perceptions and behaviours	Insights into subjective experiences within the urban environment, particularly concerning mobility and transport. This can include data gathered through surveys, focus groups, and sentiment analysis of social media.
Public satisfaction	Qualitative indicators useful for assessing public satisfaction with transport services.
Perceptions of safety	Qualitative indicators useful for assessing perceptions of safety within the urban environment.
Community engagement	Qualitative indicators useful for assessing community engagement in urban planning initiatives.
Emergent issues/societal trends	Data instrumental in identifying new issues or nascent societal trends not immediately apparent through quantitative analyses.
Scoring systems	Qualitative indicators expressed as scoring systems based on predefined criteria, used for systematic assessment of subjective qualities, such as the effectiveness of public transport services or the perceived quality of green spaces.

Table 4.2. Types of qualitative input

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Table 4.2 categorizes key types of qualitative inputs for urban mobility analysis, including human perceptions and behaviours (e.g., surveys, focus groups, social media sentiment), public satisfaction with transport services (e.g. customer satisfaction indices), perceptions of safety, community engagement in planning, emergent issues/societal trends beyond quantitative data (e.g. new mobility patterns), and scoring systems for assessing subjective qualities like public transport effectiveness or green space quality. The range of qualitative input is crucial for capturing the multifaceted nature of urban systems, offering descriptive data and expert judgments that enrich the analytical framework.

Possible sources for this qualitative data include expert judgments, interviews, and ethnographic studies, which can provide nuanced perspectives on urban resilience and sustainability. Modern technology can further enhance qualitative data collection through tools like natural language processing for sentiment analysis of social media feeds and advanced spatial analysis to contextualize subjective observations. Such subjective assessments are invaluable for uncovering changes in social and human capital, as well as complex socioeconomic dynamics, thereby preventing an overreliance on purely technical solutions to inherently social problems [40]. Furthermore, the integration of expert observations and user expressions, gathered via methods such as questionnaires and interviews, is crucial for assessing aesthetic quality, infrastructure cleanliness, and overall satisfaction with transportation services and public spaces [2]. This comprehensive approach to qualitative data ensures a robust understanding of urban dynamics, capturing both the tangible and intangible elements that contribute to a city's liveability, resilience, and antifragility.

4.3 Adaptation and revision

The Input-Process-Output-Outcome (IPOO) framework offers a robust structure for organizing the identified KPIs, providing a clear causal chain from initial inputs to ultimate outcomes [41,42]. The IPOO framework is usually applied in project management, but its principles are highly applicable to urban systems analysis. It provides a structured approach to understanding how various mobility aspects translate into quantifiable metrics, thereby enabling a comprehensive evaluation of their impact on the urban pulse. This framework delineates clear relationships between data inputs, analytical assessment processes, intermediate outputs, and the ultimate outcomes regarding urban sustainability, efficiency, resilience, and ecological enhancement. The framework ensures that the KPIs are not only well-defined but also systematically organized, facilitating their integration into the overarching urban pulse monitoring system.

The diagram of Figure 4.2 illustrates the comprehensive structure of the IPOO framework as applied to urban mobility KPIs' framework, visually representing the interdependencies between various components and their contribution to assessing the urban pulse. This structured approach aids in understanding the relevance and the effectiveness of the selected KPIs. It also highlights the input requirements, the measurement processes and the outputs and outcomes, providing a holistic view of the KPIs' framework's performance and impact.

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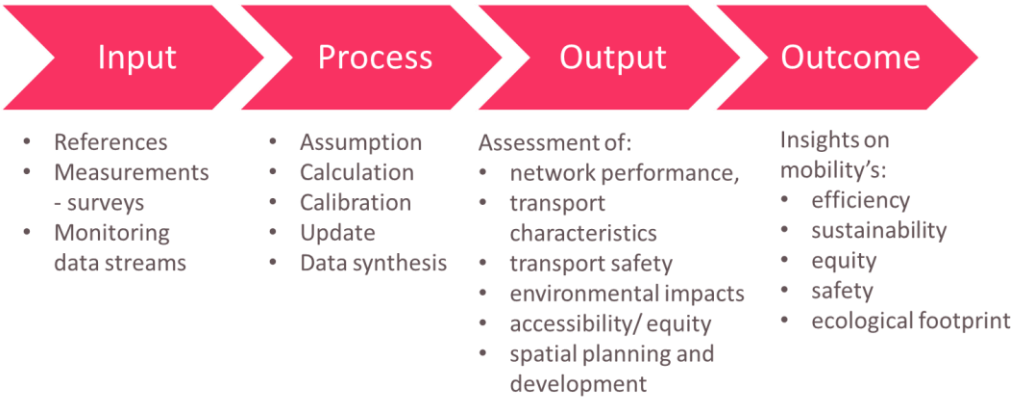

Figure 4.2. IPOO framework for the adaptivity and review of the KPIs' framework

According to Figure 4.2, there are four distinct stages within the IPOO framework, each playing a crucial role in transforming raw data into actionable insights for urban mobility management. The first stage, Input, encompasses all raw data streams and resources that feed into the system, including environmental, traffic, and infrastructure data, along with qualitative social media sentiment analysis. Naturally some considerations must be made regarding data availability, accuracy, integration, and real-time processing capabilities, as these directly influence the reliability of subsequent stages. The increase of the data acquisition and computation cost are serious considerations that need to be addressed to ensure the practicality and scalability of the urban pulse monitoring system [42]. The second stage, Process, involves the analytical transformation of these inputs into meaningful metrics through established methodologies like event elicitation from the ontology and compliance checking with regulations. Again, the complexity of the employed assessment method and the computational resources required for near real-time analysis present significant challenges [43]. Therefore, the trade-off between computational efficiency and the depth of analytical insight is a critical design consideration for the KPIs' framework's evaluation and potential review.

The third stage of the IPOO framework, Output (Figure 4.2), translates processed data into immediate, quantifiable results such as initial assessments of transportation efficiency, environmental footprint, and safety metrics. These outputs, while valuable, represent isolated measurements useful to diverse experts and operators of the urban mobility systems. It requires specific expertise to evaluate these KPIs and translate them into actionable insights for decision-makers. The final stage, Outcome, involves the synthesis of these outputs to provide a holistic understanding of the urban pulse, thereby informing strategic decisions for urban planning and policy adjustments to address potential stressors and Black Swans.

This systematic and comprehensive evaluation of the KPIs' framework is critical for validating its effectiveness and ensuring its continued relevance in mitigating urban challenges, promoting sustainability, and enhancing overall urban resilience and antifragility. It also provides a structured mechanism for incorporating feedback loops, allowing for adaptive adjustments to the framework based on real-world performance and evolving urban dynamics, such as advancements in Intelligent Transportation Systems and smart city infrastructure [4,38,44]. This iterative refinement process is crucial for operationalizing antifragility metrics and building cross-scalar dashboards that link neighbourhood-level events to city-wide flows for multi-level intervention strategies.

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4.4Architecture of the KPIs' definition tool

The KPI definition and refinement process can readily be adapted into an automated algorithm for KPI selection, leveraging expert input to refine metric choices based on their relevance, feasibility, and capacity to capture critical urban dynamics. The pseudocode for this proposed algorithm is detailed in Appendix C. This automated approach facilitates a more efficient and objective selection of KPIs, ensuring that the chosen indicators are robust, non-redundant, and directly aligned with the AntifragiCity framework's objectives.

The architecture for defining KPIs in this framework is designed to integrate both quantitative and qualitative data streams, transforming raw inputs from urban mobility practitioners and city stakeholders into actionable metrics that reflect the urban pulse. This systematic integration facilitates a comprehensive understanding of urban system performance, moving beyond singular metrics to incorporate the complex interplay of various factors that influence urban resilience and sustainability. The architecture diagram of **Figure 4.3** showcases the interconnections between data sources, processing modules, and the ultimate generation of KPIs, thereby enabling a holistic evaluation of urban functionality and stress prediction. Specifically, the tool' s architecture comprises five (5) distinct layers, corresponding to the respective steps of the proposed methodology. Each layer addresses specific computational and analytical challenges, from determining the strategic vision and setting objectives, to specifying requirements, selecting KPIs, and ultimately refining and validating them. This layered approach ensures a robust and adaptable framework for the continuous monitoring and improvement of the AntifragiCity KPIs' framework.

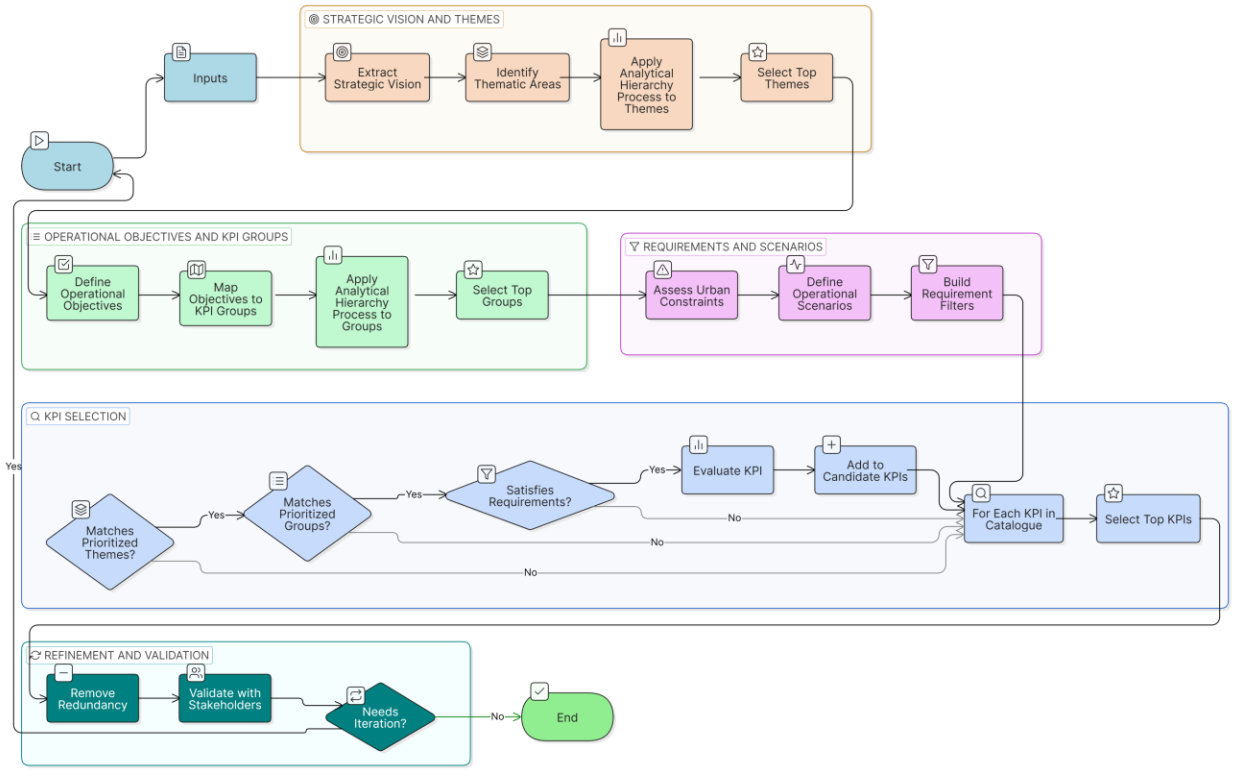

Figure 4.3. Architecture diagram of the KPIs' definition tool

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This structured methodology is also translated into an API format that facilitates seamless integration with diverse urban data platforms and enables real-time data exchange for dynamic KPI definition and calculation. The structure of the proposed API is diagrammatically presented in **Figure 4.4**. This programmatic interface allows for automated retrieval and submission of KPI-related information, promoting interoperability and scalability across various urban management systems and data streams.

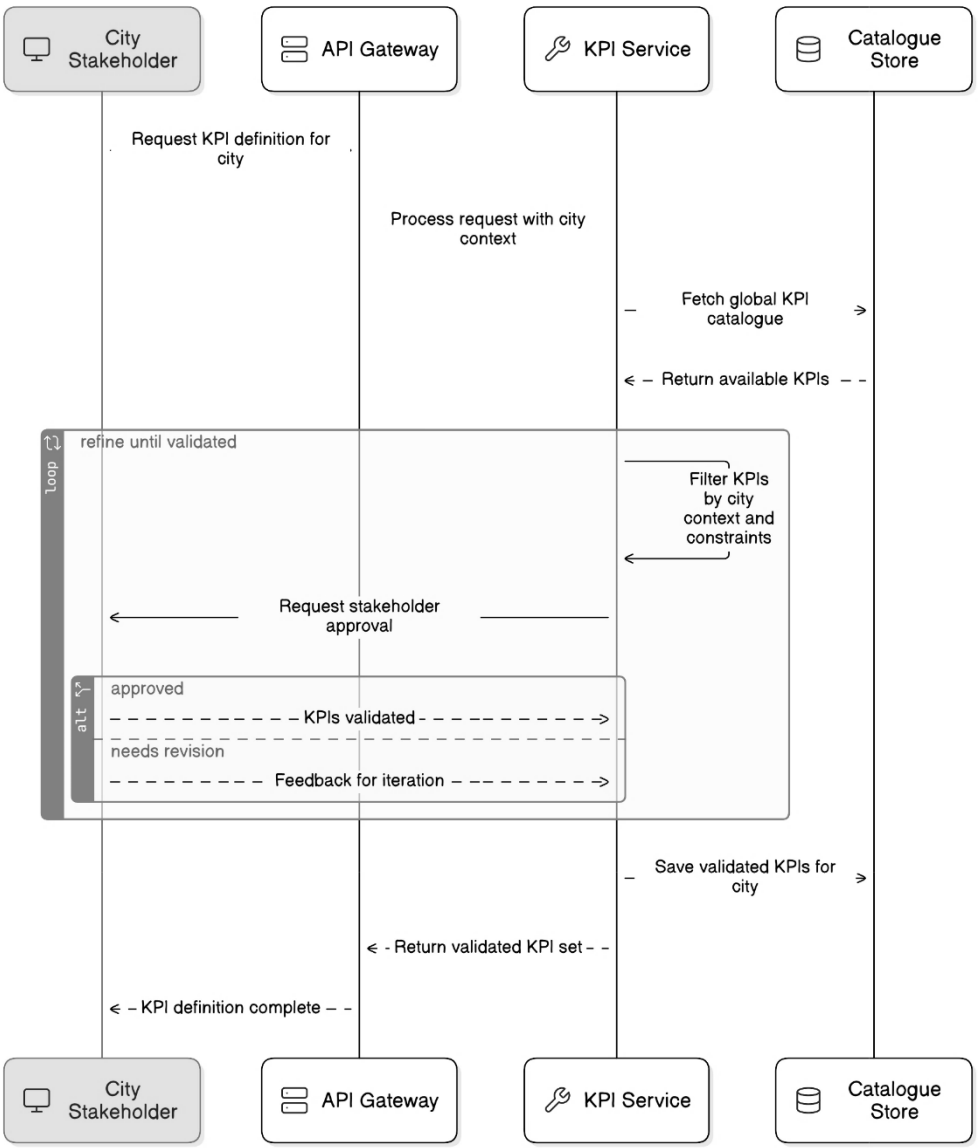

Figure 4.4. Proposed API structure

This API structure features four (4) layers the city stakeholders as the final users, the API gateway as the entry point of the software, the KPI service layer which handles selection logic and data processing, and the KPIs' catalogue store that is responsible for persistent storage and retrieval of KPI definitions and associated metadata. This structure also captures the feedback loops

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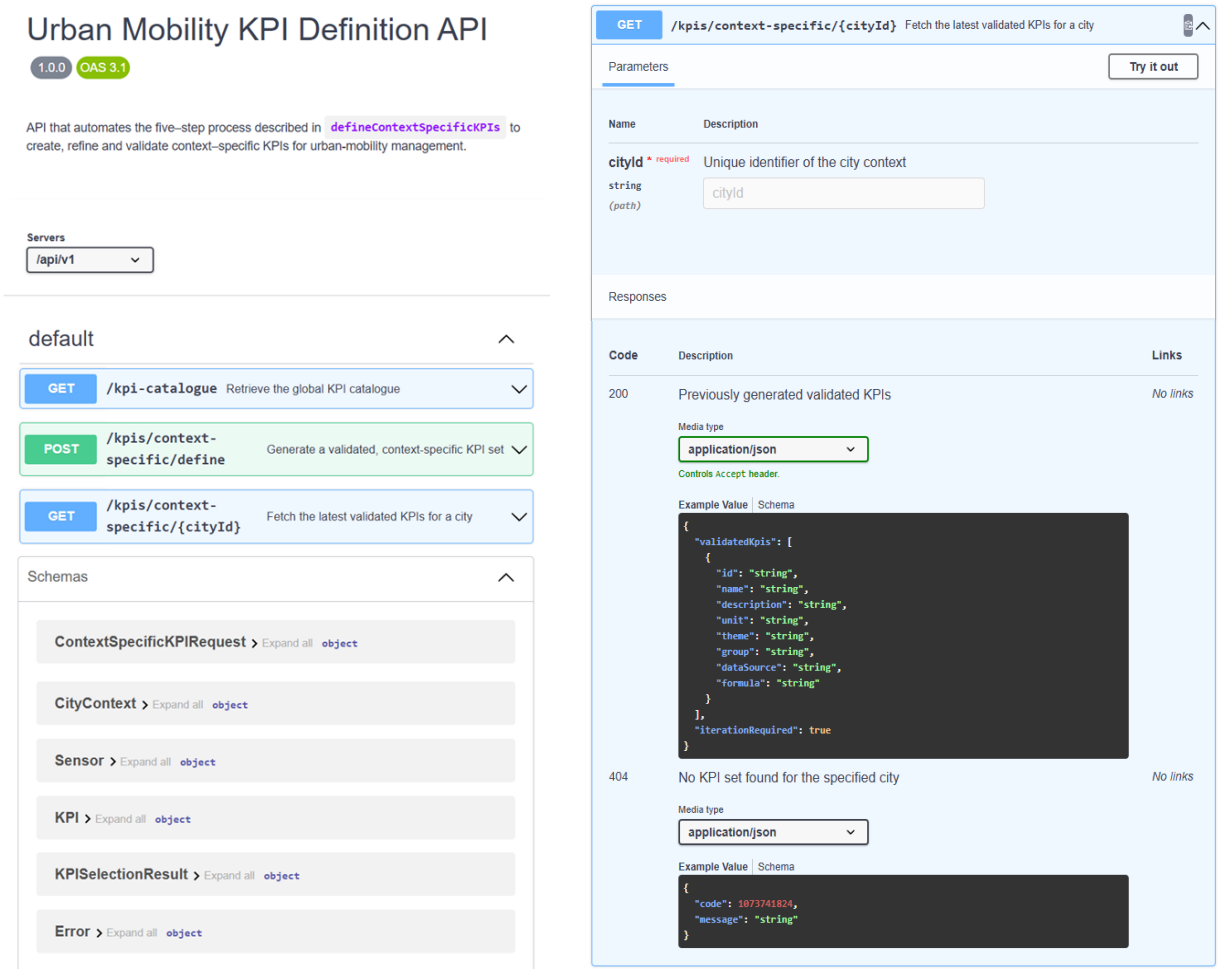
for the refinement and validation of the KPIs, ensuring their ongoing relevance and accuracy in monitoring urban pulse and predicting potential stressors. The entire script detailing the structure of the proposed API is available in a text-based and language-independent data-interchange format (JSON file) at: https://doi.org/10.5281/zenodo.18744603

This structure was tested in a simulated backend server, i.e. a mock API, which effectively mimics the real API behaviour in order to enable faster prototyping, testing of error states, and parallel frontend development. The mock API evaluates the functional integrity and performance of the system prior to full-scale deployment, ensuring that all defined KPIs are accurately calculated and accessible. Captures from this mock API are indicatively demonstrated in **Figure 4.5**.



(a) API structure overview (b) Overview of a GET request

Figure 4.5. Overview of the mock API

4.5 Connection with the Ontology Framework

The Ontology Framework (Task2.5 and D2.5) provides a structured representation of urban events, risks, and their interrelationships, which is crucial for contextualizing the quantitative and qualitative data inputs within the AntifragiCity framework. This ontological foundation allows for the

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classification of information into representative real-world categories, enabling a more granular and precise analysis of urban dynamics and the identification of related KPIs. It serves as a semantic backbone, facilitating the elicitation and analysis of sensed events from diverse data streams, thereby enriching the interpretation of urban system behaviour.

The AntifragiCity Ontology, specifically, is designed to be a dynamic and operational asset, ensuring that the theoretical constructs are directly applicable to practical urban mobility planning and resilience building. The Ontology Framework, specifically, is designed to be a dynamic and operational asset, ensuring that the theoretical constructs are directly applicable to practical urban mobility planning and resilience building. Consequently, there are numerous connections and interdependencies between the AntifragiCity Ontology and the KPIs' framework, which can be summarised in the following points:

- formal Semantic Basis for KPIs
- semantic backbone for consistent data and models
- KPI Ontology module
- integration of KPIs' concepts
- representation of KPIs in event scenarios
- support for AI analytics and decision-making
- interoperability and data integration
- monitoring and evaluation

The Ontology Framework provides the semantic foundation for the identification of KPIs from the comprehensive KPIs' catalogue, which outlines detailed information for each indicator group, including measurement units and calculation methods. The ontology structures urban events, risks, and interrelationships, enabling the precise classification of quantitative and qualitative data inputs into categories that directly inform the selection and computation of the most relevant KPIs.

It is important to note that the categorization of the KPIs initial catalogue is already integrated into the Ontology Framework as a special module to practically ensure semantic consistency and to facilitate the alignment of diverse data sources for robust analysis and decision-making. This integration enables the ontology to serve as a high-level framework for reconciling domain-specific data models and addressing data heterogeneity. In turn, it facilitates information discovery, KPIs' framework development, and agreement on an urban information model, which are essential elements for the subsequent steps of the AntifragiCity project.

Furthermore, the ontology links micro-level events to city-wide outcomes, ensuring KPIs capture interdependencies and compliance with regulations through decidable frameworks, thereby enhancing the KPIs framework's applicability to antifragility assessment. This approach ensures that the KPIs' framework can holistically represent disruptions, responses, learning processes, and equilibrium states within urban mobility systems, while also integrating heterogeneous data streams and complying with existing policies. This integration allows for the comprehensive capture of equity, vulnerability, and exposure of various user groups, fostering a more inclusive and resilient urban mobility planning paradigm.

Finally, the ontology framework enables feedback loops for refining KPIs based on real-world dynamics, promoting iterative improvements in urban pulse monitoring and cross-scalar dashboards. This quality supports the generation and the management of context-specific and valuable feedback, which ultimately informs the processes of adaptation and review of the KPIs' framework. This continuous refinement ensures the framework remains agile and responsive to emerging urban challenges and opportunities, thereby enhancing the accuracy and relevance of urban pulse monitoring.

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5 KPIs' framework integration and validation

This section delves into the methodological approaches for integrating and validating the KPIs' Framework within the broader AntifragiCity paradigm. Specifically, the framework is integrated with the lifecycle thinking principles (subsection 5.1), drawing upon methodologies from projects like ACCORD and SemanticLCA (subsection 5.2), to ensure comprehensive environmental and sustainability assessments are embedded from the outset. Additionally, the proposed framework validation is designed to rigorously evaluate its efficacy in identifying and mitigating urban stressors (subsection 5.3), as well as its compliance with international standards and established regulations (subsection 5.4).

5.1 Integration with the lifecycle thinking

The integration of Life Cycle Assessment (LCA) methods within the proposed KPIs' framework provides a robust mechanism for evaluating the long-term sustainability and resilience of urban systems and interventions, moving beyond traditional operational performance metrics. This integration explicitly considers the environmental footprint, resource consumption, and social impacts across the entire life cycle of urban mobility developments, aligning with the framework's broader objective of anticipating and mitigating stressors. This comprehensive approach facilitates a holistic assessment that transcends immediate operational efficiencies by incorporating environmental and societal dimensions often overlooked in conventional urban planning and project management. Such an integrated approach is crucial for understanding the intricate trade-offs between efficiency and vulnerability within urban systems and for defining sustainable policies [23,45].

Life cycle thinking facilitates the identification of potential environmental and socio-economic burdens and benefits associated with different urban development scenarios, thereby enabling informed decision-making for urban planners and policymakers [46,47]. This approach is based on the systematic evaluation of a process or service's environmental impact throughout its entire life cycle, from inception to completion [27].

The key aspects of the life cycle approach include a systematic evaluation and a comprehensive scope. The systematic evaluation entails a thorough analysis of all inputs and outputs across various life cycle stages, while the comprehensive scope extends to cover ecological, social, and economic dimensions, moving beyond isolated impact assessments [27,48].

The life cycle approach supports effectively the informed decision-making by providing a holistic view of urban sustainability, thereby enabling urban planners to identify optimal strategies for mitigating environmental impacts and enhancing urban infrastructure robustness [46]. This integration is particularly vital for assessing sustainable urban mobility, which, despite its positive contributions, also carries substantial negative impacts on ecosystems and human health through greenhouse gas emissions and air pollutants [27]. Moreover, the life cycle perspective, when applied to urban mobility systems, necessitates a comprehensive evaluation of the entire value chain, encompassing raw material extraction, manufacturing, distribution, use, and end-of-life considerations for vehicles and their associated infrastructure, thereby revealing environmental hotspots that might otherwise be overlooked [27].

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When integrated with KPIs frameworks, the LCA approach provides a robust mechanism for evaluating the long-term sustainability and resilience of urban systems and interventions, anticipating and mitigating shocks and stresses [45]. This methodology is crucial for understanding the intricate trade-offs between efficiency and vulnerability within urban systems and for defining sustainable policies [23]. The methodological framework of LCA, standardized by ISO 14040:2006 and 14044:2006, allows for robust assessments of various systems, including nature-based solutions and broader urban product systems, making it a reliable tool for quantifiable impact analysis [49].

The AbtifragiCity KPIs' framework will therefore leverage LCA's comprehensive environmental and economic assessment capabilities to enhance its predictive power for urban stressors, particularly concerning resource management and infrastructure sustainability [47]. This will involve the employment of region-specific environmental datasets and considering factors like GHG emission potential and raw material consumption to provide a thorough evaluation of urban interventions. This integration will allow for a more nuanced understanding of the environmental impacts associated with urban mobility and infrastructure projects, informing decisions aimed at fostering climate-resilient urban development. Specifically, the framework will utilize modular LCA approaches, which are particularly effective for assessing the sustainability of urban developments by integrating various components pertaining to the built environment and infrastructure networks [46].

The integration of LCA with the comprehensive KPIs' framework could support the identification of potential gaps and trade-offs inherent in urban interventions, thus providing a clearer picture of their environmental and economic implications over their entire life cycle. It also facilitates understanding the complex interrelations among urban issues, thereby preventing actions from working against each other in the pursuit of sustainable cities, which can anticipate and mitigate the impacts of various disruptions. Furthermore, the adoption of the lifecycle thinking can practically capture the processes of learning and adaptation essential for urban planning in response to emerging climate risks and evolving policy landscapes [46]. This holistic perspective is essential for developing resilient and antifragile urban mobility strategies that account for both current challenges and future uncertainties, ensuring that interventions are both effective and sustainable in the long term.

5.2 Alignment with LCA frameworks

There are several research studies that highlight the importance of aligning urban planning with established LCA frameworks, especially when addressing complex urban issues and striving for multi-indicator assessments [46,47]. Therefore, it is pertinent to examine the alignment of the AntifragiCity KPIs framework with established LCA approaches developed and validated in relevant research projects. Specifically, the frameworks developed by the ACCORD and SemanticLCA projects will be instrumental in refining the AntifragiCity framework's capacity for comprehensive life cycle evaluation, particularly through their emphasis on modularity and detailed data integration. Table 5.1 outlines table outlining the aspects of alignment between the proposed KPIs' framework, the ACCORD project, and the SemanticLCA project.

Aspect of Alignment	AntifragiCity KPIs' Framework	ACCORD	SemanticLCA
Integration of LCA	Explicitly integrates lifecycle thinking principles and leverages	Involves LCA for Green Building Certification within its use cases,	Focuses on devising an LCA framework weighting system, addressing

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	methodologies to embed comprehensive environmental and sustainability assessments from the outset. Enhances predictive power for urban stressors, resource management, and infrastructure sustainability through modular LCA approaches.	aiming to develop APIs for BIM-based LCA and green building certification [50].	uncertainties in environmental impact assessments, and aiming for a more inclusive and participative approach to LCA, including its social dimension [NO_PRINTED_FORM] [51].
Semantic frameworks and ontologies	Connected with an Ontology Framework that provides a structured representation of urban events, risks, and interrelationships, serving as a semantic backbone for consistent data and models. This integration ensures semantic consistency and aligns diverse data sources for robust analysis and decision-making.	Develops a semantic framework for European digital building permitting processes, regulations, data, and tools [50]. Utilizes a set of abstract ontologies defining conceptually the concepts employed by the framework [50].	While Ghoroghi et al. [51] don't explicitly mention "semantic framework" or "ontology" in relation to its core methodology, its focus on modelling complexity and dynamics and adapting to changing conditions within LCA weighting systems. Ghoroghi et al. [51] imply a structured and defined understanding of concepts that could align with semantic approaches.
Adaptability and dynamic response	Emphasizes a flexible methodological approach for KPI definition, allowing for continuous evolution and customization based on varying urban contexts and strategic objectives, and dynamic adaptation to emerging urban challenges and opportunities.	Aims to provide solutions and tools that offer consistency, interoperability, and reliability with national regulatory frameworks, processes, and standards, allowing for integration of technical solutions in dynamic lifecycle phases [50].	Utilizes Multi-Agent Systems (MAS) to adapt to changing conditions and handle uncertainties in assessing complex and ever-changing ecosystems [51]. MAS improves responsiveness to changing conditions, allowing scenario exploration and assessment of potential LCA impacts under uncertainty, enhancing resilience and sustainability [51].
Focus on urban stressors and resilience	Aims to enhance predictive power for urban stressors, particularly concerning resource management and infrastructure sustainability, ensuring a robust mechanism for evaluating long-term sustainability and	The overall aim includes improving the quality of design and construction processes to support climate-neutral buildings and advancing a sustainable built environment [50], which contributes to urban	Emphasizes the importance of monitoring, evaluation, and recalibration to ensure the continuous fitness of the MAS weighting system, enhancing the resilience and sustainability of

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	resilience of urban systems and interventions.	resilience by mitigating environmental impacts.	environmental strategies [51].
Data integration and interoperability	Integrates diverse quantitative and qualitative data source types for a comprehensive understanding of urban dynamics, and its Ontology Framework facilitates data integration and interoperability.	Focuses on integrating existing compliance tools as microservices and developing solutions and tools that ensure consistency, interoperability, and reliability with national regulatory frameworks [50]. Information flows between components use APIs and exchange data in standardized formats [50].	The MAS approach in SemanticLCA involves modelling complex interactions among various entities [51], which implicitly requires the integration of diverse data to simulate behaviours and scenarios effectively.

Table 5.1. Aspects of alignment between the AntifragiCity framework, the ACCORD project and the SemanticLCA project

Based on Table 5.1, there are five (5) discrete aspects of alignment that underscore the potential for synergistic integration between the AntifragiCity framework and established LCA methodologies. Firstly, the integration of the LCA approach, which is explicitly considered within the AntifragiCity KPIs framework, is core in ACCORD and SemanticLCA, particularly through their focus on modularity and detailed data integration. Particularly, the ACCORD project employs LCA for Green Building Certification, thereby demonstrating its practical application in assessing urban actions at the site level, whereas the SemanticLCA focuses on the weighting systems for LCA frameworks and addressing uncertainties in environmental impact assessments.

Regarding the aspect of semantic frameworks and ontologies, AntifragiCity leverages a comprehensive ontology for urban events and risks to ensure semantic consistency across diverse data sources. This aligns with the ACCORD project's development of a semantic framework for European digital building permitting processes, utilizing abstract ontologies to define conceptual elements [50]. Similarly, SemanticLCA, while not explicitly referencing *semantic frameworks*, inherently requires a structured and defined understanding of complex ecosystem interactions to model MAS and account for uncertainties in environmental impacts [51].

The aspect of adaptability and dynamic response is addressed in AntifragiCity through its flexible methodological approach to KPIs' definition, allowing continuous evolution and customization based on varying urban contexts and strategic objectives. Similarly, in ACCORD, solutions are designed for consistency and interoperability with national regulatory frameworks, processes, and standards, enabling integration of technical solutions in dynamic lifecycle phases [50]. In SemanticLCA, the Multi-Agent Systems approach is employed to adapt to changing conditions and handle uncertainties in assessing complex and ever-changing ecosystems, thereby improving responsiveness to changing conditions and enabling scenario exploration [51]. This observation underscores the shared emphasis on dynamic adaptability across all three frameworks, albeit through distinct mechanisms.

Moreover, the focus on urban stressors and resilience is paramount, with AntifragiCity aiming to enhance predictive power for urban stressors, particularly concerning resource management and infrastructure sustainability in the context of urban mobility. This objective resonates with ACCORD's commitment to improving the quality of design and construction processes

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for climate-neutral buildings, contributing to urban resilience by mitigating environmental impacts, and with SemanticLCA's emphasis on continuous monitoring and recalibration to ensure the fitness of its weighting system, thereby enhancing the resilience and sustainability of environmental strategies.

Finally, regarding data integration and interoperability, AntifragiCity integrates diverse quantitative and qualitative data sources, utilizing its Ontology Framework to facilitate semantic consistency and interoperability to ultimately leverage the development of relevant APIs. This approach parallels ACCORD's strategy of integrating existing compliance tools as microservices and leveraging APIs for standardized data exchange, and aligns with SemanticLCA's reliance on integrating diverse data to model complex interactions within its agent-based simulations. This comprehensive integration of data, semantic frameworks, and adaptive modelling techniques highlights a convergence towards robust, inter-operable systems capable of discerning complex urban dynamics and informing resilient urban planning.

5.3 Validation of applicability

The validation approach of the AntifragiCity KPIs' framework will involve rigorous testing of the framework against real-world urban data, comparing its predictions against actual outcomes to refine its accuracy and reliability. For this purpose, the framework will be applied in real use cases in the AntifragiCity's demo sites. This empirical validation will involve a detailed examination of various urban indicators, including transportation efficiency, environmental footprint, and urban safety, against the framework's projections to ensure its practical applicability and robustness in diverse urban contexts and disruption scenarios.

The framework's development process includes preparation of a pilot model, rigorous data collection, and deployment leading to its validation. This comprehensive validation strategy ensures that the AntifragiCity framework is not only theoretically sound but also practically robust and scalable for real-world urban planning and resilience assessment.

The framework's development process itself, as detailed in Section 4, involved a structured progression from the preparation and development of a pilot model to rigorous data collection, culminating in the deployment and validation of the proposed framework in the upcoming work packages of the AntifragiCity project. This iterative and stakeholder-inclusive development cycle underscores its practical and user-centric design without overlooking the scientific soundness. The detailed methodological approach for the validation process is presented below in Figure 5.1.

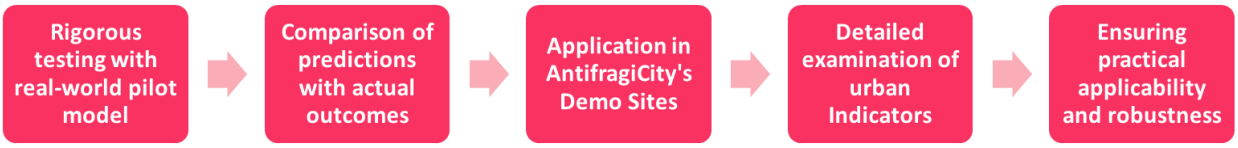

Figure 5.1. Steps for the validation of the AntifragiCity KPIs' framework

The first step of the AntifragiCity framework's validation process involves the rigorous testing with a real-world pilot model. This pilot model will undergo extensive evaluation to assess its capacity to effectively integrate operational urban data and compose a comprehensive, context-specific list of KPIs. The next step involves comparing the predictions with the actual outcomes from the pilot case. This phase will involve expert validation through workshops and structured walkthroughs with urban planners, mobility engineers, resilience experts, and data scientists,

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ensuring that the framework's outputs are aligned with real-world operational requirements and expert knowledge. This expert validation will critically assess the framework's structure and contents, confirming its accuracy and reliability in representing the real-world system.

According to Figure 5.1, the following step of the validation process involves the full-scale application of the framework in the AntifragiCity's demo sites, specifically in Bratislava, Larissa, Odessa and the AHEPA case in Thessaloniki. This broad application across multiple urban environments will allow for comprehensive assessment of the framework's adaptability and performance under varying socio-economic, environmental, and infrastructural conditions. Subsequently, the detailed examination of the proposed urban mobility indicators will determine how effectively these indicators capture the nuances of urban dynamics and contribute to an integrated understanding of the urban pulse in each specific use case. This empirical validation will incorporate feedback from demo sites' experts through structured surveys, ensuring that the framework accurately reflects real-world urban system behaviours and effectively addresses the challenges faced by city stakeholders.

The insights gained from these practical case studies, coupled with the operational experience through the use cases in demo sites aim to ensure the framework's practical applicability and robustness across diverse urban contexts and disruption scenarios. This systematic validation approach ensures the framework can effectively capture interdependent processes across environmental, technical, and societal subsystems in a holistic manner.

5.4 Compliance with standards and regulations

The AntifragiCity framework is designed to adhere strictly to relevant international and national standards and regulations governing urban planning, environmental protection, and data privacy, thereby ensuring its legal and ethical soundness. Specifically, the AntifragiCity KPIs' framework aims to align with key global and European legal frameworks, including the Global Plan for the Decade of Action for Road Safety 2021-2030, the Stockholm Declaration, the Sustainable Development Goals, the New Urban Agenda, and the Green Deal.

The AntifragiCity KPIs' framework explicitly aims to comply with these critical international guidelines and goals by integrating their principles into its comprehensive and adaptable system for monitoring and evaluating urban mobility. It achieves this by focusing on robust indicator sets that address sustainability, safety, efficiency, resilience, and equity, enabling cities to make data-driven decisions and adapt to evolving urban challenges. This compliance ensures that the framework not only provides a methodological backbone for urban assessment but also upholds the highest standards of governance and societal well-being. The main points of alignment, which illustrate how the framework's structure and proposed indicators directly map to the objectives of these foundational documents, are as follows:

- **The Global Plan for the Decade of Action for Road Safety 2021-2030:** The AntifragiCity framework aligns by prioritizing safety as a key thematic area and integrating diverse data sources to predict stressors and adapt to unforeseen challenges in urban mobility. Its focus on performance, resilience, and sustainability supports the goals of reducing road traffic deaths and fostering safer urban environments. This commitment is further demonstrated through the incorporation of safety-focused KPIs, which are instrumental in identifying high-risk areas and informing targeted interventions to enhance road safety. Similarly, the framework's emphasis on real-time data processing and event classification directly supports

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proactive measures recommended by the Global Plan, enabling rapid response to incidents and continuous improvement in urban safety infrastructure.

- **Stockholm Declaration. Third Global Ministerial Conference on Road Safety: Achieving Global Goals 2030:** The framework's commitment to holistic urban assessment, integrating social, environmental, and economic dimensions, mirrors the Declaration's call for interlinked SDG policy areas. Its focus on data-driven decision-making, monitoring, and adaptability for urban systems, including traffic management and accident reduction, contributes to achieving road safety targets. The framework's integration of diverse data streams and its capacity for near real-time event sensing further support the Declaration's emphasis on data-driven interventions for enhanced road safety and urban resilience. This alignment ensures that the AntifragiCity framework facilitates the development of robust strategies for enhancing road safety and promoting sustainable urban mobility, consistent with international mandates.

- **The Sustainable Development Goals:** The framework directly contributes to several SDGs, particularly SDG 11 and SDG 9, by providing a robust methodology for monitoring and improving urban sustainability and resilience through integrated KPIs. The framework's core objective of "sensing the urban pulse, predicting stressors," and its thematic area of resilience, align perfectly with the goals of strengthening economic, social, health, and environmental resilience, and preventing and reducing disaster risks. Its adaptable KPIs' framework supports monitoring the implementation and effectiveness of local and national strategies to enhance urban resilience against various disruptions. Moreover, by integrating indicators related to environmental footprint and resource efficiency, the framework also supports SDG 12 and SDG 13. Additionally, the framework's emphasis on integrating diverse data sources, predicting stressors, and its capacity for near real-time event sensing supports data-driven interventions is crucial for effective traffic management and accident reduction strategies, which are essential for achieving SDG 3.6. The AntifragiCity framework, through its integrated perspective on urban dynamics and its focus on adaptability and context-specificity for KPI selection, aligns with the holistic approach of fostering positive economic, social, and environmental links between urban, peri-urban, and rural areas. It provides tools for evaluating policies that aim for balanced development, considering the socio-economic and environmental aspects of urban areas.

- **The New Urban Agenda:** By incorporating principles of integrated urban planning and governance, the framework aids in developing sustainable, resilient, and inclusive urban settlements. It further supports the Agenda's call for enhancing urban resilience, particularly concerning disaster risk reduction, by providing mechanisms for proactive stressor prediction and adaptive urban management. The framework's capacity to monitor and evaluate urban performance through granular KPIs directly supports the Agenda's call for enhanced urban-rural linkages and the development of sustainable urban mobility systems. This comprehensive alignment underscores the AntifragiCity framework's potential as a critical tool for urban planners and policymakers to navigate the complexities of urban development, ensuring that cities evolve in a manner that is both sustainable and responsive to the needs of their inhabitants and the broader global community.

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- **The Green Deal:** The framework's commitment to capture environmental footprints and promote resource efficiency directly contributes to the Green Deal's objectives of achieving climate neutrality and fostering a circular economy. Specifically, by integrating LCA methods and other ecological indicators into its analytical processes, the framework supports the transition towards a more sustainable urban metabolism. Moreover, the exploration of novel mobility solutions coupled with intelligent transport systems within the AntifragiCity framework offers a pathway to significantly reduce greenhouse gas emissions from the transport sector, aligning with the Green Deal's aspirations for cleaner and more efficient mobility.

Finally, to ensure compliance with international KPI standards, the KPIs' framework has incorporated recognized indicators from bodies such as the ITF, OECD, and UN, making its assessment of urban resilience and sustainability both robust and globally comparable. The meticulous examination of the organisational reports during the early stages of AntifragiCity framework development provided a foundation for integrating comprehensive urban sustainability and resilience indicators, aligning with the principles of various international frameworks. This integration ensures that the framework's metrics for urban resilience and antifragility are both comprehensive and aligned with global best practices, supporting adaptive policies and structural changes.

Table 5.2 displays a traceability matrix that links the most selected KPIs of the proposed framework with the respective regulation framework, e.g. the SDGs, the Green Deal etc. The matrix ensures that the Key Performance Indicators align with established sustainability and policy objectives and standards.

KPI	Regulation Framework	Reference
Commuter Flow Centrality	SDG 11.2.1 focuses on accessibility to public transport, with data requirements around residential locations and destinations, which are components for flow analysis.	(Wang et al., 2022; Yu et al., 2025)
Mobility throughput	Not explicitly named as a KPI in the official SDG targets. However, the concept is closely related to SDG indicator 9.1.2, which measures "passenger and freight volumes," reflecting the movement of people and goods. Related to "Impact Assessment" for improving "personal mobility" and "traffic flow efficiency".	(AECOM, 2015; Cartolano et al., 2018; Ghoroghi et al., 2025; Sourlas et al., 2020)
GHG Emissions	EU target of 60% reduction below 1990 levels by 2050, strategies from the 2011 White Paper on transport. Renewable Energy Directive. Paris Agreement.	(AECOM, 2015; ITF, 2016; Mobility and Transport STATISTICAL POCKETBOOK 2025 EU Transport in Figures, 2025)
CO2 Emissions	Policies including carbon taxes, fuel taxes, biofuel blending mandates, CO2-based vehicle taxes, and fuel economy/emission performance standards. Measures to meet the Paris Agreement.	(AECOM, 2015; Gruetzmacher et al., 2020; ITF, 2016; Moreno & García-Álvarez, 2017; Yan et al., 2021)
Pollutant emissions from transport	EU limit values for the protection of human health. Vehicle emission standards (e.g., Euro 1/I through Euro 6/VI).	(Gillis et al., 2015; ITF, 2015; Moreno & García-Álvarez, 2017; Sdoukopoulos et al., 2019)
NOx emissions	EU steady-state emissions standards for heavy-duty diesel engines (e.g., Euro I - Euro 6). EU limit values for air quality.	(ATO Urban Database - Masterlist of Indicators, 2021; Gillis et al., 2015; ITF, 2015)

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Noise pollution	Implicitly related to EU Directives for environmental noise through models like CNOSSOS-EU for strategic noise mapping. Regulatory limits for noise levels.	(Gillis et al., 2015; Martino et al., 2021)
Population exposed to noise	Related to EU directives on environmental noise, assessing impact on human health. Regulatory limits for noise exposure.	(Gillis et al., 2015; Martino et al., 2021)
Modal split	Related to policies encouraging sustainable mobility practices, such as mobility plans required for enterprises (e.g., in France); broader transport policy scenarios aiming to influence urban mobility patterns	(ATO Urban Database - Masterlist of Indicators, 2021; Ballantyne et al., 2024; Ghoroghi et al., 2025; ITF, 2016, 2023; Martino et al., 2021)
Personal travel mode split	Related to policies encouraging sustainable mobility practices and influencing mode choice, as part of transport policy scenarios and mobility plans	(ATO Urban Database - Masterlist of Indicators, 2021; Ballantyne et al., 2024; Ghoroghi et al., 2025; ITF, 2016, 2023; Martino et al., 2021; Yang et al., 2020)
Freight mode split	Related to climate change mitigation scenarios aiming for shifts to less energy-intensive modes; national obligations to report freight transport data (e.g., EU members to Eurostat); policy portfolios for sustainable transport	(Martino et al., 2021; Muratori et al., 2017; Parraguez Ruiz & Rosati, 2019; Yang et al., 2020)
Cross-layer synergy	Related to policy initiatives addressing interconnectedness within urban transport, aiming to influence transport systems through operational and policy interventions	(Delgado et al., 2025; Ghoroghi et al., 2025; Martino et al., 2021; Sun et al., 2020)
Travel times and delays	Sustainable Urban Mobility Plans and frameworks for assessing urban mobility (e.g., SUTI, SUMI, ISO 37120); transport planning focused on accessibility and congestion relief	(Angarita-Lozano et al., 2025; Conference of Ministers of Transport, 1978; Gillis et al., 2015; ITF, 2016; Martino et al., 2021; Vidović et al., 2019)
Vehicle travel times	Related to transport policy and urban mobility assessment frameworks, including those using average vehicle speed as an indicator	(Conference of Ministers of Transport, 1978; Vidović et al., 2019)
Average commute travel time	Related to urban transport master plans (e.g., Singapore's Land Transport Master Plan 2040) setting targets for commute durations; metropolitan transportation plans using accessibility-related indicators for evaluation	(Boisjoly & El-Geneidy, 2017; Gillis et al., 2015; WHO, 2025; Yang et al., 2020)
Travel time	Metropolitan transportation plans setting accessibility objectives and indicators related to travel time; policy discussions justifying road investments based on prospective time savings	(Boisjoly & El-Geneidy, 2017; Cartolano et al., 2018; Conference of Ministers of Transport, 1978; Delgado et al., 2025; Lucas et al., 2016; Vidović et al., 2019)
Average daily trip duration per person	Related to policy scenarios (e.g., High Ambition scenario) and land-use planning aiming to influence trip lengths and reduce passenger-kilometres per trip; sustainable urban mobility indicator systems	(ITF, 2023; Lucas et al., 2016; Vidović et al., 2019; Yang et al., 2020)
Average trip duration per kilometre	Assessment frameworks for transportation efficiency where delay metrics are standardized for comparative analysis across scenarios; broader policy goals related to reducing passenger-kilometres per trip	(ITF, 2023; Sun et al., 2025; Vidović et al., 2019)

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Delays in the transport system	Sustainable Urban Mobility Plans and indicator frameworks assessing urban mobility; policy interventions and initiatives to improve public transport quality and reliability	(Angarita-Lozano et al., 2025; Conference of Ministers of Transport, 1978; Gillis et al., 2015; Gruetzmacher et al., 2020; Martino et al., 2021; Vidović et al., 2019)
Congestion and delays	Environment, Quality of life; Aligned with key EU policy goals; Sustainable Urban Mobility Plans	(AECOM, 2015; Gillis et al., 2015; Martino et al., 2021)
Transport demand	Sustainable Development Goals, particularly SDG Goal 11 to "make cities inclusive, safe, resilient and sustainable"; Sustainable development and overall system performance improvement	(Sierra Muñoz et al., 2024; Yang et al., 2020)
Non-motorized transport demand	Promotion of walking and cycling for sustainable livelihoods, better environment, social inclusion, and economic prosperity, aligning with broader European goals; Opportunity for active mobility infrastructure; Walkability and bikeability in transport-space performance	(Martino et al., 2021; WHO, 2025; Yang et al., 2020)
Motorized transport demand	Part of overall transport system performance and sustainability considerations	(Sierra Muñoz et al., 2024; Yang et al., 2020)
PRT Travel demand	Related to Sustainable Development Goals, European Green Deal/EU initiatives, Paris Agreement, and Renewable Energy Directive	(Conference of Ministers of Transport, 1978; ITF, 2016; SDG 11.2.1: Sustainable Urban Transport, 2025), (Gruetzmacher et al., 2020), (Parraguez Ruiz & Rosati, 2019; Yang et al., 2020).
PT demand	Related to Sustainable Development Goals, European Green Deal/EU initiatives, and urban mobility policies	(AECOM, 2015; Martino et al., 2021; Vidović et al., 2019; Yang et al., 2020)
Average public transport speed	Mobility system performance; Urban mobility assessment	(Vidović et al., 2019; Yang et al., 2020)
Quantity of PT trips / Number of daily PT trips	Related to Sustainable Development Goals and urban mobility policies	(Vidović et al., 2019)
Freight ton-km	Environment, Quality of life (related to GHG emissions); Sustainable Development Goals	(Martino et al., 2021; Sdoukopoulos et al., 2019)
Ridership	Sustainable Development Goals; Urban mobility policies	(AECOM, 2015; Martino et al., 2021; Yang et al., 2020)
Traffic Accidents & Injuries	Sustainable Development Goal 3.6.1 (reducing deaths and injuries from road traffic accidents); Global Plan for the Decade of Action for Road Safety 2021-2030; Quality of life	(Chan et al., 2025; Gillis et al., 2015; Lee et al., 2025; Martino et al., 2021; The Global Plan for the Decade of Action for Road Safety 2021-2030, 2021; The Global Urban Monitoring Framework, 2022)
Traffic Fatalities	Sustainable Development Goal 3.6.1; Global Plan for the Decade of Action for Road Safety 2021-2030; WHO best practice criteria for road safety legislation; Quality of life	(ATO Urban Database - Masterlist of Indicators, 2021; The Global Plan for the Decade of Action for Road Safety 2021-2030, 2021; The Global Urban Monitoring Framework, 2022; Chan et al.,

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		2025; Gillis et al., 2015; Martino et al., 2021; WHO, 2025)
Traffic Conflicts & Collision Type	Green Deal: White Paper on transport aims for improvement in road safety. Sustainable Mobility: Traffic safety is recognized as an indicator for sustainable mobility.	(Andreotti & Selpi, 2025; Gillis et al., 2015; Gruetzmacher et al., 2020; Tran et al., 2021; Yang et al., 2020)
Laws and Regulations	UN road safety conventions; National or regional legislation and systems; WHO best practice criteria for road safety legislation; Legal regulation in road traffic safety areas	(Chan et al., 2025; The Global Plan for the Decade of Action for Road Safety 2021-2030, 2021; WHO, 2025)
Transport Accessibility	Seen as an alternative to mobility-oriented planning paradigm, contributing to broader environmental, economic, and social goals; Part of transport-spaces system performance	(Boisjoly & El-Geneidy, 2017; Gillis et al., 2015; ITF, 2016; Martino et al., 2021; Yang et al., 2020)
Accessibility for Emergency Rescue	SDGs: Transportation is a vital enabler of sustainable development, enhancing urban accessibility across all 17 SDGs.	(Abouelela et al., 2025; Gillis et al., 2015; Serdar et al., 2022)
Feeling of safety during traveling	Quality of life; Perceived urban quality of life	(Gillis et al., 2015; Kutela et al., 2025; Martino et al., 2021)
Mobility satisfaction	Related to Sustainable Development Goals for enhancing human well-being and overall quality of life; Framework for evaluating urban plans considering sustainable development	(Gillis et al., 2015; Zhang et al., 2024)
General satisfaction	Related to Sustainable Development Goals, aims to improve QOL in urban areas, with safety being a key social feature. A factor in citizen satisfaction for urban mobility assessments	(Fedujwar & Agarwal, 2025; Gillis et al., 2015; Heidari et al., 2024; Karjalainen & Juhola, 2021; Kutela et al., 2025)
Overall quality of life in the city	Sustainable Transportation serving society through efficient and accessible transport systems to increase Quality of Life; Urban planning and ST significantly affect sustainability and livability of cities, vital for residents' QOL; Related to livability and sustainability of life in human societies; UN Sustainable Development Goals for more livable, sustainable cities	(Heidari et al., 2024; Kutela et al., 2025)
Satisfaction with traffic flow on major city streets	Implicitly linked to urban quality of life and efficient transportation systems, contributing to sustainable development. Congestion and delays are important targets for improving mobility sustainability	(Gillis et al., 2015; Kutela et al., 2025)
Satisfaction with on-street bicycle accessibility	Related to policies that improve the safety, convenience, and feasibility of cycling; Providing a safe and comfortable cycling environment (objective of on-street bike lanes)	(Kutela et al., 2025; Parraguez Ruiz & Rosati, 2019; Tran et al., 2021)
Passenger waiting costs	Related to efficient transportation systems, a component of sustainable development. Travel cost and financial sustainability are critical economic factors for public transportation operations.	(Al-lami & Török, 2025; Gillis et al., 2015; Heidari et al., 2024; ITF, 2016)
Vehicle running and	Policies that increased direct costs (vehicle registration taxes, green taxes); CO2 Tax Special taxes for environmental pollution	(Conference of Ministers of Transport, 1978; Fanati Rashidi et al., 2025; Parraguez Ruiz & Rosati, 2019)

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depreciation costs		
Personnel costs	Efficiency and optimal resource utilization are components of sustainable development. Operational costs, including personnel costs, are critical for financial sustainability and efficiency in public transportation operations	(Conference of Ministers of Transport, 1978; GLOBAL URBAN MOBILITY INDICATORS PUBLIC TRANSPORT METRICS FROM 53 CITIES WORLDWIDE IN 2023, 2023; Karam et al., 2022; Mahdinia et al., 2018)
Running costs	Taxes and levies are operating or running costs; Specific taxes and levies for the use of common infrastructures	(Chung, 2017; Conference of Ministers of Transport, 1978; Karam et al., 2022)
Travel costs	Package of complementary transport and land-use policies that have increased both direct and indirect costs of car use; Policies to reduce emissions associated with passenger road transport should include pricing measures to capture external costs of car use	(Eurostat, 2024; Ghoroghi et al., 2025; ITF, 2016, 2021, 2023; Lucas et al., 2016; Mobility and Transport STATISTICAL POCKETBOOK 2025 EU Transport in Figures, 2025; Parraguez Ruiz & Rosati, 2019; Sierra Muñoz et al., 2024; Vidović et al., 2019)
Parking fees	Parking pricing and restrictions regulations to control availability and price; Parking management policies; Pricing measures or congestion charges to reduce vehicle traffic; Parking pricing to capture external costs of space consumption by static cars	(ITF, 2016, 2021, 2023; Parraguez Ruiz & Rosati, 2019; Sierra Muñoz et al., 2024; Vidović et al., 2019; WHO, 2025)
Fuel price	ITF Policies on fuel prices, fuel taxes; Carbon pricing maintained for ICE vehicles	(Eurostat, 2024; ITF, 2016, 2019, 2023; Karam et al., 2022; Parraguez Ruiz & Rosati, 2019; Sdoukopoulos et al., 2019; Vidović et al., 2019; Yan et al., 2021)
Daily parking ticket costs	Regulation of zonal minimum wage by Governmental Decrees (though specific to public transport, it indicates governmental regulation of transport-related costs); Regulatory measures affecting parking prices	(ITF, 2021; Vidović et al., 2019)
Residents' parking ticket expenditure	Urban parking policies; Parking management policies; Sustainable Urban Mobility Planning; ITF Regulations to control parking prices	(ITF, 2021, 2023; Martino et al., 2021; Parraguez Ruiz & Rosati, 2019; Vidović et al., 2019)
Expenditure for public transportation	Sustainable Development Goals; High Ambition Scenario for decarbonizing transport (ITF)	(ITF, 2023; Naganathan & Chong, 2017; Vidović et al., 2019)
Private transportation cost	Policies increasing direct costs (vehicle registration taxes, green taxes) and indirect costs (slower speeds, less parking, congestion) of car use; Pricing policies or policies regulating vehicle use	(ITF, 2016; Lucas et al., 2016; Parraguez Ruiz & Rosati, 2019; Vidović et al., 2019)
Annual infrastructural investments	Sustainable Development Goals; "Decide and provide" approach to infrastructure provision; Sustainable urban mobility policies in High Ambition scenario (ITF); Strategic Transport Plan	(ITF, 2023; Martino et al., 2021; Vidović et al., 2019)
Annual transportation	Sustainable Development Goals, particularly Indicator 1.a.2 on government expenditure for essential services and Indicator 1.b.1 on national spending plans for	(Abouelela et al., 2025; Bachok et al., 2015; ITF, 2016;

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expenditures per capita	poverty reduction and gender-sensitive strategies; Sustainable transportation performance evaluation; Nationally Determined Contributions in transport policies	Naganathan & Chong, 2017; Vidović et al., 2019)
Annual public transportation revenues per public transportation expenditures	Financial sustainability; Sustainable development goals	(Chung, 2017; Gillis et al., 2015; Miller et al., 2016; Yang et al., 2020)
Infrastructure spatial features	Urban planning to improve urban habitat and the efficiency of the urban system; Network planning and development	(Wang et al., 2022; Yu et al., 2025)
Length of total operational roads	Sustainable mobility system (implicit via land use taken by transport modes); Infrastructure supply in quantity and quality	(2024 Inland Transport Statistics FOR EUROPE AND NORTH AMERICA, 2024; ATO Urban Database - Masterlist of Indicators, 2021; Gillis et al., 2015; ITF, 2016)
Ratio of roadways of specific standards	Design standards; Technical standard and star rating target for all designs; Specifications in Highway Design Manual	(AECOM, 2015; ATO Urban Database - Masterlist of Indicators, 2021; The Global Plan for the Decade of Action for Road Safety 2021-2030, 2021; Tran et al., 2021)
Network density / Total roads length per area	Green road transportation management and environmental sustainability; Urban planning to improve urban habitat and the efficiency of the urban system	(ATO Urban Database - Masterlist of Indicators, 2021; Din et al., 2023; Gillis et al., 2015; Wang et al., 2022)
Public transport stops	SDG 11.2: Sustainable Cities and Communities (Target 11.2)	(GLOBAL URBAN MOBILITY INDICATORS PUBLIC TRANSPORT METRICS FROM 53 CITIES WORLDWIDE IN 2023, 2023; SDG 11.2.1: Sustainable Urban Transport, 2025; The Global Urban Monitoring Framework, 2022)
Public transport route network	SDG 11.2: Sustainable Cities and Communities (implied by inclusive urban mobility)	(GLOBAL URBAN MOBILITY INDICATORS PUBLIC TRANSPORT METRICS FROM 53 CITIES WORLDWIDE IN 2023, 2023)
Public transport efficiency	EU climate and sustainability goals, European Green Deal (implied)	(2024 Inland Transport Statistics FOR EUROPE AND NORTH AMERICA, 2024; Gruetzmacher et al., 2020)
Public transport availability	SDG 11.2: Sustainable Cities and Communities (Target 11.2.1)	(SDG 11.2.1: Sustainable Urban Transport, 2025; The Global Urban Monitoring Framework, 2022)
Total highways length per area	Sustainable land use and urban planning, European Green Deal (implied by environmental impact)	(Din et al., 2023; Gillis et al., 2015)
Urban roads length per area	Sustainable land use and urban planning, SDG 11.3.1 (implicitly through land consumption rate)	(ATO Urban Database - Masterlist of Indicators, 2021; The Global Urban Monitoring

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		Framework, 2022; Gillis et al., 2015)
Total highways length per capita	Sustainable land use and urban planning	(ATO Urban Database - Masterlist of Indicators, 2021; Gillis et al., 2015)
Bus route length per area	SDG 11.2 (inclusive urban mobility), Urban mobility planning	(ATO Urban Database - Masterlist of Indicators, 2021; ITF, 2016)
Fossil fuel consumption in transportation	Paris Agreement, SDG 12.c (rationalize inefficient fossil-fuel subsidies), Policies for 'well-to-wheel' emissions	(Gruetzmacher et al., 2020; ITF, 2019; Parraguez Ruiz & Rosati, 2019)
Energy consumption in transportation	SDG 7.3 (energy efficiency), Policies aiming for transport efficiency, Sustainable Development goals	(Gillis et al., 2015; ITF, 2019)
Density of public services within all area services	Inclusive cities, accessibility objectives in metropolitan transportation plans of the Global Monitoring Framework	(ATO Urban Database - Masterlist of Indicators, 2021; The Global Urban Monitoring Framework, 2022)
Greening coverage of built-up areas	Environmental protection and greening policies, Strategies to increase urban green areas, Urban planning for sustainable cities	(ATO Urban Database - Masterlist of Indicators, 2021; Petit-Boix et al., 2017; Yang et al., 2020)
Green space area	Urban planning authorities, Environmental sustainability	(The Global Urban Monitoring Framework, 2022)
Green space area per capita	Urban planning authorities, Improve quality of life and air quality, Environmental sustainability policies	(ATO Urban Database - Masterlist of Indicators, 2021; The Global Urban Monitoring Framework, 2022; Gillis et al., 2015)
GDP per capita	ITF Economic development goals, Income growth and transport demand, Economic profile of cities	(ITF, 2016, 2023)
Urban growth generated by transport sector	Sustainable Urban Mobility Plans, Land Use and Transport Integration models (ITF)	(ITF, 2015, 2016; Yang et al., 2020)
Net public finance	Financial sustainability of urban mobility systems, Funding for public transport infrastructure, Fiscal policies	(Heidari et al., 2023; ITF, 2023; Sdoukopoulos et al., 2019; WHO, 2025)

Table 5.2. Traceability matrix of KPIs and regulation and policy frameworks

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6 KPIs' framework application

This section details the application of the developed KPIs' framework to capture the urban pulse, including the identification of potential stressors. The framework supports the articulation of various simulation scenarios (subsection 6.2), as well as the assessment of their impacts on urban systems and the emergence of antifragile properties (subsection 6.3). Moreover, this section elucidates the framework's contribution to AntifragiCity's SUMA API (subsection 6.3.1) and its potential applications beyond the AntifragiCity project (subsection 6.4).

6.1 Outline and use

The AntifragiCity framework for KPIs serves as a critical instrument for evaluating and enhancing resilience and sustainability of urban mobility systems by monitoring a comprehensive set of indicators. This framework facilitates the selection of contextually relevant indicators to assess diverse urban mobility solutions in response to diverse events, thereby enabling practitioners to make informed decisions for resilient and antifragile urban development. **Figure 6.1** presents graphically the outline of use for the AntifragiCity KPIs' framework.

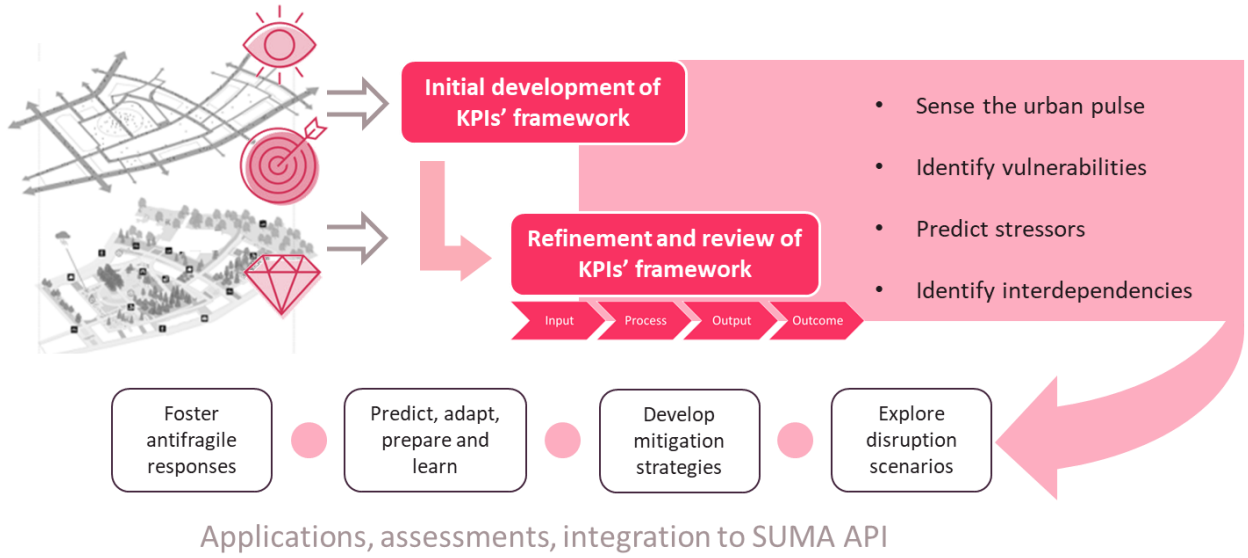

Figure 6.1. Outline of use for the KPIs' framework

The principal purpose of the proposed KPIs' framework is to sense the urban pulse, predict potential stressors, and evaluate the efficacy of various interventions aimed at improving urban resilience and sustainability. Its comprehensive and context-specific nature enables a nuanced understanding of the urban pulse across environmental, social, and economic dimensions. This holistic approach is crucial for capturing intricate interdependencies within urban systems and identifying potential vulnerabilities before they escalate into significant disruptions. Furthermore, the framework's predictive capabilities derived from analysing both quantitative and qualitative data streams that enable proactive measures to mitigate risks and enhance urban system adaptability. It

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also supports systematic evaluation of urban mobility plans and transport infrastructure designs, empowering practitioners and policymakers to select locally relevant indicators for assessing scenarios that advance sustainable development and overall system performance. Ultimately, this fosters transformative improvements that allow systems to withstand diverse disruptions, adapt to evolving conditions, and even benefit from stressors, cultivating an antifragile urban environment.

The AntifragiCity KPIs' framework is designed to move beyond traditional static indicator-based measurement to embrace a dynamic, adaptive, and predictive approach to urban performance assessment, thereby fostering urban systems that not only withstand shocks but also evolve and improve through disturbance. The customized algorithmic development and the iterative loops of review and adaptivity are central to the framework's ability to continuously learn and reconfigure in ways that enhance its effectiveness, relevance, and accuracy over time. The dynamic evolution of the KPIs' framework along with the specificities of the contextual data input, ensures that the system is continually refined to better capture the nuances of urban dynamics and reflect the urban pulse.

An illustrative application of the framework features the AHEPA Hospital in Thessaloniki, which employs records of emergency vehicle response times and public transport disruptions. By leveraging an adaptive and predictive approach, and integrating diverse quantitative data streams, such as records of traffic speed and volume on specific routes, the framework could effectively "sense the urban pulse" and predict potential stressors. For example, specific meteorological events, like the severe snowstorm of 2017 correlates with an increased traffic congestion due to road closures in the vicinity of the AHEPA hospital, thereby causing significant delays for incoming patients. The AntifragiCity KPI framework would use these insights to define specific indicators that capture the increasing traffic congestion around the hospital, as well as indicators that reflect potential alternatives for the hospital' s accessibility, like adjustments to public transport routes or the deployment of additional emergency service vehicles during similar meteorological events, thereby fostering antifragility in urban systems. The monitoring of these KPIs would allow for real-time assessment of interventions and provide valuable feedback for continuous improvement of urban resilience strategies. In combination with the Mobility Triage Handbook (WP3) and the SUMA API (WP5), which could practically support the refinement of the KPIs' selection based on quantifiable effects of diverse disruption scenarios and recovery strategies, the framework provides a comprehensive toolkit for urban planners and policymakers to work on real-data driven simulation scenarios and proactively address vulnerabilities and foster sustainable, resilient, and antifragile urban environments.

6.2 Development of application scenarios

The KPIs' framework supports effectively the generation of hypothetical yet plausible disruptive events. The various events, such as extreme weather conditions, infrastructure failures, or significant socio-economic shifts, to rigorously test the framework's predictive capabilities, have already been identified in the Urban Events Mapping (D2.1) and semantically analysed in the Ontology Framework (D2.5). However, the KPIs' framework facilitates the integration of relevant parameters and metrics to model the impact of these events on urban systems, particularly on mobility networks, and to assess the resilience and antifragility of proposed solutions. Moreover, the context-specific development of the KPIs' framework enables the adaptation of current performance assessment practices to capture expected impacts or refine methodological approaches. This systematic approach allows for a granular understanding and quantifiable evidence on how different urban systems respond to their challenges, facilitating the development of robust and adaptable strategies.

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Each one of the AntifragiCity demo sites is expected to develop its own scenario in accordance with its unique urban characteristics, specific challenges and of course performance assessment practices using its custom KPIs' catalogue. Some application examples for the respective Demo Sites are presented hereby to illustrate the adaptability of the framework to diverse contextual requirements.

6.2.1 Application examples in Demo Sites

Bratislava

The application scenario for the city of Bratislava focuses on flood-related disruptions to bridges and riverside corridors, assessing antifragile routing strategies. Given the city's current performance assessment practices, the authorities already employ various KPIs across all the domains, but there is a strong emphasis on the transport network's capacity, particularly public transportation systems. Consequently, the specific KPIs' framework is expected to include more indicators relevant to public transport performance, such as ridership fluctuations, service reliability under stress, and accessibility of alternative routes during disruption, informed by its minimal viable knowledge base. Simultaneously, indicators relevant to accessibility, land use and infrastructure, and urban development will be monitored to gauge the broader societal impacts and recovery potential.

Larissa

The application scenario for Larissa, conversely, focuses on two distinct events, specifically a heatwave and a flood that may occur in quick succession even within a few weeks' interval. Based on the responses of the KPIs' survey, currently the authorities in Larissa are using only a limited number of indicators to assess the performance of their urban systems, primarily related to urban transport and growth-related metrics rather than environmental and social factors. Therefore, the AntifragiCity KPIs' framework for Larissa will prioritize the integration of environmental and social indicators, such as urban heat island effect, air quality, and green space accessibility, alongside mobility and infrastructure resilience, to address these critical gaps. This approach will allow for a comprehensive evaluation of heatwave and flood impacts on vulnerable populations, land use and infrastructure, ecological systems, highlighting interdependencies within the urban setting.

Odessa

The Odessa scenario addresses port-related disruptions and their cascading effects on urban mobility and logistics, incorporating security, governance, and legal constraints into the analysis. Given the current geopolitical context and operational status of the city of Odessa, the local authorities monitor only few urban performance KPIs. Therefore, the AntifragiCity framework for Odesa will integrate a more comprehensive set of indicators, encompassing economic, social, environmental, and governance dimensions to provide a holistic understanding of urban resilience and antifragility in the face of complex disruptions. This comprehensive integration will be crucial for developing robust and adaptable strategies that account for the multifaceted nature of urban challenges in Odessa.

AHEPA Hospital (Thessaloniki)

The AHEPA Hospital in Thessaloniki provides a final case study, focusing on traffic flow disturbances impacting the serviceability of a tertiary hospital and the operations of a densely populated urban centre. The city of Thessaloniki employs a small range of KPIs in its current practice that primarily assess traffic congestion, public transport ridership and transport safety, often overlooking broader aspects of urban resilience such as environmental impact or social equity. The KPI framework

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developed for the special case of the AHEPA Hospital will therefore emphasize indicators related to emergency vehicle access, hospital operational continuity, and public health outcomes, reflecting the critical role of healthcare infrastructure in urban resilience. This tailored approach ensures that each city's unique context and existing data infrastructure are considered, thereby enhancing the applicability and relevance of the framework's outputs in real-world urban planning and policy-making.

6.3 Support of assessment processes

The AntifragiCity KPIs' framework is designed to significantly enhance urban mobility assessment processes by offering a dynamic, adaptive, and predictive approach. It aims to not only help urban systems withstand disruptions but also to improve and grow stronger in response to them, achieving antifragility. The AntifragiCity KPIs will be used in the prioritisation of disruptions in urban mobility, along with the respective response strategies that enhance the entire system's resilience and antifragility.

Specifically, the KPIs' framework will become an integral part of the mobility triage processes of WP3, which aim to assess the impacts of disruptions, prioritise their response measures, and facilitate informed decision-making to mitigate their adverse effects on urban mobility and overall city functioning. The KPIs will demonstrate quantitatively the expected impacts of various urban systems, providing critical insights into how cities can better prepare for and respond to challenges such as climate change, population growth, and infrastructure stress. The mobility triage is expected to rely on modelling and simulation-based tools for disruption anticipation, response, and learning, integrating multi-modal, multi-source mobility data from the selected KPIs.

This integration will allow for comprehensive analyses of cascading effects across various urban systems, thereby enabling the development of a robust, evidence-based decision support tool for city authorities, operators, and citizens in the pilot cities. The KPIs will thus facilitate the measurement and monitoring of antifragility across diverse urban systems, while remaining familiar to local authorities and operators. The final objective of WP3 is the development of a novel Decision Support System (DSS) that is capable to continuously assess the level of resilience of an urban area and helps identify and characterize potential stressors and their severity using an event ontology. The DSS will simulate disruption scenarios and evaluate short-term response strategies, by simulating "what-if" scenarios to understand the implications of sudden changes in transport availability. Moreover, the DSS that integrates the AntifragiCity KPIs' will evaluate societal well-being and mobility outcomes from policy and planning interventions, focusing on who benefits from urban mobility changes and assessing ecological sustainability, economic, outcomes. This comprehensive framework will be tested and validated in the case studies of the AntifragiCity's demo sites.

The AntifragiCity KPIs' framework also supports the development of antifragile traffic control strategies in WP4. It aims to progress beyond current disruption response solutions by integrating real-time data processing, predictive analytics, and adaptive response mechanisms, ensuring safety and sustainability are central to transportation infrastructure. The KPIs will practically support the articulation and refinement of innovative antifragile traffic control processes. Simultaneously, the KPIs will make the proposed framework more contextually consistent, adaptable, and meaningful for local practitioners. Furthermore, the AntifragiCity KPIs will support the determination of the antifragility assessment methodology, leveraging metrics that indicate a system's capacity to not only withstand but also improve from disruptive events. This involves establishing success criteria and thresholds to analyse technical, environmental, and socio-economic aspects, considering the trade-offs between efficiency and vulnerability within urban transport networks.

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6.3.1 Integration into the SUMA API

The Simulator for Urban Mobility Antifragility (SUMA) API offers a novel and versatile application of the proposed KPIs within the project context. The AntifragiCity KPIs' framework is crucial in supporting the SUMA API by providing the core metrics, monitoring capabilities, and analytical foundation necessary for assessing and enhancing urban mobility systems. The SUMA API is designed to include quantifiable metrics for evaluating urban mobility resilience. These metrics incorporate the full range of indicators—both established and novel—developed within the AntifragiCity framework, which assess a system's ability to improve under disruptions. The SUMA platform utilizes these KPIs to continuously sense the urban metabolism, relying on both quantitative and qualitative data streams, including traffic, weather, and social media sentiment analysis. This allows the SUMA platform to provide real-time insights into urban dynamics, thereby facilitating proactive decision-making and adaptive management of urban infrastructure. In fact, the mobility triage DSS, which is accessible through the SUMA platform and helps in adapting and reconfiguring transport systems in response to stressors.

The AntifragiCity KPIs' framework supports the SUMA API's capacity for scenario analysis. It facilitates the generation of hypothetical disruptive events and models their impact on urban systems, allowing for the assessment of resilience and antifragility of proposed solutions. The successful deployment and refinement of the SUMA API in demonstration cities will be measured using the AntifragiCity KPIs. This includes evaluating improvements in urban mobility resilience, public engagement, and the adoption of the API's recommendations. Moreover, the SUMA API aims to provide services for urban mobility management that embody antifragile principles, so the KPIs are central to its continuous processing and analysis of urban environment data. In essence, the AntifragiCity KPIs' framework provides the essential data-driven insights and evaluation mechanisms that enable the SUMA API to function as a comprehensive tool for urban mobility management, allowing cities to not only withstand disruptions but also to emerge stronger from them.

6.4Applications beyond the AntifragiCity project

The AntifragiCity KPIs' framework can be employed as a stand-alone tool for assessing and enhancing urban resilience, offering a versatile solution that extends beyond the immediate scope of the project. It provides a comprehensive solution for evaluating current conditions and forecasting potential stressors. For this purpose, the entire KPIs' catalogue and the pseudocode for the development and the refinement of context-specific frameworks will be available to urban planners, policymakers, and researchers as an open-source resource, promoting wider adoption and adaptation across diverse urban contexts. This open-source approach will foster collaborative innovation, enabling continuous improvement and expansion of the framework's applicability through community contributions and feedback.

The methodology for the development and validation of the AntifragiCity KPIs offers a transferable framework for assessing and enhancing urban resilience across a wider spectrum of urban domains, extending beyond mobility to areas such as energy, water, and public health infrastructure. For instance, the framework could be applied to evaluate the resilience of energy grids against climate-induced disruptions or to monitor the stability of public health systems during pandemics by tracking key metrics and identifying potential vulnerabilities. This broader applicability underscores the framework's potential as a foundational tool for integrated urban

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management, facilitating a holistic understanding of urban systems' capacity to respond and adapt to disruptions.

Consequently, the AntifragiCity KPIs' framework provides a flexible and adaptable system that can help cities globally to not only understand their urban dynamics but also proactively build more resilient and antifragile urban environments. In turn, this extended applicability underscores its potential as a universal framework for urban resilience and sustainability quantification, capable of incorporating diverse urban typologies and local specificities.

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7 Conclusions

The AntifragiCity KPIs' framework for urban mobility, aiming to sense the "urban pulse," predict stressors, and ultimately foster antifragile urban environments requires a comprehensive approach to ensure its adaptability, inclusiveness, and robustness. The proposed framework synthesizes insights from extensive literature reviews and practical experience, identifying both the critical benefits and persistent challenges in implementing KPI-driven evaluation frameworks for urban mobility systems.

The core benefits of KPI-driven frameworks in urban mobility are multifaceted. They are instrumental for comparing mobility levels and impacts across different cities, facilitating international benchmarking, particularly for monopolistic systems. These frameworks identify crucial success criteria and indicators for assessing techno-economic, environmental, and social impacts, which is vital for sustainable urban development. By integrating diverse data sources and methodologies, they generate comprehensive, transparent, and comprehensible indicator sets that enhance communication among stakeholders and support project design and implementation. KPIs enable both pre-emptive impact assessment and post-construction monitoring, allowing for dynamic adaptation to challenges and trends, thereby optimizing performance and achieving long-term sustainability goals. Continuous monitoring also aids in establishing realistic targets, identifying best practices, and guiding resource allocation.

However, the implementation of these frameworks faces significant challenges. Data acquisition, standardization, and the dynamic nature of urban environments pose considerable hurdles. Limitations often stem from the availability, measurability, reliability, and feasibility of data, compounded by a lack of robust collection systems and fragmented data ownership. There is also a tendency to focus on public transportation and private motorized vehicles, frequently overlooking active modes like walking and cycling, leading to a limited scope that neglects comprehensive sustainability. The high variability in indicator use and fragmentation across the field hinder knowledge accumulation and the development of universally applicable tools. Operationalizing certain indicators can be overly complex or specific, leading to the exclusion of qualitative aspects like social equity and well-being, while many existing frameworks lack governance dimensions or citizen perceptions, limiting their generalizability. These challenges highlight a critical need for more adaptable, inclusive, and holistic frameworks.

A systematic literature review, encompassing academic articles, EU-funded research projects, and grey literature from international organizations, formed the foundation for the proposed framework. This review processed a total of 2,540 indicators, which were subsequently refined into context-specific groups due to overlaps. The analysis revealed a complex interplay and frequent overlaps among key thematic areas such as efficiency, safety, sustainability, equity, and ecological footprint.

The review identified a predominant focus on sustainability (53.24% of indicators, covering environmental, economic, and social dimensions) and efficiency/performance (26.44%). Conversely, thematic areas like safety, resilience (3.09%), and equity (4.26%) were found to be significantly underrepresented, suggesting critical gaps in comprehensive urban sustainability assessments. Text-data mapping highlighted commonly used terms such as "number," "vehicle," and "time," indicating a persistent emphasis on quantifiable traffic flow, infrastructure metrics, and mobility patterns. However, terms like "person," "population," and "proportion" also emerged, pointing to a growing recognition of human-centric aspects.

Distinct prioritization emerged across different source categories:

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- Academic Literature: Tended to offer theoretically robust frameworks with sophisticated methodologies, prioritizing environmental sustainability while under-representing social sustainability beyond equity. There was a focus on technical aspects and a preference for indicator systems over composite indices.

- Research Projects' Reports: Emphasized practical application, technical performance, and the social dimension of sustainability, with a notable inclusion of resilience indicators. Cost, vehicle, and driver were frequently occurring terms.

- International Organizations' Reports: Prioritized overall performance, efficiency, and safety, often with a balanced exploration of sustainability, leaning slightly towards the social dimension. They showed less emphasis on resilience and ecological footprint.

These findings collectively underscore the need for a framework that can integrate these diverse perspectives and bridge the existing gaps.

The proposed KPI framework designed to be comprehensive, flexible, and adaptable, moving beyond static measurements to a dynamic, adaptive, and predictive approach. Its development involves a three-stage mixed-methods approach:

1. **Systematic Literature Review:** Synthesizing existing knowledge to form a foundational pool of potential indicators.

2. **Practitioner Survey:** Eliciting requirements and perspectives from urban mobility practitioners across eight European cities to ensure practical relevance and viability. This survey revealed that only 31% (64 out of 209) of the available KPIs were commonly selected by practitioners, highlighting a tailored, context-specific approach rather than a one-size-fits-all adoption. Certain KPI groups, like "Urban development and growth," "Transport modes, travel times and characteristics," and "Ecological impacts," consistently garnered more attention.

3. **Integration of Inputs:** Combining theoretical insights with practical feedback to finalize a robust and comprehensive set of indicators.

The methodological approach for defining and refining context-specific KPIs is systematic and cyclical, involving five steps: delineation of strategic vision through identifying and prioritizing relevant thematic areas; establishing operational objectives by defining detailed, context-specific targets and benchmarks; requirements specification by examining current practices, constraints, and operational limitations to select relevant and actionable indicators; systematic identification and selection by drawing pertinent indicators from the KPIs' catalogue based on relevance, feasibility, and potential for integration; and refinement or extension by evaluating selected indicators for redundancy and interdependence, with options for extending the set to ensure long-term

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applicability and adaptability. This process can be adapted into an automated algorithm, which is included in the form of pseudocode.

The framework leverages an Input-Process-Output-Outcome (IPOO) structure to organize KPIs, ensuring a clear causal chain from raw data to actionable insights. This involves input from diverse quantitative and qualitative data, which undergoes analytical transformation through methodologies like event elicitation and compliance checking to produce immediate, quantifiable results, ultimately yielding a holistic understanding of the urban pulse that informs strategic decisions and adaptation.

A critical component is the AntifragiCity Ontology Framework, which provides a structured, semantic representation of urban events, risks, and their interrelationships. This ontology serves as a semantic backbone for consistent data and models, facilitates event elicitation, integrates KPI concepts, and supports AI analytics and decision-making. It ensures semantic consistency, addresses data heterogeneity, and links micro-level events to city-wide outcomes, enhancing the framework's applicability to antifragility assessment.

The framework also integrates Life Cycle Assessment methods to evaluate the long-term sustainability and resilience of urban systems, considering environmental footprint, resource consumption, and social impacts across the entire life cycle of urban mobility developments. This alignment with LCA frameworks (i.e., ACCORD and SemanticLCA projects) ensures a comprehensive environmental and sustainability assessment, supporting climate-resilient urban development and understanding trade-offs.

The validation approach for the AntifragiCity KPIs' framework involves rigorous testing against real-world urban data in the project's demo sites. This empirical validation will assess the framework's adaptability and performance under varying socio-economic, environmental, and infrastructural conditions, incorporating expert feedback to refine its accuracy and reliability.

Furthermore, the framework is designed to comply strictly with relevant international and national standards and regulations, including the Global Plan for the Decade of Action for Road Safety 2021-2030, the Stockholm Declaration, the Sustainable Development Goals (especially SDG 11 and 9), the New Urban Agenda, and the European Green Deal. This ensures its legal, ethical, and global relevance, promoting robust indicator sets that address sustainability, safety, efficiency, resilience, and equity. Recognized indicators from bodies like the ITF, OECD, and UN are incorporated to ensure global comparability.

The AntifragiCity KPIs' framework serves as a critical instrument for evaluating and enhancing the resilience and sustainability of urban mobility systems. It facilitates the selection of contextually relevant indicators for assessing diverse urban mobility solutions, enabling practitioners to make informed decisions for antifragile urban development. It will be integral to the mobility triage processes (assessing disruption impacts, prioritizing response measures), supporting the development of a Decision Support System for anticipating, responding to, and learning from disruptions in WP3. It also supports the development of antifragile traffic control strategies in WP4. The framework additionally provides the core metrics and analytical foundation for the SUMA API, which aims to offer quantifiable metrics for evaluating urban mobility resilience, provide real-time insights into urban dynamics, and support scenario analysis.

Finally, beyond the project, the AntifragiCity KPIs' framework is envisioned as a stand-alone, open-source tool, transferable to other urban domains like energy, water, and public health infrastructure. Its flexible, adaptable system allows cities globally to understand their urban dynamics and proactively build more resilient and antifragile urban environments, contributing to a universal framework for urban resilience and sustainability quantification.

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9 Appendix

9.1 Appendix A: Sources of literature review

The tables of this section demonstrate the complete lists of sources that were included in the review of the EU-funded research projects (Table 9.) and the international organisations (Table 9.2), respectively.

id	Acronym	Project title	Project URL	Start	End
1	**MultiModX**	Integrated Passenger-Centric Planning of Multimodal Transport Networks	https://cordis.europa.eu/project/id/101114815	Jul-23	Dec-25
2	**MOCO**	Motion Control Systems of Multi-Actuated Ground Vehicles	https://cordis.europa.eu/project/id/101183051	Jan-25	Dec-28
3	**HARMONY**	Holistic Approach for Providing Spatial & Transport Planning Tools and Evidence to Metropolitan and Regional Authorities to Lead a Sustainable Transition to a New Mobility Era	https://cordis.europa.eu/project/id/815269/results	Jun-19	Feb-23
4	**Move2CCAM**	Methods and tools for comprehensive impact Assessment of the CCAM solutions for passengers and goods	https://cordis.europa.eu/project/id/101069852/reporting	Sep-22	Jun-25
5	**Cities-4-People**	People Oriented Transport & Mobility	https://cordis.europa.eu/project/id/723194/reporting	Jun-17	Nov-20
6	**MORE**	Multi-modal Optimisation for Road-space in Europe	https://cordis.europa.eu/project/id/769276/reporting	Sep-18	Feb-22
7	**SELFY**	SELF assessment, protection & healing tools for a trustworthy and resilient CCAM	https://cordis.europa.eu/project/id/101069748/results	Jun-22	May-25
8	**ICT4CART**	ICT Infrastructure for Connected and Automated Road Transport	https://cordis.europa.eu/project/id/768953/results	Sep-18	Feb-22
9	**IMOVE**	Unlocking Large-Scale Access to Combined Mobility through a European MaaS Network	https://cordis.europa.eu/project/id/723314/results	Jun-17	Nov-19
10	**Baseline**	Programme Support Action (PSA) to support Member States in collecting Key Performance for road safety	https://baseline.vias.be/	2020	2022
11	**MOST**	Sustainable Mobility Center / EU / Italian Ministry for Universities and Research, National Recovery and Resilience Plan	Perazzini et al., 2023	2022	2026
12	**MOBSTER**	Initiative for strategic planning of transport systems and network integration measures for more accessible and attractive sustainable mobility	https://www.progettomobster.eu/en/home-page-en/	2019	2022
13	**SPROUT**	Sustainable Policy Response to Urban mobility Transition	https://cordis.europa.eu/project/id/814910/results	May-19	Feb-23
14	**RESLOUTE**	RESilience management guidelines and Operationalization appLied to Urban Transport Environment	https://cordis.europa.eu/project/id/653460/results	May-15	Apr-18

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15	**RESIST**	RESilient transport InfraSTructure to extreme events	https://cordis.europa.eu/project/id/769066	Sept-18	Jun-22

Table 9.1. Research projects included in the KPIs-oriented literature review

Organization	Report/Initiative Title	Year(s)	Type of Transportation KPIs
International Transport Forum	ITF Transport Outlook reports	2017 onwards (e.g., ITF Transport Outlook 2017)	Transport developments, projections, CO2 emissions, logistics performance measurement methodologies
	Key Transport Statistics	2025 (2024 data)	data for more than a dozen selected indicators
	International Transport Forum Policy Papers	2016 onwards	Various transport policy indicators and analyses
	International Traffic Safety Data and Analysis Group (IRTAD)	annual	Road fatalities, serious injuries, safety performance indicators by mode and region
	Road Safety Annual Report	2024	road safety performance
United Nations	Sustainable Development Goals	Launched 2015	Includes indicators for monitoring progress in sustainable transportation, urban accessibility, and related areas
United Nations Economic Commission for Europe (UNECE)	Infocards / Transport Statistics Publications	2025 (2024 data)	Key transport indicators for each member State.
United Nations ESCAP	Sustainable Urban Transport Index (SUTI)	*from 2017*	*Accessibility, safety, affordability, reliability, emissions, land-use integration, governance indicators*
International Energy Agency	CO2 Emissions Statistics	After 2015	CO2 emissions from transportation
	Fuel Economy in Major Car Markets: Technology and Policy Drivers	2019 (covering 2005-2017)	Fuel economy in car markets
	The Future of Trucks	2017	Implications for energy and the environment from trucks
	Global EV Outlook	2019	Electric mobility scaling and outlook
World Bank	Logistics Performance Index	Updates include 2016, 2018; data processed for 2019-2021	Benchmarking trade logistics performance, including customs efficiency, infrastructure quality, and timeliness
	data.worldbank.org/indicator		transport indicators for investment, infrastructure, environment, trade
European Union	Measuring the progress towards a resource-efficient European Union under the Europe 2020 strategy	2017	Indicators related to ensuring efficient mobility and reducing greenhouse gas emissions from transport
	Environmental Performance Assessment of the Transport Sector in the European Union	2020 (covering 2015-2018)	Environmental performance indicators for the transport sector in EU countries
European Commission –	Statistical pocketbook 2025	2025	transport performance, infrastructure, safety, energy, environment

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Directorate-General for Mobility and Transport (DG MOVE)			
Eurostat	Transport Database	2025	infrastructure, economic performance, safety, environment
International Association of Public Transport (UITP)	Global Urban Mobility Indicators	2023	33 indicators for urban mobility
TomTom Traffic Index	Global Congestion Index Reports	Annual	Congestion levels, average travel time, peak vs. off-peak delays, efficiency trends
Asian Transport Observatory (ADB & ITF)	Urban Transport Indicators Database	from 2023	Traffic congestion, accessibility, public transport use, active mobility share
Asian Development Bank (ADB)	Asian Transport Outlook (ATO)	from 2021	Urban accessibility, congestion, modal share, CO_2 emissions, gender and inclusion KPIs.

Table 9.2. Organisational reports included in the KPIs-oriented literature review

9.2 Appendix B: KPIs' catalogue from literature review

Table 9.3 presents the entire KPIs' catalogue, organized according to thematic groups and subgroups.

No.	KPI	Definition
A	**Network/System performance**	
A.1	**Centrality**	
1	Betweenness Centrality	*Describes to what extent a particular node lies between other nodes in the network. A node with high betweenness centrality is on the shortest paths connecting many nodes, indicating its criticality.*
2	Closeness Centrality	*The inverse of the average shortest path distance from the node to any other node in the network.*
3	Degree Centrality	*A first centrality measure, defined as the number of links incident upon a node.*
4	Eigenvector Centrality	*The eigenvector centrality of a node is proportional to the sum of the centralities of the nodes it connects to.*
5	Commuter Flow Centrality	*The total number of commuters through a node per hour.*
A.2	**Capacity**	
1	Network Reserve Capacity	*Used to account for both route choice behaviour of travellers and congestion effects on networks.*
2	Transportation Network Resiliency Index	*Uses a fuzzy inference system to assess transportation resilience at a system level (Sun et al., 2020)*

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3	Network Resilience metric	*The expected value of the post-event demand divided by the pre-event demand for the network (Zhou et al., 2019)*
4	Friability	*Calculated using an undirected graph to represent roadway networks (related to resilience) (Faturechi & Miller-Hooks, 2014)*
5	Network Travel Time Resilience	*The ratio of travel time under normal conditions to the travel time in a disrupted scenario (Sun et al., 2020)*
6	System recovery rates	*One of the measures used to evaluate the recoverability of freight transport systems. These rates quantify the recovery speed and efficiency of system functionality, often expressed as the percentage of network functionality returning to its original or an acceptable level after a disruption (Sun et al., 2020; Zhou et al., 2019)*
7	Demand recovery	*This indicator assesses how quickly and completely the demand for transportation services returns to its operational levels following a disruption.*
8	Rapidity	*The speed of returning to the normal state, used to measure the resilience of transportation networks.*
9	Mobility throughput	*Mobility throughput: average multi-modal trips per capita per hour*
10	Network redundancy	*Redundancy ratio: dormant links / active links (pre and post- event scenarios)*
11	Community adaptation velocity	*Adaptation velocity: ∂(modal share diversity)/∂t based on household surveys*
12	Network entropy of mobility flows	*Shannon entropy of OD flows from OD matrix, normalised by maximum*
13	Event disturbance index	*Domain/recency/source weighted disturbance index*
14	Stress index	*Aggregates congestion and capacity strain; Reduced performance (speed) of the network links in a given area of interest considering their initial capacity*
B	**Ecological impacts**	
B.1	**Emissions**	
1	GHG Emissions	*The primary environmental impact being studied, referring to greenhouse gas emissions from the freight sector, mainly CO2. (Also listed as 'GHG Emissions' for Scope 1, 2, 3 and 'Annual greenhouse gases by transportation per capita', 'Annual greenhouse gases by transportation per area', 'Annual greenhouse gases by transportation per total energy used by transportation', 'Annual on-road greenhouse gases per total annual VKT', 'GHG emissions in transportation per capita', 'Share of GHG emissions in the transportation sector from total GHG emissions')*

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2	CO2	*Undesirable output variable, consumption calculated using CO2 emission factor. (Also listed as 'Carbon dioxide emissions', 'CO2 emissions from transportation', 'Carbon emissions per capita', 'Greenhouse gas emissions from transport per capita', 'Per capita emissions of greenhouse gases', 'Carbon emissions', 'Carbon dioxide gas emissions', 'Transport CO2 Emissions')*
3	Pollutant emissions from transport	*Analyses transport emissions of nitrogen oxides, non-methane volatile organic compounds, and particulate matter. (Also listed as 'Nitrogen oxide emissions', 'Total motor vehicle pollutant emissions', 'Air polluting emissions', 'Annual air pollution emissions by transportation per capita', 'Annual air pollution emissions by transportation per area', 'Annual air pollution emissions by transportation per total energy used by transportation', 'Annual on-road air pollution emission per total annual Vehicle Kilometres Travelled', 'Per capita emissions of global air pollutants', 'Air pollution', 'Air pollution rate')*
4	Average CO2 emissions per kilometre from new passenger cars	*Measures the average emissions of carbon dioxide per kilometre for new passenger cars.*
5	Energy consumption and polluting emissions	*Amount of energy used and pollutants released.*
6	Average density of pollutants generated by transport	*Concentration of pollutants from transport.*
7	Average exposure to transport-generated pollutants	*Average contact with pollutants originating from transport.*
8	Average exposure to overall pollutants	*Average contact with all types of pollutants.*
9	Average duration of the exposure to the pollutants.	*Average length of time exposed to pollutants*
10	Total polluting emissions	*Overall amount of pollutants released*
11	NOx emissions	*Emissions of nitrogen dioxide pollution*
B.2	**Noise pollution**	
1	Noise pollution	*Sound levels, often related to traffic. (Also listed as 'Noise emissions', 'Noise hindrance', 'Traffic noise pollution index', 'Average of traffic arterial noise', 'Control on Noise pollution')*
2	Population exposed to noise	*The percentage of the population or number of people affected by noise levels that exceed a certain limit value (Also listed as 'Number of people exposed to traffic noise', 'People exposed to noise >55 dB').*
B.3	**Ecological degradation**	
1	Ozone Layer Depletion	*The thinning of the stratospheric ozone layer, which leads to increased penetration of harmful UV-B radiation to the Earth's surface*

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2	Ecological footprint	*The amount of biologically productive land and water required to generate the resources consumed and to assimilate the waste produced, using contemporary technologies. This metric is calculated in hectares per capita and the results are often presented in logarithmic form.*
3	Water pollution index	*A metric used to quantify the level of contamination within water bodies. For example, the Wastewater Quality Index serves as an indicator for assessing sanitary effluents from industries, aiding in reliable and sustainable water reuse.*
4	Waste from transport sector	*The amount of solid waste produced by the transportation sector.*
5	Preservation of habitats / biodiversity	*Measures such as ensuring an ecosystem's capacity to support ecological processes and diverse communities of organisms, often quantified by an 'Intactness' or 'biodiversity' score or index.*
C	**Transport Modes, Travel Times and Characteristics**	
C.1	**Modal split**	
1	Modal split	*Personal travel mode split. Also, the proportion of different transport modes used for travel.*
2	Personal travel mode split	*Share of non-motorized, automobile, public transport.*
3	Freight mode split	*The percentage share of each inland mode of transport in total transport, expressed in ton-kilometres across different transportation modes such as truck, rail, ship, air.*
4	Cross-layer synergy	*Graph-based proximity between modal layers (e.g., bike–rail hubs); sum of inverted transfer distances*
C.2	**Travel times and delays**	
1	Vehicle travel times	*Often defined as the total time a vehicle spends on a journey, including driving, idling, and waiting, from origin to destination*
2	Passenger waiting times	*The duration passengers spend awaiting a vehicle, with stakeholders recommending metrics like an average of 20 minutes during peak hours to assess service frequency*
3	Waiting time savings	*The reduction in the amount of time passengers spend awaiting a vehicle*
4	In-vehicle time	*Time spent by passengers inside the vehicle.*
5	Average commute travel time	*The average actual commuting travel time for commuters across all available transport modes. This measure should encompass the total door-to-door travel time, including delays, parking time for cars, and for public transport, it covers travel to a transit stop, waiting time, on-board travel, transfers, and travel from the stop to the destination*
6	Travel time	*The total duration a vehicle spends on a journey, encompassing driving, idling, and waiting from origin to destination.*

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7	Average daily trip duration per person	*The average time a person spends travelling per day.*
8	Average trip duration per kilometre	*The average time taken to travel one kilometre.*
9	Average trip duration per vehicle	*The average time a vehicle spends on a trip.*
10	Average travel time using public transport system to the desired point of interest	*The average time it takes to reach a destination using public transport.*
11	Average travel time using road network to the desired point of interest	*The average time it takes to reach a destination using the road network.*
12	Time required to switch between transport modes	*The duration needed to transfer from one mode of transport to another.*
13	Delays in the transport system	*The amount of time lost due to transportation delays.*
14	Network delays indicator	*Measures delays within the transport network.*
15	Lost time	*Time not productively used due to transport issues.*
16	Congestion and delays	*Calculates the impact of congestion and delays in the urban mobility system, which can be measured by time lost in congestion or the extent of congestion in time or space.*
17	Congestions in the transport system	*The duration of traffic congestion within the transport system.*
18	Time-to-Collision	*Defined and calculated only when the speed of a trailing vehicle (ego vehicle) exceeds that of the leading vehicle, representing the time remaining until a collision if their current speeds and trajectories are maintained*
19	Time Delay Centrality	*Describes the extent of delay to people affected through a linear combination of the additional time required by every trip.*
20	DelayFlow Centrality	*It is a time-based centrality measure that accounts for the contribution of commuter flow to time delay within a network (Sun et al., 2020).*
21	Passenger Delay	*A performance indicator used to quantify the resilience of mass railway transportation systems. Defined as the average difference between the actual travel time and the scheduled travel time for passengers*
22	Average Yearly Commuter Delay	*Average amount of additional time or delays experienced by individual commuters.*
23	Delay	*The average difference between the actual and scheduled travel time, or more generally, the additional time experienced by commuters due to various factors like congestion.*
24	Travel Time Reliability	*Quantifies the additional time needed to complete a trip compared to normal conditions.*
C.3	**Travel demand**	

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1	Transport demand	*The overall need or desire for the movement of people and goods within a transportation system. It is often measured by the volume of passenger and freight travel, influenced by factors such as population, economic activities, and societal needs.*
2	Non-motorized transport demand	*The overall need or desire for travel accomplished through active, non-motorized modes such as walking, cycling, skateboarding, and manual wheelchair use.*
3	Motorized transport demand	*The overall need for movement of people and goods that relies on engine-powered vehicles, differentiating it from non-motorized transport. This indicator typically quantifies passenger travel (e.g., in passenger-kilometres) and freight movement (e.g., in ton-kilometres) carried by motor vehicles.*
4	Demand for Freight Services	*Overall need for goods transportation.*
5	Inter-City Passenger Demand	*Relates to the demand side of the intercity passenger transport system.*
6	PRT Travel demand	*The total number of trips made using private vehicles per day, reflecting individuals' reliance on personal modes of transport for their mobility needs. It is commonly measured by the daily number of trips per person or vehicle-kilometres, encompassing travel by privately owned cars and motorcycles.*
7	PT demand	*The overall volume of passenger travel utilizing public transportation modes like buses, trains, and metros. It is commonly measured by indicators such as passenger-kilometres or ridership numbers, reflecting the community's need for and reliance on shared transit services.*
8	Demand management	*Composite index in [0,1] from time-varying fares via min–max by mode/time, aggregated with ridership weights*
C.4	**Transport characteristics**	
1	Expected Fraction of Demand Satisfied in Post-Disaster Network	*The proportion of demand that can be met by the transportation network after a disaster, considering specific recovery costs. This indicator typically involves comparing the maximum demand satisfied for an origin-destination pair in the post-disaster network () against the demand satisfied in the pre-disaster network () (Zhou et al., 2019).*
2	Average public transport speed	*The average commercial speed of public transport, typically measured in kilometres per hour, based on the scheduled operation.*
3	Average distance covered by daily PT trips	*The average distance travelled per day by individuals using public transport modes.*
4	Quantity of PT trips / Number of daily PT trips	*The total number of journeys or boardings made by individuals using public transportation modes.*
5	Total number of passenger and ton kilometres	*The cumulative distance travelled by passengers and goods.*

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6	Daily average distance travelled by motorised and non-motorised trips	*The typical distance covered each day by individuals across all modes of transport. This includes distances travelled by vehicles with engines (motorised) as well as active forms like walking and cycling (non-motorised).*
7	Total vehicle-km travelled	*The total distance covered by vehicles. It is defined as the movement of one vehicle the distance of one kilometre, representing the actual distance run, and includes movements of empty road motor vehicles.*
8	Transit passenger km travelled	*The total distance covered by passengers using public transport, calculated by multiplying the number of passengers by the distance each travels.*
9	Freight ton-km	*A unit of measure for goods transport, representing the movement of one tonne of goods over a distance of one kilometre.*
10	Vehicle Occupancy / Passenger Load	*The ratio of the number of passengers to the sitting capacity of the vehicle, often estimated as the Load Factor.*
11	More than One Person per Vehicle Index	*An indicator used to measure car occupancy, reflecting the number of people traveling together in a single vehicle (Vidović et al., 2019). This metric helps assess vehicle utilization and can be differentiated by factors such as time of day or trip purpose (Gillis et al., 2015).*
12	Ridership	*The total number of passengers utilizing a public transportation service. It is a crucial indicator for assessing the volume of passenger usage and the overall performance and impact of public transport systems.*
13	Trip/Travel Rate	*The frequency of travel, representing the number of trips an individual or household makes over a defined period. It can also be expressed as the average number of daily trips per person or per transportation mode.*
D	**Transport safety**	
D.1	**Traffic accidents**	
1	Traffic accidents	*Annual number of traffic crashes and fatalities per 100,000 residents. (Also listed as 'Annual number of traffic crashes and fatalities per 100,000 residents')*
2	Traffic conflicts	*Number of collisions between motor vehicles and non-motorized travellers.*
3	Injury Severity	*KABCO Scale Outcomes, a classification system for injury severity.*
4	Accident rate	*Typically measured in various ways, such as fatalities per capita, per vehicle-mile travelled, or per registered vehicles, and is influenced by factors like traffic volume and car ownership*
5	Fatalities per billion passenger journeys	*The number of deaths relative to the total distance travelled by passengers*
6	People dead in road accidents	*Number of fatalities resulting from road accidents.*

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7	Annual traffic fatalities per capita	*It measures the mortality rate of a population due to road accidents, expressing the number of road deaths relative to the total number of inhabitants. This metric quantifies the average citizen's overall risk of being killed in traffic and allows for comparisons with other causes of death*
8	Crash Count	*Number of crashes or accidents occurring within a given period and area.*
9	Traffic related injuries and fatalities	*The number of casualties resulting from traffic accidents.*
10	Casualties of traffic accidents	*Individuals injured or killed in traffic incidents.*
D.2	**Security**	
1	Intentional security	*Number of violent incidents; Number of thefts from vehicles.*
2	Unintentional security	*Accident rate for 10,000 vehicles; Traffic accident casualties per capita.*
3	Security	*The perceived protection against various threats, including offenses against property, passengers, and operatives, and intentional attacks like physical or cyberattacks. Based on a population survey for public perception.*
D.3	**Safety control**	
1	Engineering safety control	*Incorporating safety features into design.*
2	Laws and regulations	*Enacting and enforcing specific laws and regulations to govern safety.*
3	Education (for road safety)	*Its aim is to enhance awareness of road safety signs, regulations, hazards, and risks, and to extend this training to the broader community. Based on a population survey.*
4	Residents' perception of traffic safety	*Residents' perception on the statement: I feel safe traveling with other drivers on the road. Based on a population survey.*
5	Residents' perception of safety within their neighbourhood	*Residents' perception on the statement: I feel safe in my neighbourhood during the day/night. Based on a population survey.*
6	Residents' perception of safety in downtown	*Residents' perception on the statement: I feel safe walking alone downtown during the day/night. Based on a population survey.*
7	Safety and Efficiency Index	*A dimensionless indicator ranging from 0 (unsafe and inefficient) to 1 (safe and efficient) that integrates efficiency with a safety component based on Time-to-Collision (Andreotti & Selpi, 2025).*
8	Road cargo turnover	*A control variable quantifying the total amount of cargo transported over a certain distance within a given period.*
9	Per capita car ownership	*The number of vehicles owned per person within a population, serving as an indicator of private vehicle prevalence.*
10	Alcohol/Drug Use	*A critical indicator in crash severity studies, reflecting a key risk factor in road safety.*
11	Vehicle Characteristics/Type/Year	*Characteristics of vehicles involved in crashes.*

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12	Collision Type	*Manner in which a collision occurred.*
13	Type of Violation/Accident Reason	*Cause or contributing factor to an accident.*
14	ICT uptime	*Percentage uptime of telecommunication- control systems critical to transport and mobility operations*
E	**Accessibility, costs, and satisfaction**	
E.1	**Accessibility**	
1	Transport accessibility	*Share of population living within 300–500 m from public transport stations. (Also listed as 'Accessibility to public transport station', 'Public space accessibility', 'Accessibility by walking', 'Accessibility by cycling', 'Accessibility to PT', 'Access to transit', 'Access to public transport', 'Accessibility - Transit', 'Accessibility - Pedestrian / Bicycle', 'Access to Destinations', 'Access to jobs and activities', 'Equitable access', 'Job Access', 'Change in the number of jobs accessible by public transport within 45 minutes travel time', 'Percentage of population in the 10 percent most deprived areas of London within 45 minutes travel time of international and metropolitan centres', 'Bus accessibility index', 'Accessibility to employment', 'Accessibility to public services', 'Access to mobility services', 'Accessibility for mobility impaired groups').*
2	Land use accessibility	*Number of public spaces within 10-min walk.*
3	Accessibility by car-driving	*This indicator can be measured by assessing the percentage of the population with access to a car or the travel time required to reach specific locations by car.*
4	Accessibility for emergence rescue	*It measures the timeliness of disaster relief efforts and the ability to access necessary emergency infrastructure within a given spatial distribution.*
5	Destination Accessibility	*Total number of direct destinations accessible.*
6	Equity penalty	*Stratified Theil-T of accessibility minutes across income or neighbourhood groups*
E.2	**Public perception and satisfaction**	
1	Feeling of safety during traveling	*The proportion of individuals who feel protected from crime, threatening behaviour, and general safety concerns, typically measured through surveys and public perception studies.*
2	Perception of comfort	*This indicator evaluates the physical and mental comfort experienced by individuals within a system or environment, and it is typically measured through reported satisfaction in surveys.*
3	Perceived beauty	*The subjective aesthetic appreciation individuals have for urban spaces and environments, assessed by conducting surveys.*
4	Satisfaction of the public spaces	*Perception to gauge contentment with public areas assessed by conducting surveys.*
5	Mobility satisfaction	*An indicator used to assess individuals' contentment with their mobility experiences, often measured through surveys using a five-point Likert scale.*

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6	Infrastructure accessibility satisfaction	*How content individuals are with the ease and quality of reaching destinations and services via available infrastructure. This indicator is typically evaluated through surveys, taking into account factors like physical accessibility, maintenance, and the specific needs of diverse user groups, such as those with disabilities. Assessed using a five-point Likert scale.*
7	General satisfaction	*An indicator that assesses an individual's overall contentment or happiness with a given system, service, or experience. It is typically evaluated through surveys and public perception studies to gauge the general public's overall sentiment.*
8	Quality of public area	*How the public perceives and evaluates various aspects of public spaces, such as aesthetics, cleanliness, and functionality, to gauge overall satisfaction and liveability based on a population survey for public perception.*
9	Population unsatisfied with the transport-related emission and noise levels	*The proportion of residents who express discontent regarding the environmental impacts of transportation, specifically concerning noise and emissions.*
10	Population that considers the transport – related consumption of land to be appropriate	*The proportion of residents who deem the land used for transportation infrastructure as suitable.*
11	Population that considers the sustainable energy sources available	*The proportion of people who believe sustainable energy sources are accessible.*
12	Overall quality of life in the city	*The perceived well-being, encompassing their satisfaction with physical, material, social, and emotional aspects of urban living (Kutela et al., 2025). It is a complex, multidimensional indicator reflecting happiness, health, and personal fulfilment, often influenced by the social impacts of mobility, including safety and health (Gillis et al., 2015; Heidari et al., 2024).*
13	Satisfaction with lighting	*Residents' satisfaction with lighting conditions*
14	Satisfaction with sidewalks	*Residents' satisfaction with the condition of sidewalks*
15	Satisfaction with street conditions	*Residents' satisfaction with overall street conditions.*
16	Satisfaction with traffic flow on major highways	*Residents' satisfaction with traffic flow in highways.*
17	Satisfaction with traffic flow on major city streets	*Residents' satisfaction with traffic flow on major city streets.*
8	Satisfaction with on-street bicycle accessibility	*Residents' satisfaction with bicycle lanes.*
19	Satisfaction with timing of traffic signals on city streets	*Residents' satisfaction with traffic signal timing.*
E.3	**Transport costs and affordability**	
1	Passenger waiting costs	*A cost-related indicator that quantifies the financial or economic disutility incurred by passengers due to time spent waiting for public transport.*

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2	Vehicle running and depreciation costs	*Total costs associated with vehicle operation and wear.*
3	Personnel costs	*Expenses related to staff. These costs are generally considered part of the operating or running costs of a transport system, encompassing wages and social insurance contributions .*
4	Running costs	*Operational expenses, encompass the direct costs associated with operating a transport system. These typically include expenditures such as wages, social insurance contributions, fuel, materials, supplies, and various taxes and levies.*
5	Total Recovery Effort	*The resource expenditures and costs attributed to recovery activities within resilience assessment, particularly for infrastructure systems. This indicator is utilized to quantify the expenses incurred during the restoration phase, often calculated by summing the costs of individual repair tasks (Ahmadi et al., 2021).*
6	Net Present Value	*A quantitative indicator that assesses the financial or economic performance of an investment alternative. It calculates the discounted net monetary value of an investment, reflecting the difference between discounted total benefits and costs over time.*
7	Travel costs	*The total financial expenditures incurred for a journey, encompassing direct expenses like fuel, fares, and tolls. These costs can also include indirect expenses such as vehicle depreciation, insurance, and maintenance, reflecting the overall economic impact of transportation.*
8	Parking fees	*Charges imposed for the use of parking spaces, serving as an economic instrument within urban mobility systems.*
9	Parking costs of the employees	*The expenses individuals incur for parking their vehicles, typically associated with their commute to work or during work-related activities.*
10	Household expenditure	*Spending by households. This includes costs such as the purchase and operation of transport equipment, as well as payments for transport services. It can also be viewed as a share of income spent on transport, which can indicate "transport poverty" if it exceeds a certain percentage.*
11	Fuel price	*Cost of fuel. (Also listed as 'Fuel costs', 'Gasoline price', 'Annual total cost spend for gasoline price including taxes per total annual VKT', 'Annual total cost spend for gasoline price including taxes per capita')*
12	Daily parking ticket costs	*Expenses for daily parking tickets.*
13	Residents' parking ticket expenditure	*Spending by residents on parking tickets.*
14	Fixed costs of a car allocated to urban mobility	*Fixed expenses related to car ownership and urban travel.*

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15	Expenditure for public transportation	*The financial outlays made for public transit systems, encompassing operational costs and infrastructure investments. These expenditures often include subsidies necessary to cover service costs not entirely met by fare revenues.*
16	Cost operation	*The expenses incurred in the day-to-day functioning of a system, contributing to its overall operational costs. These costs are essential for maintaining financial efficiency and can encompass various expenditures related to services and infrastructure.*
17	Cost of property damage and treatment	*The financial expenditures incurred due to damage to vehicles and infrastructure, as well as the medical costs associated with injuries resulting from incidents such as traffic accidents or disasters.*
18	Private transportation cost	*The financial outlays associated with a private transportation subsystem, such as expenses for fuel consumption and vehicle maintenance.*
19	Environmental Cost Indicators	*Converts environmental impacts into a single weighting factor expressed in monetary value.*
20	Price	*Typically refers to the Manufacturer's Suggested Retail Price or the annual lease price of a vehicle, representing the monetary value associated with acquiring or using a product or service.*
21	Total Cost of Ownership	*Includes various expenses (depreciation, taxes, fuel, insurance, etc.).*
22	Annual infrastructural investments	*Financial outlay for infrastructure projects.*
23	Annual transportation expenditures per capita	*Also listed as 'Annual transportation expenditures per GDP per capita', 'Annual public transportation expenditures per capita', 'Annual public transportation expenditures per total unlinked passenger trips by public transportation'*
24	Annual public transportation revenues per public transportation expenditures	*The financial self-sufficiency of public transportation systems by comparing the total revenue generated from operations against the total expenditures incurred for providing those services.*
F	**Land use and Infrastructure**	
F.1	**Transport infrastructure**	
1	Infrastructure spatial features	*The physical characteristics of the transportation network, such as the total length of operational roads. This indicator can be measured in kilometers to quantify the extent of the road infrastructure.*
2	Length of total operational roads	*the total physical extent of all roads within a given area that are currently in use. This indicator is typically measured in kilometers and quantifies the overall size and reach of the road infrastructure.*

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3	Ratio of roadways of specific standards	*An indicator that quantifies the proportion of a road network that meets certain defined criteria, such as safety ratings or specific design standards. This ratio can be expressed as a percentage of the total road network length or traffic volume that adheres to particular quality or functional classifications, such as motorways or urban roads*
4	Network density / Total roads length per area	*The extent of a network, typically expressed in kilometers per square kilometer (Km/km2).*
5	Public transport stops	*Further measured by passenger shelter, cleanness, seats, lighting, CCTV, timetable, real-time display, Wi-Fi, etc.*
6	Public transport route network	*Further measured by multimodal compatibility, the system of ticket sale and distribution, distance to a bus stop, etc.*
7	Public transport efficiency	*The occupancy rate of buses, reflecting how effectively public transport vehicles are utilized. It quantifies the proportion of available capacity being used by passengers.*
8	Public transport availability	*The extent to which public transportation is accessible to a population, often measured by the proportion of people living within a certain range of public transport stops.*
9	Total highways length per area	*The total physical length of highways within a defined geographical region, typically expressed in kilometers per square kilometer (Km/km^2).*
10	Urban roads length per area	*The total physical length of roads within a defined urban expanse, typically expressed in kilometers per square kilometer (Km/km^2).*
11	Total highways length per capita	*The total extent of highway infrastructure available per individual within a given area.*
12	Total urban roads length per capita	*The total physical length of urban roads within a defined urban area, relative to its population.*
13	Share of exclusive bus transit route over bus routes	*Percentage of exclusive and controlled right-of-way motor bus transit route per total motor bus transit route length*
14	Share of exclusive bus transit route over total road network	*Percentage of total motor bus transit route length per total roads length*
15	Bus route length per area	*The extent to which an area is covered by public transportation infrastructure, often expressed as kilometers of bus routes per square kilometer.*
F.2	**Land use and space consumption**	
1	Land use diversity / mix	*An entropy-based measure on the variety of different land uses.*
2	Proportion of land devoted to transport facilities	*The proportion of land utilized by all urban transport modes, covering direct infrastructure like roads and rails, and indirect uses such as parking spaces. This indicator quantifies the physical space consumed by the transportation sector, commonly measured in square meters per capita.*

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3	Ratio of agriculture land used for transport facilities to total land take	*The proportion of land initially designated for agricultural purposes that has been converted or occupied by transportation infrastructure, relative to the overall land area consumed by all transport facilities.*
4	Ratio of green spaces used for transport facilities to total land take	*The proportion of green land area that has been converted or occupied for transportation infrastructure, relative to the total land area consumed by all transport facilities.*
5	Space consumption in terms of the transport infrastructure / Transportation infrastructure land	*The land area utilized by all urban transport modes, encompassing direct infrastructure like roads and rails, and indirect uses such as parking spaces.*
6	Urban compactness	*The concentrated spatial arrangement and mixed-use development of an urban area.*
7	Population density	*The concentration of people per unit area.*
8	Per capita road area	*The amount of land dedicated to road infrastructure per individual resident, reflecting the coverage of the urban road network.*
9	Travelling- and parking-related public space consumption	*Amount of public space utilized for travel and parking.*
10	Consumption of public space for mobility needs	*Public space utilized for movement and travel.*
11	Consumption of public space for parking needs	*Public space utilized for vehicle parking.*
12	Transport-related land consumption per inhabitant	*The total land area utilized by all urban transport modes, such as roads, rails, and parking, normalized by the number of residents.*
13	Portion of land paved for transport facilities	*The percentage of a city's total land area that is covered by constructed transportation infrastructure. This includes paved surfaces like roads, streets, and parking areas, quantifying the physical footprint of the transport system.*
G	**Urban development and growth**	
G.1	**Energy consumption**	
1	Energy Utilization	*It is typically quantified in units such as gigawatt-hours, electrification network percentage, watt-hours per seat per kilometer, and megajoules.*
2	Renewable Energy Utilization	*Represents the consumption of renewable energy as a percentage of the total final energy consumption. (Also listed as 'Utilisation of the sustainable energy sources', 'Share of energy from renewable sources in transport', 'Annual renewable energy consumption by transportation per capitac', 'Annual renewable energy consumption per total transportation energy consumptionc')*

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3	Energy efficiency	*How effectively energy inputs are utilized to achieve transport outputs, commonly measured as total energy consumed per passenger-kilometer or ton-kilometer across various modes.*
4	Fossil fuel consumption in transportation	*The amount of energy derived from fossil fuels (such as oil, gasoline, and natural gas) used to power various transport modes.*
5	Energy consumption in transportation	*The total energy utilized by various transport modes, encompassing both fossil fuels and renewable sources. This indicator is typically quantified in units like megajoules or kilowatt-hours, and can be measured per capita, per passenger-kilometer, or per ton-kilometer to assess the efficiency and environmental impact of the mobility system.*
6	Fuel consumption per each car	*The amount of fuel a single vehicle uses over a given distance, typically measured in units like liters per 100 kilometers or miles per gallon.*
7	Per capita transport energy consumption	*The total energy utilized by various transport modes per individual resident, reflecting the energy intensity of mobility within a population.*
8	Energy consumption per ton – kilometre	*Energy used to transport one ton over one kilometre.*
9	EV-grid resilience	*Composite of uptime and spare capacity for EV charging during events*
G.2	**Urban and spatial planning**	
1	Urban density	*The concentration of population in urban areas, which is a key indicator for assessing land resource utilization and urban planning aspects.*
2	Density of public services within all area services	*The concentration and availability of public services throughout a given geographical area. This indicator assesses how well essential services are distributed and accessible to the population within urban environments.*
3	Greening coverage of built-up areas	*Proportion of green space within built-up area.*
4	Green space area	*Total land cover of green space within the urban areas. (Also listed as 'Total area of green spaces and parks /Green areas as a share of the total urban area').*
5	Green space area per capita	*The total amount of green space within a city divided by its population.*
6	Spatial openness	*The number of entrances of visibility along the street. This indicator, measured in entrances per meter, quantifies the extent of open sightlines within a given urban space.*
7	Land use compactness	*The percentages of built-up areas within urban growth boundaries. This indicator assesses how efficiently urban land is utilized, reflecting the concentration of development within a defined urban footprint.*
G.3	**Economic growth**	
1	GDP per capita	*Gross Domestic Product per person.*

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2	Urban growth generated by transport sector	*Share of GDP contributed by the transportation sector; Employment in transportation. (Also listed as 'Share of GDP contributed by transport sector/Share of GVA generated by transport sector')*
3	Net public finance	*The net result of government's revenues from transport-related taxes and charges minus costs, per unit of GDP.*

Table 9.3. KPIs' catalogue

9.3 Appendix C: Pseudocode for the development of the AntifragiCity KPIs

Table 9.4 provides the pseudocode for the automated selection and refinement process, detailing the algorithmic steps involved in filtering, scoring, and ultimately selecting. It is a Python-like pseudocode followed by some targeted step-purpose comments.

```
function defineContextSpecificKPIs(city_context, kpi_catalogue):

  # STEP 1: Define strategic vision and prioritize thematic areas
  # Purpose: Ensure KPI selection aligns with long-term urban goals
  strategic_vision = extractStrategicVision(city_context)
  thematic_areas = identifyThematicAreas(strategic_vision)
  weighted_themes = applyAHP(thematic_areas)
  prioritized_themes = selectTopThemes(weighted_themes)

  # STEP 2: Define operational objectives and KPI groups
  # Purpose: Translate strategic vision into measurable objectives
  operational_objectives = defineOperationalObjectives(prioritized_themes)
  kpi_groups = mapObjectivesToKPIGroups(operational_objectives)
  weighted_groups = applyAHP(kpi_groups)
  prioritized_groups = selectTopGroups(weighted_groups)

  # STEP 3: Specify requirements and operational scenarios
  # Purpose: Filter KPIs based on feasibility and context constraints
  constraints = assessUrbanConstraints(city_context)
  scenarios = defineOperationalScenarios(city_context)
  requirements = buildRequirementFilters(constraints, scenarios)

  # STEP 4: Select relevant KPIs from the catalogue
  # Purpose: Build a tailored KPI set aligned with goals and feasibility
  candidate_kpis = []
  for kpi in kpi_catalogue:
    if matchesThemes(kpi, prioritized_themes) and
      matchesGroups(kpi, prioritized_groups) and
      satisfiesRequirements(kpi, requirements):
       score = evaluateKPI(kpi, strategic_vision, constraints)
       candidate_kpis.append((kpi, score))
```

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```
selected_kpis = selectTopKPIs(candidate_kpis)

# STEP 5: Refine, validate, and extend KPI set
# Purpose: Remove redundancy, ensure robustness, and enable adaptability
refined_kpis = removeRedundancy(selected_kpis)
validated_kpis = validateWithStakeholders(refined_kpis)

if needsIteration(validated_kpis, city_context):
   # Restart process to adapt to new goals or data
   return defineContextSpecificKPIs(city_context, kpi_catalogue)

return validated_kpis
```

Table 9.4. Pseudocode for the development of the AntifragiCity KPIs

9.3.1 Key Requirements & Constraints Analysis

Core requirements

- A multi-step, structured methodology for defining context-specific KPIs
- Five consecutive but optionally cyclic steps
- Alignment with:
    - Strategic urban vision
    - Prioritized thematic areas and KPI groups
    - Contextual constraints (data, resources, feasibility)
- Use of AHP-like weighting for prioritization at multiple stages
- Filtering KPIs from a catalogue based on requirements
- Final refinement, validation, and iteration

Constraints

- Data availability and quality
- Operational and technical feasibility
- Local urban context variability
- Need for adaptability over time

9.3.2 Data Structures Used

- Lists
    - Thematic areas, KPI groups, KPIs' catalogue

- Dictionaries / Objects

    - City context (vision, constraints, data availability)
    - KPI metadata (theme, group, feasibility, data source)

- Tuples
    - (kpi, score) pairs for ranked selection
- Graphs or Matrices (conceptual)

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- Used internally by AHP-like weighting methods

These structures balance flexibility, traceability, and scalability.

9.3.3 Algorithm & Thought Process Explanation

The methodology follows a top-down decision refinement approach:

1. Start with strategic intent
2. Gradually increase granularity and specificity
3. Apply multi-criteria decision-making (AHP) at each prioritization layer
4. Enforce real-world constraints before final selection
5. Validate and iterate to ensure long-term relevance

This mirrors real urban planning workflows while remaining algorithmically structured.

9.3.4 Step Purpose (Embedded in Pseudocode)

Each step is explicitly commented to explain:

- Why it exists
- What decision it supports
- How it filters or refines KPIs

This ensures transparency and ease of implementation.

9.3.5 Time Complexity Analysis

Let:

- T = number of thematic areas
- G = number of KPI groups
- K = number of KPIs in the catalogue

Overall complexity

- AHP prioritizations: $O(T^2 + G^2)$ (pairwise comparisons)
- KPI filtering and scoring: $O(K)$
- Refinement and redundancy checks: $O(K^2)$ in worst case

Total worst-case time complexity:
$O(T^2 + G^2 + K^2)$

In practice, T and G are small, making performance dominated by KPI refinement.

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